

Every Countable Turing Ideal as Coarse Common Information

Exact spectra and Turing-cone embeddings with a fixed core;
effective exact pairs, layered recovery, and prescribed jumps

Merged from three AI-assisted research manuscripts prepared for
Vladimir Reshetnikov, all dated 28 September 2026 (batch 37, manuscripts 12, 08, 10)

Abstract

For a set $X \subseteq \mathbb{N}$, let $\text{Core}(X)$ be the sets computable from every coarse description of X . We prove that the possible cores are exactly the countable Turing ideals. More precisely, a positive-density column code for a sequence (A_i) has core equal to the ideal generated by that sequence, and its entire representative spectrum is characterized by a single computable array whose rows eventually stabilize as whole reals. The quantifier order in this characterization is essential; merely computing each A_i separately is not enough.

We then separate common information from the difficulty of producing a representative. If Y can be approximated arbitrarily closely in upper-density distance by computable sets, adjoining Y preserves the core of every X . For the classical dyadic code $\mathcal{R}(H)$ this gives the exact operation

$$\text{Core}(X \oplus \mathcal{R}(H)) = \text{Core}(X), \quad \text{Spec}(X \oplus \mathcal{R}(H)) = \text{Spec}(X) \cap \{\mathbf{b} : H \leq_{\mathbf{T}} B'\}.$$

Here B is any representative of $\mathbf{b}$. Relative jump inversion turns this identity into a join-preserving order embedding of the Turing cone above $\deg_{\mathbf{T}}(X')$ into both the uniform and the nonuniform coarse degrees with core $\text{Core}(X)$, above the coarse degree of X . Every fixed-core collection therefore contains continuum many distinct representative spectra and has no countable cofinal subset. The proofs are explicit, including the analytic locality lemma, the coding estimates, and the jump-inversion argument.

Merged report. Part I is the manuscript above (source 12, the base). Part II (source 08) realizes every countable ideal by a two-level code whose descriptions are Medvedev equivalent to arrays of finite variants of the generators, builds a perfect family of pairwise exact descriptions with jumps below $\bigoplus_m (\bigoplus_{n < m} A_n)'$ and one common effective null envelope, and constructs, inside every core fiber, exact pairs with any prescribed common jump above the jump of the presentation. Part III (source 10) realizes every countable ideal by a layered factorial-block carrier, proves an ideal-protected fusion theorem giving a second proof without cone-avoiding compactness, and locates each level of the ideal in a robustness filtration within a factor four in radius. Results proved by more than one source are printed once and credited; genuinely different proofs are kept as marked second routes. The core-realization theorem was already proposed, with a proof sketch, as target C4 of the programme's research plan, and its column mechanism has a published antecedent in Hirschfeldt–Jockusch–Schupp [4].

Scope and status. This manuscript answers the core-realization question left open in the inspected ProvelT research synthesis. Its coding ingredients have substantial published antecedents, which are credited below. The paper supplies proofs of the stated conclusions,

but makes no claim of established worldwide priority, independent refereeing, or Lean verification. In particular, it does not resolve the general effective-dense least-representative question or classify all possible coarse spectra.

[merge] Status of the merged report. This is research report 12 of the ProveIt CoarseDegrees programme, built from three AI-assisted manuscripts (sources 12, 08 and 10; Appendix C). It is unrefereed and **not formalized**: none of its new theorems has a Lean or Rocq proof. The repository's Lean library `Computability/TuringDegrees/Lean/CoarseDegrees` proves some finite coding ingredients (Section 1.4). All three sources describe core realization as answering a question left open in the programme's synthesis; the programme's research plan, at the same pinned commit, had already stated the realization theorem as target C4 with a proof sketch [7], so what is new here is the verification of that sketch, its independent proofs, and the structural results built on it.

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Part I

Source 12: ideal realization, whole-row spectra, and fixed-core embeddings

1 The target and the contribution

1.1 The repository question

The ProvelIt synthesis [1] studies the failure of least Turing-degree representatives in coarse equivalence classes. Its final subsection, “What is not settled”, specifically asks which countable ideals can occur as cores and which upward-closed sets of degrees can occur as representative spectra. We answer the first question completely and give a family of exact constructions for the second. The repository was inspected at commit 1a1396d4d3a2; the full commit identifier and source path appear in Appendix B.

The distinction between these two questions is fundamental. The core records what *every* description can compute. The spectrum records which oracles can produce *some* description. A lower bound common to all descriptions need not itself produce a description. A principal core is therefore not a certificate of a least representative. The repository already demonstrates this for every principal ideal [1]; we do not count that observation as a new result.

[merge] The synthesis is not the only place the question appears. At the same pinned commit the programme’s research plan [7] lists it as target C4, “Is every countable Turing ideal a core?”, rated “V: proof sketch supplied; novelty unchecked”. Its Proposition (proposed) uses the column code $X(2^k(2t+1)-1) = \mathcal{J}(A_k)(t)$, which is the shorter code of Remark 5.2 below, and its sketch is the two-inclusion argument of Theorem 5.1 with the robust radius taken from the synthesis. None of the three sources cites the plan. The plan’s C4 tasks are addressed here as follows: the sketch is checked (Theorem 5.1, and independently in Parts II and III); the literature search finds the antecedent [4, Lemma 3.10 and Theorem 3.11] (Section 1.3); for which sequences the class has a least degree is decided for the code $X_{\vec{A}}$ by Theorem 6.2 with Corollary 2.5; and the uniform core defined with a single functional is trivial (Proposition 18.1).

Gerdes’s Question 7 [6] places least representatives and spectral classification in a broader comparison of robust reducibilities. The present paper concerns binary coarse descriptions, where erroneous values are allowed on a density-zero set. It does not silently transfer its constructions to error-free partial descriptions.

1.2 Four statements proved here

The central results are the following, with all notation defined in the next section.

Realization. For every countable Turing ideal $\mathcal{I}$ there is a binary set X with $\text{Core}(X) = \mathcal{I}$. The construction works from any sequence of generators and gives $X \equiv_T \bigoplus_{i \in \mathbb{N}} A_i$ for that presentation (Theorem 5.1). Since every core is a countable ideal, this is an exact classification of the possible cores.

Presentation-sensitive spectra. For our code $X_{\vec{A}}$, an oracle B computes a coarse description exactly when it computes a total binary array $f(i, s, n)$ satisfying

$$(\forall i)(\exists s_i)(\forall s \geq s_i)(\forall n) \quad f(i, s, n) = A_i(n).$$

The stabilization is uniform over the entire real in each fixed row, but the threshold need not be uniform over the rows (Theorem 6.2). A concrete example explains why the ideal alone cannot determine this spectrum.

Preserving the core while changing the spectrum. For arbitrary X, H , the operation $T_X(H) = X \oplus \mathcal{R}(H)$ preserves $\text{Core}(X)$ and intersects its spectrum with the jump cone determined by H (Theorem 8.4). This is a special case of a general invariance theorem for joins with sets of coarse computability bound 1 (Theorem 8.2).

Universality within a fixed core. For $H, K \geq_T X'$,

$$H \leq_T K \iff T_X(H) \leq_{uc} T_X(K) \iff T_X(H) \leq_{nc} T_X(K).$$

The induced maps preserve binary joins. Thus an entire Turing upper cone lives above every coarse degree without changing its core (Theorem 9.2). This is stronger than producing a pair of inequivalent examples with the same common information.

1.3 What comes from earlier work

The cone-avoiding compactness theorem of Hirschfeldt, Jockusch, Kuyper, and Schupp [2, Theorem 3.7] is the source of the robust-radius principle used here. The prefix-density proof in Section 3 follows the locality approach in the repository synthesis [1]. The classical dyadic coarse-computability criterion is due to Jockusch and Schupp [3, Theorem 2.19]; we use its relativization and give the proof.

The positive-density column construction is not new. Hirschfeldt, Jockusch, and Schupp [4, Lemma 3.10 and Theorem 3.11] use columns of robust codes of the members of a countable ideal, together with finite-column truncations, in studying closed subspaces of the density metric. Our exact core identity follows by combining that mechanism with the radius principle. The repeated robust code used for the sharper spectrum statement is a variant of the uniform coarsenings in [2, Section 2]; compare also the robust coding framework in [5].

Accordingly, the ideal-realization theorem should be understood as a precise resolution of the repository’s stated target from established ingredients, not as evidence that a difficult published open problem had escaped all previous methods. The whole-row characterization and fixed-core structural theorems are proved as research extensions here. Their global novelty would require a more exhaustive literature review than the targeted source check performed for this manuscript.

1.4 How this report is assembled

[merge] This subsection and the next were written for the merge.

This report merges three manuscripts delivered together in batch 37 of ProveIt’s intake (manuscripts 12, 08 and 10 of that batch), all written against ProveIt commit 1a1396d4d3a2ac6812692df520517d86ba3a4785 and all dated 28 September 2026. They share one spine:

- (i) every countable Turing ideal is the core of a binary set;
- (ii) joining the dyadic code $\mathcal{R}(H)$ keeps the core and cuts the spectrum by the jump cone of H ;
- (iii) relativized Friedberg inversion turns (ii) into an order embedding of a Turing upper cone into a single core fiber, for both $\leq_{uc}$ and $\leq_{nc}$.

Part I (Sections 1–11 and Appendices A–B) is source 12, the base. It is the most general of the three: its core-neutrality theorem (Theorem 8.2) and embedding theorem (Theorem 9.2) hold for an *arbitrary* base set X , where sources 08 and 10 prove the analogues only for their own carriers; it accepts an arbitrary generator sequence, where source 10 needs an increasing cofinal chain; it alone has the exact whole-row spectrum (Theorem 6.2) and the fiber cofinality result (Corollary 9.4); and it proves the robust radius (Theorem 3.1) instead of importing it. **Part II** (Sections 12–19) is source 08, and **Part III** (Sections 20–29) is source 10. Each part keeps its own code and its own hypotheses. Appendix C records what each source contributed and where the merge had to choose.

The three sources realize an ideal by three *different* codes, so the classification (i) is proved here four times:

- (a) by the robust radius, proved from Baire category (Theorem 5.1, source 12);
- (b) by cone-avoiding compactness [2, Theorem 3.7] used as a published black box (Section 16.1, source 08; Section 22, source 10);
- (c) by row-finite finite-extension forcing, which yields a perfect family of pairwise exact descriptions with jump control (Section 16, source 08);
- (d) by ideal-protected fusion, which yields a perfect family of pairwise exact descriptions close in the prefix metric (Section 23, source 10).

Routes (c) and (d) use neither the Baire category theorem on $CD(X)$ nor [2, Theorem 3.7]. Statements proved in more than one source (the elementary properties of cores and spectra, the column estimates, the limit lemma, the dyadic criterion, uniform dyadic transport, and relative jump inversion) are printed once, in Part I, with a merge note naming the other sources; where their proofs coincide in substance they are not repeated. Source 10’s embedding theorem is a special case of Theorem 9.2 and is stated as such in Section 25. Source 08’s embedding (Theorem 17.3) is a different construction, with the parameter in rows inside the code, and is printed in full with its own proofs; a merge observation (Remark 17.7) shows that its classes coincide, up to uniform coarse equivalence, with those of Theorem 9.2 for $X = Q(\vec{A})$.

Formal status. Nothing in this report is formalized. The ProveIt Lean library Computability/TuringDegrees/Lean/CoarseDegrees proves, among other things, the following finite ingredients (the library README and the file `Audit.lean` there record their axioms; this report did not rebuild them):

- `card_tail` (`Columns.lean`): the exact tail count $|T_k \cap [0, N]| = \lfloor N/2^k \rfloor$ of (4.2);
- `densityZero_of_bounded_columns` (`Columns.lean`): a set meeting every column in a bounded set has density zero, the last sentence of Lemma 4.1; and `errCnt_div_tendsto` (`Majority.lean`): density-zero errors have relative density zero in each column, the forward half of that lemma in the form used for column majorities;
- `blockCode_decode` (`Block.lean`): every coarse description of $\mathcal{J}(A)$ computes A . The library’s `blockCode` is exactly the code $\mathcal{J}$ of (4.4) (and the plan’s J); it is not source 08’s shifted code

$\mathcal{J}^{\text{sh}}$ or source 10's factorial code $\mathcal{J}^!$;

- `description_iff_limit` (`Majority.lean`) and `spectrum_Rc` (`Spectrum.lean`): Theorem 7.2 in *limit form*, “ B computes a description of $\mathcal{R}(A)$ iff A is the limit of a B -computable approximation”. The library has no jump operator, so Lemma 7.1, which converts the limit form into $A \leq_T B'$, is not formalized;
- `exists_description_of_degree` (`BlockChar.lean`): the sparse-coding step of Proposition 2.3, with markers $\{2^k - 1\}$ in place of $\{2^n\}$;
- `isLeastNC_iff_blockCode` (`BlockChar.lean`): a coarse class has a representative of least Turing degree iff it is the class of some $\mathcal{J}(B)$ (synthesis, Theorem 3.6), the background for the principal example after Theorem 5.1. The library README says nothing is admitted in it; the line printing its axioms in `Audit.lean` sits under the heading that expects `sorryAx`. This report did not run the audit to decide which is current.

None of the theorems of this report that go beyond these ingredients is formalized: not the robust radius (the library has no prefix-density metric and no statement of [2, Theorem 3.7]), not ideal realization, the whole-row spectrum, core neutrality, the jump filter, jump inversion or the embedding, and none of the constructions of Parts II and III, which would need a use-bounded model of Turing functionals that the library does not have. The finite Python checks shipped with the report (Appendix C) test finite arithmetic only.

1.5 Notation across the three sources

[merge] The three sources use the same letters for different objects. The merge fixes one convention, used from here on in all three parts; the table lists every symbol renamed from a source. No normalization changed: every renamed object is the source's object under a new name.

Codes and carriers.

Here	Meaning; source symbols
$p(i, m) = 2^i(2m + 1) - 1$; C_i, T_k	position m of dyadic column i ; the column; the tail $\bigcup_{i \geq k} C_i$. Source 08 writes P_n for C_n ; source 10 writes $c_n(k)$ for $p(n, k)$.
$\mathcal{R}(H)$	dyadic code, $\mathcal{R}(H)(p(i, m)) = H(i)$ (source 08: R ; source 10: $\mathcal{R}$).
$\mathcal{J}(B)$	block code of Part I: $\mathcal{J}(B)(0) = 0$, $\mathcal{J}(B)(t) = B(\lfloor \log_2 t \rfloor)$, blocks $[2^k, 2^{k+1})$. Equal to the plan's J and to the Lean <code>blockCode</code> .
$\mathcal{J}^{\text{sh}}(U)$	source 08's J : $\mathcal{J}^{\text{sh}}(U)(t) = U(\lfloor \log_2(t + 1) \rfloor)$, blocks $I_k = [2^k - 1, 2^{k+1} - 1)$. Not $\mathcal{J}$: $\mathcal{J}^{\text{sh}}(U)(t) = \mathcal{J}(U)(t + 1)$.
$\mathcal{J}^!(A)$	source 10's $\mathcal{J}$: factorial blocks $I_m^! = [(m + 1)!, (m + 2)!) carrying the bit with label \lambda(m) = v_2(m + 1) (source 10: I_m, h_m = (m + 1)!, \ell(m)); every label recurs infinitely often.$
$\mathcal{E}(A) = \mathcal{J}(\mathcal{R}(A))$	Part I's whole-real code.
$X_{\vec{A}}$	Part I's ideal code, $\mathcal{E}(A_i)$ on column C_i , any sequence.
$\mathcal{Q}(\vec{A})$	source 08's two-level code, $\mathcal{J}^{\text{sh}}(A_n)$ on column C_n , any sequence.
$\Lambda_{\vec{A}}$	source 10's layered carrier, $\mathcal{J}^!(A_n)$ on column C_n , for an increasing cofinal chain. Source 10 calls it $X_{\vec{A}}$: it is not Part I's $X_{\vec{A}}$.
$T_X(H) = X \oplus \mathcal{R}(H)$	Part I's filter. Source 10's $F_{\vec{A}, B}$ is $T_{\Lambda_{\vec{A}}}(H)$ here (its parameters B, C are H, K here).
$F_H = \mathcal{Q}(\vec{W}^H)$	source 08's <i>interleaved</i> code F_C (its C, E are H, K here): the parameter sits in constant rows <i>inside</i> the code. Not $T_X(H)$.

Here	Meaning; source symbols
$\Lambda_{\vec{A}, < s}, U_s(H),$ $\Lambda_{\vec{A}, < s}^H$	source 10's truncations $T_s, U_s(B)$ and V_s ; here $\Lambda_{\vec{A}, < s}^H = \Lambda_{\vec{A}, < s} \oplus U_s(H)$.

Oracles, ideals, spectra and other letters.

Here	Meaning; source symbols
$A_\oplus = \bigoplus_n A_n,$ $A_{< m} = \bigoplus_{n < m} A_n$	presentation oracle and its finite initial joins. Source 12 writes S for $A_\oplus$ (Corollary 6.3); source 08 writes B and B_m ; source 10 writes Z .
$\vec{A}, \mathcal{I}_{\vec{A}}$	a presentation (A_n) and its generated ideal $\bigcup_k \text{Low}(A_{< k})$ (source 08: $A, \mathcal{I}_A$; source 10 writes $\mathcal{I} = \bigcup_n \text{Low}(A_n)$ for an increasing chain, the same ideal).
$\Theta_{\vec{A}} = \bigoplus_m A'_{< m}$	source 08's construction oracle H_A (and H_{W^c} is $\Theta_{\vec{W}^H}$).
$\mathcal{Q}_H$	the jump cone $\{\deg_T(B) : H \leq_T B'\}$ (source 08: $\mathcal{J}(\deg_T H)$).
$\mathcal{F}_{\vec{A}}$	source 08's row-finite mass problem; not the fiber $\mathfrak{F}_r(I)$.
$\mathfrak{F}_r(I)$	fixed-core collection ("fiber"; source 08 writes "fibre").
$\text{Spec}(X)$	representative spectrum (source 10 writes $\Sigma(X)$).
$\mathcal{K}_r(X)$	source 10's information robust at error below r ; not a parameter K .
P, Q	branch parameters in $2^{\mathbb{N}}$ (source 08 and source 10). P_s (source 10) are protected regions and P in the proof of Proposition 2.3 is $\{2^n\}$.
$\tilde{A}_s$	source 10's eventually correct sets in Lemma 21.2 (source 10: A_s , which clashes with the generators).
G	the common function in Lemma 23.2 (source 10: H).

Other letters are local and are listed where they first appear: V is source 08's common null envelope, R_n its finite error sets (not the code $\mathcal{R}$), $M(r)$ its density modulus (not the majority M_D of Part I), K_m its error counts (not a parameter K), and Γ, Δ its coding and decoding functionals; W is source 10's bounding oracle (not the array $\vec{W}^H$), and $\text{Res}(\sigma, s)$ its reservoirs. The tempting false readings are recorded in the table: $X_{\vec{A}}$ of source 10 is $\Lambda_{\vec{A}}$, and F_H is not $T_X(H)$.

2 Descriptions, spectra, and common lower bounds

2.1 Conventions

We work in classical mathematics (ZFC) with standard oracle Turing machines. We identify a set of natural numbers with its characteristic function. Write $A \leq_T B$ for Turing reducibility, A' for the Turing jump, and $\deg_T(A)$ for the Turing degree of A . Binary join means

$$(A \oplus B)(2n) = A(n), \quad (A \oplus B)(2n+1) = B(n).$$

An infinite join uses a fixed computable pairing function. Different standard pairings give uniformly Turing-equivalent encodings.

For $n \geq 1$ and $E \subseteq \mathbb{N}$, put

$$\rho_n(E) = \frac{|E \cap [0, n)|}{n}, \quad \bar{\rho}(E) = \limsup_{n \rightarrow \infty} \rho_n(E), \quad \delta(X, Y) = \bar{\rho}(X \triangle Y).$$

The pseudometric δ measures upper-density disagreement. We write $X \approx Y$ when $\delta(X, Y) = 0$, and

$$\text{CD}(X) = \{D \in 2^{\mathbb{N}} : D \approx X\}.$$

Every description in this paper is total and binary.

Definition 2.1. Write $X \leq_{\text{nc}} Y$ if every $D \in \text{CD}(Y)$ computes some member of $\text{CD}(X)$. Write $X \leq_{\text{uc}} Y$ if a single Turing functional maps every member of $\text{CD}(Y)$ to a member of $\text{CD}(X)$. The corresponding equivalence relations are denoted $\equiv_{\text{nc}}$ and $\equiv_{\text{uc}}$.

Set

$$\text{Spec}(X) = \{\deg_{\text{T}}(B) : (\exists D \leq_{\text{T}} B) D \in \text{CD}(X)\}, \quad \text{Core}(X) = \bigcap_{D \in \text{CD}(X)} \text{Low}(D),$$

where $\text{Low}(D) = \{A : A \leq_{\text{T}} D\}$.

We use ideals of *sets*: a Turing ideal is a nonempty collection closed under Turing reducibility and binary join. It is automatically saturated under Turing equivalence. Equivalently, one can work with its collection of degrees. Countability has the same meaning in either formulation, because a Turing degree contains only countably many sets.

Lemma 2.2 (Elementary properties). *For every X , $\text{Core}(X)$ is a countable Turing ideal. It is the common lower cone of $\text{Spec}(X)$:*

$$A \in \text{Core}(X) \iff (\forall \mathbf{b} \in \text{Spec}(X)) \deg_{\text{T}}(A) \leq \mathbf{b}.$$

Both Core and Spec are invariants of nonuniform coarse equivalence, and hence of uniform coarse equivalence.

Proof. Downward closure is immediate. If two sets are computable from every $D \in \text{CD}(X)$, their binary join is too. Every computable set belongs to the intersection. Since $X \in \text{CD}(X)$, the core is contained in the countable collection $\text{Low}(X)$.

If B computes a description of X , every core member is below B . Conversely, each description itself supplies a degree in the spectrum. This proves the common-lower-cone identity. Equivalence preserves which oracles can compute a description, by composition of Turing computations, and then preserves the core. $\square$

[merge] Source 08 (its Lemma 2.2) and source 10 (its Proposition 2.3) prove the first assertion by the same argument. Both add monotonicity: if $X \leq_{\text{nc}} Y$, then $\text{Core}(X) \subseteq \text{Core}(Y)$. For $Z \in \text{Core}(X)$ and $D \in \text{CD}(Y)$ a reduction gives some $E \leq_{\text{T}} D$ in $\text{CD}(X)$, whence $Z \leq_{\text{T}} E \leq_{\text{T}} D$; no uniform choice of the reduction index is required (source 08).

Proposition 2.3 (The three representative spectra). *$\text{Spec}(X)$ is upward closed and equals each of*

$$\{\deg_{\text{T}}(D) : D \approx X\}, \quad \{\deg_{\text{T}}(Y) : Y \equiv_{\text{uc}} X\}, \quad \{\deg_{\text{T}}(Y) : Y \equiv_{\text{nc}} X\}.$$

Furthermore,

$$X \leq_{\text{nc}} Y \iff \text{Spec}(Y) \subseteq \text{Spec}(X). \tag{2.1}$$

Proof. Suppose $D \approx X$ and $D \leq_T B$. On the computable density-zero set $P = \{2^n : n \in \mathbb{N}\}$ replace $D(2^n)$ by $B(n)$, and leave all other positions unchanged. The resulting E satisfies $E \approx X$ and $E \equiv_T B$: B computes E using its computation of D , while the values on P recover B . This proves the first equality and upward closure. If $E \approx X$, the identity functional gives reductions in both coarse directions, since $\text{CD}(E) = \text{CD}(X)$.

If $Y \equiv_{\text{nc}} X$, applying $X \leq_{\text{nc}} Y$ to Y itself shows that Y computes a description of X . This proves the remaining equalities. Finally, the forward implication of (2.1) follows by composition. For the reverse, a description D of Y has degree in $\text{Spec}(Y)$, so the assumed inclusion says exactly that D computes a description of X . $\square$

[merge] Source 08 (Lemma 2.3) and source 10 (Lemma 2.1) prove the same identities by the same sparse coding on $\{2^n\}$. Source 08 adds that the common lower bounds of the spectrum are exactly $\text{Core}(X)$ (Lemma 2.2); source 10 states (2.1) and $\text{Core}(X) = \bigcap_{d \in \text{Spec}(X)} \{A : \deg_T(A) \leq d\}$ in the same lemma. The sparse-coding step is formalized, with markers $\{2^k - 1\}$ (`exists_description_of_degree`; Section 1.4).

This proposition, including sparse degree coding, is background already present in [1]. The reversal of inclusion in (2.1) is worth noting: a more difficult coarse problem has fewer solving degrees.

Proposition 2.4 (Binary joins). *For all X, Y ,*

$$\text{Spec}(X \oplus Y) = \text{Spec}(X) \cap \text{Spec}(Y). \quad (2.2)$$

Binary join is the least upper bound for both coarse reducibilities. Moreover,

$$\delta(X \oplus Y, U \oplus V) \leq \frac{1}{2}\delta(X, U) + \frac{1}{2}\delta(Y, V). \quad (2.3)$$

The upper-density error of either half is at most twice the error of the join.

Proof. At a prefix of even length $2n$, the error count is the sum of the two length- n error counts. Odd endpoints change the fractions by a quantity tending to zero. This proves the density statements. Splitting a description gives descriptions of both halves; conversely, joining two descriptions gives a description of the join. These operations are uniform. The spectrum equality follows, allowing the two reductions from a fixed solving oracle to have different indices. $\square$

[merge] Source 10 (Lemma 2.2) proves the spectrum identity and the least-upper-bound property in the same way. It notes that only two computations are combined, so no question of uniformity over an infinite sequence arises, and it also records that $X \leq_{\text{nc}} Y$ implies $\text{Core}(X) \subseteq \text{Core}(Y)$.

Corollary 2.5 (Infimum versus least element). *$\text{Spec}(X)$ has an infimum in the full partial order of Turing degrees if and only if $\text{Core}(X)$ is principal. If $\text{Core}(X) = \text{Low}(A)$, that infimum is $\deg_T(A)$. It is a least element precisely when $\deg_T(A) \in \text{Spec}(X)$.*

Proof. By Lemma 2.2, the degrees of the core are exactly all the lower bounds of the spectrum. They have a greatest member exactly when the ideal is principal. A greatest lower bound is a least member precisely when it belongs to the spectrum itself. $\square$

[merge] Source 08 (Corollary 2.4) states the same: a least member of $\text{Spec}(X)$ exists if and only if the core is principal and its generator belongs to the spectrum. Source 10 proves the direction used in its no-least arguments as Lemma 26.1.

3 A locality theorem for common information

The following theorem is an equivalent form of the cone-avoiding compactness principle in [2, Theorem 3.7]. We include a proof to expose exactly where nonuniform finite information enters. The prefix-metric approach is developed in the repository's category section [1].

Theorem 3.1 (Robust radius). *For all $A, X \subseteq \mathbb{N}$,*

$$A \in \text{Core}(X) \iff (\exists \eta > 0)(\forall Y \in 2^{\mathbb{N}})[\delta(X, Y) < \eta \implies A \leq_T Y]. \quad (3.1)$$

3.1 The complete metric used in the proof

The pseudometric δ vanishes throughout $\text{CD}(X)$, so it cannot be used for Baire category on that space. Instead put

$$d(U, V) = \sup_{n \geq 1} \rho_n(U \triangle V).$$

This is a genuine metric. It controls short prefixes as well as asymptotic behavior, and $\delta(U, V) \leq d(U, V)$.

Lemma 3.2. *The metric space $(\text{CD}(X), d)$ is nonempty and complete. Every finite-prefix cylinder is open in it. If $\delta(Y, Z) < a$, there is a finite modification $\tilde{Y}$ of Y with $d(\tilde{Y}, Z) < a$.*

Proof. A disagreement at position k forces $d(U, V) \geq 1/(k+1)$, which proves both separation and openness of cylinders. The triangle inequality follows from the corresponding inequality for error counts at each prefix.

Let (D_j) be d -Cauchy in $\text{CD}(X)$. Each coordinate eventually stabilizes; call the coordinatewise limit D . For a given $\varepsilon > 0$, choose j_0 so that $d(D_j, D_k) < \varepsilon/2$ whenever $j, k \geq j_0$. Fix $j \geq j_0$ and a prefix length n . Taking k sufficiently large that $D_k \upharpoonright n = D \upharpoonright n$ gives $\rho_n(D_j \triangle D) < \varepsilon/2$. Consequently $d(D_j, D) \leq \varepsilon/2$, and $D_j \rightarrow D$ in d . Since

$$\delta(D, X) \leq d(D, D_j) + \delta(D_j, X) = d(D, D_j),$$

we have $D \in \text{CD}(X)$. Nonemptiness follows from $X \in \text{CD}(X)$.

Finally choose a' strictly between $\delta(Y, Z)$ and a . For all sufficiently large n , $\rho_n(Y \triangle Z) < a'$. Replace the corresponding finite initial segment of Y by that of Z . The new error count is zero at short prefixes and no larger than the old error count at long prefixes. The resulting distance is at most $a' < a$. $\square$

The finite modification in the last paragraph is Turing equivalent to Y . Its finite values and its length can be incorporated as constants into a program. No effective procedure for finding those constants is asserted.

For a string τ and a set Z , let $\tau \triangleright Z$ agree with τ below $|\tau|$ and with Z afterwards. Then

$$d(\tau \triangleright Z, Z) = \max_{1 \leq m \leq |\tau|} \frac{|\{j < m : \tau(j) \neq Z(j)\}|}{m}, \quad (3.2)$$

with the empty maximum interpreted as 0. Beyond the end of the string the numerator is fixed and the denominator only increases.

3.2 Decoding from a neighborhood

Fix a standard enumeration (Φ_e) of oracle programs, and define

$$S_{e,A} = \{D \in \text{CD}(X) : \Phi_e^D \text{ is total and equals } A\}.$$

For a finite oracle string, a halting computation is accepted only when every oracle query lies within that string.

Lemma 3.3 (Local decoder). *Suppose $S_{e,A}$ is dense in a nonempty d -ball $B_d(Z, r)$ inside $\text{CD}(X)$, where $r > 0$ is rational. Every $Y \in 2^{\mathbb{N}}$ satisfying $d(Y, Z) < r/4$ computes A . A decoding program can be chosen using only e and r , not the center Z .*

Proof. On input n , search for a finite string τ such that $\Phi_e^\tau(n)$ halts and

$$\max_{1 \leq m \leq |\tau|} \frac{|\{j < m : \tau(j) \neq Y(j)\}|}{m} < r/2. \quad (3.3)$$

Dovetail the halting tests. The displayed condition is a finite rational test computable from Y . Output the value of the first successful computation.

The search terminates. Density supplies $D_* \in S_{e,A} \cap B_d(Z, r/4)$, so $d(D_*, Y) < r/2$. A sufficiently long prefix of D_* witnesses the search condition and the halting computation at n .

Every accepted value is correct. For an accepted τ , put $H = \tau \smallfrown Z$. It belongs to $\text{CD}(X)$ because it is a finite modification of Z . At prefixes of length at most $|\tau|$, the error against Z is less than $r/2 + r/4$. By (3.2), $H \in B_d(Z, r)$. The cylinder determined by τ , intersected with this ball, is therefore a nonempty open set. It contains a member of $S_{e,A}$. On that member the same finite computation must output $A(n)$, so the accepted output is $A(n)$. $\square$

Proof of Theorem 3.1. The right-to-left direction holds because every coarse description has distance zero from X . Conversely, let $A \in \text{Core}(X)$. Then $\text{CD}(X) = \bigcup_e S_{e,A}$. The Baire category theorem and Lemma 3.2 imply that some $S_{e,A}$ is not nowhere dense. Thus its closure contains a nonempty ball $B_d(Z, r)$ with rational $r > 0$, and it is dense in that ball.

Let $\eta = r/4$, and suppose $\delta(X, Y) < \eta$. As $Z \approx X$, $\delta(Y, Z) < \eta$. By the finite-patch part of Lemma 3.2, a finite modification $\tilde{Y} \equiv_T Y$ satisfies $d(\tilde{Y}, Z) < r/4$. Lemma 3.3 yields $A \leq_T \tilde{Y} \equiv_T Y$. $\square$

Corollary 3.4 (Finite-stage capture). *If $\delta(X, X_k) \rightarrow 0$ and $X_k \leq_T B_k$ for every k , then*

$$\text{Core}(X) \subseteq \bigcup_k \text{Low}(B_k).$$

Indeed every member of the core is below B_k for all sufficiently large k .

Proof. Choose the radius in Theorem 3.1 for a fixed core member and use convergence. No effective rate and no uniform list of the reductions $X_k \leq_T B_k$ are needed. $\square$

[merge] Sources 08 and 10 do not prove Theorem 3.1. Both use [2, Theorem 3.7] as a published black box (Theorems 16.1 and 22.1); source 10 obtains (3.1) as its contrapositive, noting that strict and non-strict error tolerances are interchangeable after shrinking a positive radius. Source 10's ideal approximation lemma

(Lemma 22.2) is the ideal-valued form of Corollary 3.4: if a Turing ideal contains sets arbitrarily close to X in δ , it contains $\text{Core}(X)$.

This is the locality principle behind ideal realization: information common to all exact coarse descriptions is already unavoidable at some positive error scale. It can therefore be captured by a sufficiently accurate finite-stage approximation.

4 Coding geometry and a uniform robust code

4.1 A partition with computably small tails

Define

$$p(i, m) = 2^i(2m + 1) - 1, \quad C_i = \{p(i, m) : m \in \mathbb{N}\}. \quad (4.1)$$

Every $n \in \mathbb{N}$ lies in exactly one column, whose index is $v_2(n + 1)$. Here $v_2(t)$ is the exponent of 2 dividing the positive integer t . The density of C_i is 2^{-i-1} .

Lemma 4.1 (Column estimates). *Let $T_k = \bigcup_{i \geq k} C_i$. Then*

$$T_k = \{n : 2^k \mid n + 1\}, \quad |T_k \cap [0, N)| = \left\lfloor \frac{N}{2^k} \right\rfloor. \quad (4.2)$$

If E has density zero, then $E_i = \{m : p(i, m) \in E\}$ has density zero for every i . Conversely, if every E_i has density zero, then E has density zero. In particular, a set meeting each column in only finitely many positions has density zero.

Proof. The first assertions follow by factoring $n + 1$. For the forward density implication, the first M positions of C_i lie below $2^{i+1}M$, so

$$\frac{|E_i \cap [0, M)|}{M} \leq 2^{i+1} \rho_{2^{i+1}M}(E) \longrightarrow 0.$$

For the reverse implication, fix k . On each of the finitely many columns $i < k$, the count of errors below N is $o(N)$, because the number of positions of that column below N is asymptotic to $2^{-i-1}N$. The remaining errors lie in T_k . Hence

$$\limsup_N \rho_N(E) \leq 2^{-k}.$$

Now let $k \rightarrow \infty$. □

[merge] Source 08 adds the exact column count $|C_n \cap [0, N)| = \lfloor (N + 2^n)/2^{n+1} \rfloor$ (equation (13.2)), and source 10 the quantitative restriction bound $\delta(X_n, D_n) \leq 2^{n+1}\delta(X, D)$ (Lemma 21.1). The tail count, the bounded-column criterion and the restriction of density-zero errors to a column are formalized as `card_tail`, `densityZero_of_bounded_columns` and `errCnt_div_tendsto` (Section 1.4).

The last argument is a finite-head/small-tail argument, not an appeal to closure of density-zero sets under arbitrary countable unions. Such closure is false. It is the uniform tail bound (4.2) that makes this particular countable assembly safe.

4.2 Three different codes

We need to distinguish three operations. The dyadic repetition code is

$$\mathcal{R}(A)(p(i, m)) = A(i). \quad (4.3)$$

Its decoding from a coarse description generally costs a jump. In contrast, write

$$\mathcal{J}(B)(0) = 0, \quad \mathcal{J}(B)(t) = B(\lfloor \log_2 t \rfloor) \quad (t \geq 1). \quad (4.4)$$

Thus $B(k)$ is repeated on the entire interval $[2^k, 2^{k+1})$. Finally define

$$\mathcal{E}(A) = \mathcal{J}(\mathcal{R}(A)). \quad (4.5)$$

All three exact codes are uniformly Turing equivalent to their inputs. The composition in (4.5) provides the stronger decoding property required for our spectrum theorem.

Lemma 4.2 (Block majority). *For a binary oracle D , let $M_D(k)$ be the majority bit of D on $[2^k, 2^{k+1})$, with ties assigned 0. If $D \approx \mathcal{J}(B)$, then $M_D(k) = B(k)$ for all sufficiently large k .*

Proof. A wrong majority on a block of length 2^k requires at least 2^{k-1} errors on that block. Therefore

$$M_D(k) \neq B(k) \implies \rho_{2^{k+1}}(D \triangle \mathcal{J}(B)) \geq \frac{1}{4}.$$

The implication remains valid for $k = 0$, where any wrong bit gives an even larger error fraction. Infinitely many wrong block majorities would contradict density-zero disagreement. $\square$

[merge] The consequence that every coarse description of $\mathcal{J}(B)$ computes B is `blockCode_decode` in the Lean library, whose `blockCode` is exactly $\mathcal{J}$. Sources 08 and 10 prove majority lemmas for their own block codes: Lemma 13.1 for $\mathcal{J}^{\text{sh}}$ (error fraction above $1/4$ at the block endpoint $2^{k+1} - 1$) and Lemma 21.2 for $\mathcal{J}^!$ (recovery from every D with $\delta(D, \mathcal{J}^!(A)) < 1/2$).

Proposition 4.3 (Whole-real recovery in the limit). *There is a single oracle-computable procedure which, from any $D \in \text{CD}(\mathcal{E}(A))$, produces a total binary function $F_D(s, n)$ with*

$$(\exists s_0)(\forall s \geq s_0)(\forall n) \quad F_D(s, n) = A(n). \quad (4.6)$$

In particular, every coarse description of $\mathcal{E}(A)$ computes A .

Proof. Set

$$F_D(s, n) = M_D(p(n, s)).$$

By Lemma 4.2, there is a K such that $M_D(k) = \mathcal{R}(A)(k)$ whenever $k \geq K$. Since $p(n, s) \geq s$ for every n, s , all $s \geq K$ satisfy

$$F_D(s, n) = \mathcal{R}(A)(p(n, s)) = A(n) \quad \text{for every } n.$$

This proves (4.6). To obtain an individual Turing reduction of A to D , use one such stable s as a fixed program constant. $\square$

Remark 4.4 (What is uniform, and what is not). The procedure computing the entire two-variable array F_D is uniform in D . Selecting a stable row is not asserted to be uniform. Confusing these two claims would invalidate later arguments. A single output real computed uniformly from all descriptions is a stronger demand than the uniform array in (4.6).

The repeated-code method is a concrete instance of uniform coarsening as used in [2, Definition 2.1 and Proposition 2.3]. The choice of exponentially sized blocks rather than another standard rapidly growing partition affects neither the conclusions nor the underlying computability notion.

5 Realizing exactly the countable Turing ideals

For a sequence $\vec{A} = (A_i)_{i \in \mathbb{N}}$, define

$$X_{\vec{A}}(p(i, m)) = \mathcal{E}(A_i)(m), \quad \mathcal{I}_{\vec{A}} = \bigcup_{k \in \mathbb{N}} \text{Low} \left(\bigoplus_{i < k} A_i \right). \quad (5.1)$$

The empty finite join is computable. No computability assumption is placed on the sequence of inputs; $\bigoplus_{i \in \mathbb{N}} A_i$ is the oracle presenting that sequence.

Theorem 5.1 (Ideal realization). *For every sequence $\vec{A}$,*

$$\text{Core}(X_{\vec{A}}) = \mathcal{I}_{\vec{A}}, \quad X_{\vec{A}} \equiv_{\text{T}} \bigoplus_{i \in \mathbb{N}} A_i. \quad (5.2)$$

Consequently the cores of binary sets are exactly the countable Turing ideals.

Proof. Let $D \approx X_{\vec{A}}$ and define the i th column oracle by $D_i(m) = D(p(i, m))$. By Lemma 4.1, $D_i \approx \mathcal{E}(A_i)$. Proposition 4.3 gives $A_i \leq_{\text{T}} D_i \leq_{\text{T}} D$. Thus every finite join of the A_i is computable from D , and so is every member of $\mathcal{I}_{\vec{A}}$. This proves $\mathcal{I}_{\vec{A}} \subseteq \text{Core}(X_{\vec{A}})$.

For the opposite inclusion let $X_{<k}$ agree with $X_{\vec{A}}$ on columns $i < k$ and be zero on all later columns. Then

$$X_{<k} \leq_{\text{T}} \bigoplus_{i < k} A_i, \quad \delta(X_{<k}, X_{\vec{A}}) \leq 2^{-k}$$

by (4.2). Corollary 3.4 now gives the desired reverse inclusion. This is the step excluding information accidentally encoded across infinitely many columns from becoming common to all coarse descriptions.

The infinite join uniformly computes $X_{\vec{A}}$ by evaluating the codes. In the other direction, $X_{\vec{A}}$ uniformly supplies each exact oracle $\mathcal{E}(A_i)$, and each $A_i(n)$ can be read at a computably specified position of that exact code. For example,

$$A_i(n) = X_{\vec{A}}(p(i, 2^{p(n, 0)})).$$

Hence the exact code computes the entire presentation, proving the Turing equivalence in (5.2).

Finally let $\mathcal{I}$ be a countable Turing ideal and enumerate its members as (B_i) . Take $A_i = \bigoplus_{j \leq i} B_j$. These sets generate exactly $\mathcal{I}$, so the construction realizes it. Conversely every core is a countable Turing ideal by Lemma 2.2. $\square$

[merge] The classification in Theorem 5.1 is target C4 of the research plan [7], proposed there with essentially this proof sketch for the shorter code of Remark 5.2, and its column mechanism is [4, Lemma 3.10 and Theorem 3.11]. The same classification is proved in Part II for source 08's code $Q(\vec{A})$, by row-finite forcing (Theorem 12.1) and from [2, Theorem 3.7] (Section 16.1), and in Part III for source 10's layered carrier $\Lambda_{\vec{A}}$, from [2, Theorem 3.7] (Theorem 20.1) and by ideal-protected fusion (Theorem 23.4). Neither source 08 nor source 10 cites [4]; that antecedent bears on their column constructions as well.

Remark 5.2 (A shorter code suffices for the core alone). Replacing $\mathcal{E}(A_i)$ by $\mathcal{J}(A_i)$ in (5.1) gives the same core identity: block majority recovers A_i modulo finitely many bits, and those bits can be supplied as finite constants. This is closely aligned with the coding construction in [4, Theorem 3.11]. We use $\mathcal{E}$ because its uniform approximation by whole reals also yields the exact spectrum in the next section. The extra repetition is not needed to solve the bare core-realization problem.

5.1 Two examples and an effectivity boundary

Example 5.3 (A principal ideal). For the constant sequence $A_i = A$, the exact presentation is Turing equivalent to A . Theorem 5.1 gives core $\text{Low}(A)$; since the code itself has degree $\deg_T(A)$, that degree is the least member of its spectrum.

Example 5.4 (Exactly the arithmetical sets). Let $A_i = \emptyset^{(i)}$, the i th finite jump. Then

$$\text{Core}(X_{\vec{A}}) = \bigcup_i \text{Low}(\emptyset^{(i)}), \quad X_{\vec{A}} \equiv_T \emptyset^{(\omega)} := \bigoplus_{i \in \mathbb{N}} \emptyset^{(i)}.$$

The displayed ideal is not principal. If it had a greatest degree, that degree would be below some finite jump, whereas the next jump is a strictly larger member of the ideal. Thus the representative spectrum of $X_{\vec{A}}$ has *no infimum at all*, by Corollary 2.5. Every description still computes every individual finite jump. It need not compute their infinite join: such a claim would make $\emptyset^{(\omega)}$ a core member, contradicting the formula above.

[merge] Source 08 (Section 16.3) and source 10 (Example 27.2) treat the same ideal with their own codes and add exact pairs: two descriptions of $\mathbf{Q}(\vec{A})$ whose common lower cone is the arithmetical ideal and whose jumps are below $\emptyset^{(\omega)}$, and two descriptions of $\Lambda_{\vec{A}}$ computable in $A'_\oplus$ with the same common lower cone, at least one of which does not compute $A_\oplus$.

The construction is uniform from a *given presentation*; it does not provide a computable enumeration of an arbitrary countable ideal. Nor does countability make the choices of indices for the separate reductions $A_i \leq_T D$ uniformly computable. These distinctions are precisely what the spectrum calculation measures.

6 An exact spectrum theorem for the ideal code

6.1 Eventually correct whole rows

Definition 6.1. For a fixed sequence $\vec{A}$, let $\text{Row}(\vec{A})$ be the set of Turing degrees $\mathbf{b}$ for which an oracle $B \in \mathbf{b}$ computes a total binary function $f(i, s, n)$ satisfying

$$(\forall i)(\exists s_i)(\forall s \geq s_i)(\forall n) \quad f(i, s, n) = A_i(n). \quad (6.1)$$

The degree property does not depend on the chosen representative B . The parameter s_i may depend arbitrarily on i . However, once $s \geq s_i$, the *entire* function $n \mapsto f(i, s, n)$ is correct; this is not merely pointwise convergence as s grows.

Theorem 6.2 (Whole-row spectrum). *For the code in (5.1),*

$$\text{Spec}(X_{\vec{A}}) = \text{Row}(\vec{A}). \quad (6.2)$$

This is simultaneously the spectrum of the density-zero class, the uniform coarse class, and the nonuniform coarse class of $X_{\vec{A}}$.

Proof. Suppose B computes $D \approx X_{\vec{A}}$. The column oracles $D_i(m) = D(p(i, m))$ are uniformly computable from B and satisfy $D_i \approx \mathcal{E}(A_i)$. Apply the single array-producing procedure of Proposition 4.3 on each column:

$$f(i, s, n) = M_{D_i}(p(n, s)).$$

For each fixed i , only finitely many block majorities are wrong. The proof of that proposition supplies a threshold s_i valid for every n , so (6.1) holds.

Conversely assume B computes such an f . Construct a total binary set D columnwise. Put $D(p(i, 0)) = 0$. At $m \geq 1$, put

$$D(p(i, m)) = f(i, m, v_2(\lfloor \log_2 m \rfloor + 1)). \quad (6.3)$$

Indeed the target bit is

$$\mathcal{E}(A_i)(m) = \mathcal{R}(A_i)(\lfloor \log_2 m \rfloor) = A_i(v_2(\lfloor \log_2 m \rfloor + 1)).$$

For fixed i , (6.3) is therefore correct at every $m \geq \max(1, s_i)$, and is also correct at $m = 0$ by definition of $\mathcal{E}$. The error set is finite on each column. Lemma 4.1 gives $D \approx X_{\vec{A}}$, and $D \leq_T B$ by construction. The last sentence follows from Proposition 2.3. $\square$

The theorem is an exact effective condition rather than a characterization solely in terms of the generated ideal. It removes the density quantifiers from membership in this particular spectrum and replaces them by a uniform array with the specified stabilization property.

Corollary 6.3 (Necessary information bounds). *Let $A_\oplus = \bigoplus_{i \in \mathbb{N}} A_i$. Then*

$$\{\mathbf{b} : A_\oplus \leq_T B\} \subseteq \text{Spec}(X_{\vec{A}}) \subseteq \{\mathbf{b} : (\forall i) A_i \leq_T B \text{ and } A_\oplus \leq_T B'\}. \quad (6.4)$$

[merge] Source 12 writes S for $A_\oplus$ here (Section 1.5).

Proof. The left inclusion follows either from $X_{\vec{A}} \equiv_T A_\oplus$ or by taking $f(i, s, n) = A_i(n)$. For the right inclusion, a stable row computes A_i from B using its threshold as a constant. Also $f(i, s, n)$ is a B -computable pointwise convergent approximation to the characteristic function of $A_\oplus$, so the relativized limit lemma gives $A_\oplus \leq_T B'$. A proof of that limit lemma is included in Section 7. $\square$

6.2 Computable rows can have a nontrivial collective spectrum

Example 6.4 (The presentation itself carries information). Given any $H \subseteq \mathbb{N}$, let A_i be the constant real with value $H(i)$. Every A_i is computable, so $\mathcal{I}_{\vec{A}} = \text{Comp}$. Nevertheless

$$X_{\vec{A}} \approx \mathcal{R}(H), \quad \text{Spec}(X_{\vec{A}}) = \{\mathbf{b} : H \leq_T B'\}. \quad (6.5)$$

To see the first assertion, $\mathcal{E}(A_i)$ agrees with the constant real A_i at all positions except possibly 0. Hence discrepancies from $\mathcal{R}(H)$ can occur only at $p(i, 0) = 2^i - 1$, a density-zero set. The spectral assertion is Theorem 7.2 below.

Taking $H = \emptyset''$ makes the spectrum omit the computable degree, even though every individual input row is computable. Replacing this presentation by the constant zero presentation gives the same ideal Comp but a spectrum containing every Turing degree. Thus the condition $(\forall i) A_i \leq_T B$ is not a substitute for (6.1).

[merge] The same phenomenon for the other two codes: source 08 (Section 16.4) and source 10 (Proposition 27.3) take A_n constant with value $H(n)$. Source 10's carrier then differs from $\mathcal{R}(H)$ only on $\{2^n - 1\}$, as here; source 08's code is then exactly $\mathcal{R}(H)$ (a merge observation: every row is constant).

This example is also a warning about the phrase “uniformly computable sequence”. A sequence all of whose entries are computable can encode an arbitrarily complicated choice of those entries. Here that choice is exactly H .

7 The dyadic jump spectrum and uniform transport

This section gives the classical ingredients needed for the fixed-core construction. They are not claimed as new: the basic dyadic criterion is the relativization of [3, Theorem 2.19], and is also used in [1, 4].

Lemma 7.1 (Relativized limit lemma). *$H \leq_T B'$ if and only if there is a total B -computable binary function $h(i, s)$ such that $h(i, s) \rightarrow H(i)$ for every i .*

Proof. For one direction, B' can decide whether there exists a $t > s$ with $h(i, t) \neq h(i, s)$, since this is a computably enumerable condition relative to B . Search for a stage with no subsequent change and return its value. The search terminates because the binary sequence converges.

For the other direction, approximate B' by finite stages of its standard B -computable enumeration. On input (i, s) , run the fixed reduction of $H(i)$ to B' for s steps, using that stage's finite approximation as the oracle. Return its binary answer if it halts, and 0 otherwise. The genuine halting computation uses only finitely many oracle answers. Eventually all those answers are correct and the time bound is long enough, so the simulation returns $H(i)$ at all later stages. $\square$

[merge] Source 08 (its Appendix A) and source 10 (its Lemma 3.4) prove the same lemma by the same argument. Source 08 notes that the proof is included to expose both uniformity directions used in its spectrum factorization (Theorem 17.1).

Theorem 7.2 (Dyadic jump-cone criterion). *For every H ,*

$$\text{Spec}(\mathcal{R}(H)) = \{\mathbf{b} : H \leq_T B'\}. \quad (7.1)$$

Moreover, for every $\varepsilon > 0$ there is a computable C such that $\delta(\mathcal{R}(H), C) < \varepsilon$.

Proof. Suppose $D \approx \mathcal{R}(H)$ and $D \leq_T B$. On each column the errors have density zero. Thus the majority of the first $s + 1$ column values,

$$h(i, s) = \text{Maj}\{D(p(i, 0)), \dots, D(p(i, s))\},$$

is eventually $H(i)$. This is a total B -computable pointwise approximation, so Lemma 7.1 gives $H \leq_T B'$.

Conversely, take the approximation h supplied by that lemma and define $D(p(i, m)) = h(i, m)$. On each column D differs from $\mathcal{R}(H)$ at only finitely many positions. The column lemma gives $D \approx \mathcal{R}(H)$.

For the final claim, copy the finitely many bits $H(i)$ on columns $i < k$ and put zero on columns $i \geq k$. The result C_k is computable, because those finitely many bits can be included as program constants, and $\delta(C_k, \mathcal{R}(H)) \leq 2^{-k}$. Choose k with $2^{-k} < \varepsilon$. $\square$

[merge] Source 10 (its Theorem 3.5) proves the same criterion with the same proof (majority of the first s column entries, stage-zero majority 0), citing [3, Theorem 2.19] for the unrelativized case and the synthesis for the relativized spectrum formulation. Source 08 uses the constant-row case inside its own code (Section 17). In limit form the criterion and the spectrum are formalized (`description_iff_limit`, `spectrum_Rc`); Lemma 7.1 is not, since the Lean library has no jump operator.

The last assertion does not say that the sequence of indices for C_k is computable independently of H . It also does not imply the existence of a single computable coarse description. For example, when $H = \emptyset''$, (7.1) rules out such a description.

Proposition 7.3 (Uniform transport along Turing reductions). *If $H \leq_T K$, then $\mathcal{R}(H) \leq_{uc} \mathcal{R}(K)$. In addition,*

$$\mathcal{R}(H \oplus K) \equiv_{uc} \mathcal{R}(H) \oplus \mathcal{R}(K). \quad (7.2)$$

Proof. Let $D \approx \mathcal{R}(K)$. Column majorities give, uniformly in D , a total array $k(i, s)$ with pointwise limit $K(i)$. Fix an index for the reduction $H \leq_T K$. At stage s simulate that program on input i for s steps with oracle $j \mapsto k(j, s)$; return 0 if no binary answer is obtained. The resulting $h(i, s)$ is total and converges to $H(i)$, because the true computation has a finite halting trace. Now output $h(i, m)$ at $p(i, m)$. The converse construction in Theorem 7.2 shows this is a coarse description of $\mathcal{R}(H)$. Every step is one fixed oracle procedure once the reduction index is fixed.

For (7.2), a description of the right side splits into descriptions of $\mathcal{R}(H)$ and $\mathcal{R}(K)$. Their column-majority approximations can be interleaved into one approximation to $H \oplus K$, and then converted into a description of $\mathcal{R}(H \oplus K)$. Conversely, the even and odd entries of a pointwise approximation to $H \oplus K$ approximate H and K . Applying the same conversion to both arrays and joining the outputs proves the other uniform reduction. $\square$

[merge] Source 10 (its Lemma 3.6) proves the same monotonicity and join equivalence by the same bounded simulation against pointwise approximations.

8 Changing a spectrum without changing its core

8.1 A general invariance principle

Definition 8.1. A set Y is *approximable by computable sets in density distance* if

$$(\forall \varepsilon > 0)(\exists C \text{ computable}) \quad \delta(Y, C) < \varepsilon. \quad (8.1)$$

Equivalently, its coarse computability bound $\gamma(Y) = 1 - \inf_{C \text{ computable}} \delta(Y, C)$ equals 1.

The fact that (8.1) implies $\text{Core}(Y) = \text{Comp}$ is the theorem credited to Igusa in [2, Theorem 4.3]. The following statement establishes invariance under adjoining Y to an *arbitrary* base problem.

Theorem 8.2 (Core neutrality under joins). *If Y satisfies (8.1), then for every X ,*

$$\text{Core}(X \oplus Y) = \text{Core}(X). \quad (8.2)$$

Proof. A description of $X \oplus Y$ computes a description of X by taking its even half, so $\text{Core}(X) \subseteq \text{Core}(X \oplus Y)$.

For the reverse inclusion let $A \in \text{Core}(X \oplus Y)$ and take its radius $\eta > 0$ from Theorem 3.1. Choose a computable C with $\delta(Y, C) < \eta/2$. If $\delta(X, U) < \eta/2$, then

$$\delta(X \oplus Y, U \oplus C) \leq \frac{1}{2}\delta(X, U) + \frac{1}{2}\delta(Y, C) < \eta.$$

The radius property yields $A \leq_T U \oplus C \equiv_T U$. Thus $\eta/2$ is a robust radius witnessing $A \in \text{Core}(X)$. $\square$

This proof uses more than $\text{Core}(Y) = \text{Comp}$. It uses arbitrarily accurate *computable approximations* to Y . The theorem does not assert that every trivial-core set is neutral under joins.

8.2 An exact jump filter

Definition 8.3. For a base set X and a parameter H , put

$$T_X(H) = X \oplus \mathcal{R}(H), \quad \mathcal{Q}_H = \{\deg_T(B) : H \leq_T B'\}.$$

Theorem 8.4 (Core-preserving spectral filter). *For all sets X, H ,*

$$\text{Core}(T_X(H)) = \text{Core}(X), \tag{8.3}$$

$$\text{Spec}(T_X(H)) = \text{Spec}(X) \cap \mathcal{Q}_H. \tag{8.4}$$

Also $X \leq_{\text{uc}} T_X(H)$ and $T_X(H) \equiv_T X \oplus H$. For every sequence $\vec{A}$ this specializes to

$$\text{Core}(T_{X_{\vec{A}}}(H)) = \mathcal{I}_{\vec{A}}, \quad \text{Spec}(T_{X_{\vec{A}}}(H)) = \text{Row}(\vec{A}) \cap \mathcal{Q}_H. \tag{8.5}$$

Proof. Theorem 7.2 shows that $\mathcal{R}(H)$ satisfies (8.1); hence (8.3) follows from Theorem 8.2. Equations (2.2) and (7.1) give (8.4). Projection to the even half is the asserted uniform reduction. The exact Turing equivalence follows from $\mathcal{R}(H) \equiv_T H$. Finally use Theorems 5.1 and 6.2. $\square$

[merge] For source 10's carrier, $T_{\Lambda_{\vec{A}}}(H)$ is source 10's $F_{\vec{A}, B'}$, and Theorem 24.1 states (8.3) and (8.4) for it. Its proof is a second route: it approximates $T_{\Lambda_{\vec{A}}}(H)$ by sets *in the ideal* and applies the ideal approximation lemma instead of core neutrality; a third proof comes from ideal-protected fusion (Theorem 20.3). Source 08 attaches the jump cone differently, by constant rows inside its code (Theorem 17.1).

The parameter H can be arbitrarily complicated. Its effect is visible in which jumps occur above solving degrees, while the intersection of their lower cones stays fixed. Thus the entire common-information ideal can remain unchanged under substantial increases in the difficulty of producing a coarse description.

Corollary 8.5 (Strict extensions and unattained infima). *If $H \not\leq_T X'$, then*

$$X \leq_{\text{uc}} T_X(H), \quad T_X(H) \not\leq_{\text{nc}} X,$$

and $\text{Spec}(T_X(H))$ has no least element. If $\text{Core}(X) = \text{Low}(A)$, its infimum is $\deg_T(A)$ and is not attained. If $\text{Core}(X)$ is nonprincipal, its spectrum has no infimum.

Proof. The degree of X belongs to $\text{Spec}(X)$ but, by (8.4), not to $\text{Spec}(T_X(H))$. Formula (2.1) rules out $T_X(H) \leq_{\text{nc}} X$.

Suppose $\mathbf{b}$ were the least member of $\text{Spec}(T_X(H))$. Every representative of $\mathbf{b}$ would lie in the core of $T_X(H)$, which equals $\text{Core}(X) \subseteq \text{Low}(X)$. Thus $\mathbf{b} \leq \deg_T(X)$. Membership in the filtered spectrum would give $H \leq_T B' \leq_T X'$, a contradiction. The two infimum assertions now follow from Corollary 2.5 and (8.3). $\square$

For example, $H = X''$ always satisfies the hypothesis. This gives a strict extension inside the same core above every base set X , not merely above the special ideal codes.

Example 8.6 (A transparent principal-core spectrum). Take $X = \mathcal{E}(A)$. Proposition 4.3 and the exact equivalence $X \equiv_T A$ give $\text{Spec}(X) = \{\mathbf{b} : A \leq_T B\}$. Consequently

$$\text{Core}(\mathcal{E}(A) \oplus \mathcal{R}(H)) = \text{Low}(A), \quad \text{Spec}(\mathcal{E}(A) \oplus \mathcal{R}(H)) = \{\mathbf{b} : A \leq_T B \text{ and } H \leq_T B'\}.$$

When $H \not\leq_T A'$, the infimum is $\deg_T(A)$ and is unattained. The existence of prescribed unattained infima is already in [1]; the displayed spectral formula is the useful additional information supplied by this form of the construction.

9 Turing-cone universality inside a fixed core

9.1 The classical inversion ingredient

We include the relative jump-inversion argument to make the separation part of the embedding theorem independent of any unproved construction in the repository. This is the classical Friedberg jump-inversion method, not a new inversion theorem. Its use in density-metric arguments is also discussed in [4, Section 3].

Lemma 9.1 (Relative jump inversion). *If $K \geq_T X'$, there is a set $B \geq_T X$ such that $B' \equiv_T K$.*

Proof. We construct a binary real Y and put $B = X \oplus Y$. Start with the empty string σ_0 . Given σ_e , ask whether there is a finite extension $\tau \supseteq \sigma_e$ such that

$$\Phi_e^{X \oplus \tau}(e) \downarrow,$$

where every query to the Y -part must be answered within τ . This is a computably enumerable question relative to X , hence decidable using X' . If the answer is yes, choose the first witness found by a fixed X -effective dovetailed search. If the answer is no, put $\tau = \sigma_e$. In either case set

$$\sigma_{e+1} = \tau \frown K(e).$$

Let $Y = \bigcup_e \sigma_e$. Each stage adds at least one bit, so Y is total. The construction is computable from K , because K computes X' and X .

The answer to the stage- e question is exactly the characteristic value $B'(e)$. A yes answer fixes a finite halting computation forever. A no answer excludes halting on every infinite continuation of σ_e . Thus K computes B' by reconstructing the stages, and $B' \leq_T K$.

To decode K from $Y \oplus X'$, reconstruct the same sequence of stages. At stage e , X' determines the yes/no answer and the deterministic choice of τ . Then the next coding bit is $Y(|\tau|) = K(e)$. Hence $K \leq_T Y \oplus X'$. Since $Y \leq_T B \leq_T B'$ and $X' \leq_T B'$, we obtain $K \leq_T B'$. The opposite reduction was already proved. $\square$

[merge] Source 08 (its Appendix B) and source 10 (its Appendix A) prove exactly this lemma, with base oracle $A_\oplus$, by the same finite-extension argument. Source 10 cites Friedberg’s original article [11]. Source 08 adds that modern work continues to develop variants of Friedberg inversion in weak systems, for example [12]; that citation is context, not a dependency.

The requirement $B \geq_T X$ matters. An arbitrary jump inverse of K need not solve the base coarse problem X ; the relative construction guarantees that it does.

9.2 The embedding theorem

For $r \in \{\text{uc}, \text{nc}\}$ and a countable ideal $\mathcal{I}$, define the *fixed-core collection*

$$\mathfrak{F}_r(\mathcal{I}) = \{[Y]_r : \text{Core}(Y) = \mathcal{I}\},$$

ordered by the corresponding coarse reducibility. It is well defined by Lemma 2.2. This is the fiber of the core invariant; no closure under arbitrary joins of its members is assumed.

Theorem 9.2 (Fixed-core upper-cone embedding). *Fix X and let $\mathbf{x} = \deg_T(X)$. For $H, K \geq_T X'$,*

$$H \leq_T K \iff T_X(H) \leq_{\text{uc}} T_X(K) \iff T_X(H) \leq_{\text{nc}} T_X(K). \quad (9.1)$$

For either $r \in \{\text{uc}, \text{nc}\}$, the map

$$\mathbf{h} \mapsto [T_X(H)]_r \quad (\mathbf{h} \geq \mathbf{x}')$$

is therefore a well-defined order embedding into $\mathfrak{F}_r(\text{Core}(X))$, with every image element above $[X]_r$. It preserves binary joins:

$$T_X(H \oplus K) \equiv_{\text{uc}} T_X(H) \oplus T_X(K). \quad (9.2)$$

Distinct input degrees also have distinct representative spectra.

Proof. If $H \leq_T K$, split a description of $T_X(K)$ into its two halves. Keep its X -description, and apply Proposition 7.3 to its description of $\mathcal{R}(K)$. Joining the outputs gives a description of $T_X(H)$ by one fixed functional. This proves the first forward implication. Uniform reducibility always implies nonuniform reducibility.

For the converse, suppose $T_X(H) \leq_{\text{nc}} T_X(K)$. By Lemma 9.1, choose $B \geq_T X$ with $B' \equiv_T K$. The oracle B computes the exact set X and a coarse description of $\mathcal{R}(K)$, the latter by Theorem 7.2. Thus

$$\deg_T(B) \in \text{Spec}(T_X(K)).$$

The assumed nonuniform reduction and (2.1) put the same degree in $\text{Spec}(T_X(H))$. Formula (8.4) now gives $H \leq_T B' \equiv_T K$, as required.

The map is well defined on Turing degrees by the already proved forward implication in both directions, and is injective by the converse. Its core and its position above $[X]_r$ are given by Theorem 8.4.

For binary joins, rearrange

$$(X \oplus \mathcal{R}(H)) \oplus (X \oplus \mathcal{R}(K))$$

to two copies of an X -description and one description of each dyadic code. Either copy of the X -description suffices; in the reverse direction one can duplicate it. These finite splitting, rearranging,

and duplicating operations preserve density-zero error. Proposition 7.3 replaces the two dyadic descriptions by one of $\mathcal{R}(H \oplus K)$, or conversely. This proves (9.2) with uniform reductions.

Finally, equality of two image spectra implies nonuniform equivalence by (2.1), and (9.1) then forces equality of the input Turing degrees. $\square$

[merge] Taking $X = \Lambda_{\vec{A}}$ and restricting to $H, K \geq_T A'_{\oplus}$, a subcone because $\Lambda_{\vec{A}} \leq_T A_{\oplus}$ gives $\Lambda'_{\vec{A}} \leq_T A'_{\oplus}$, gives source 10's embedding theorem, restated as Theorem 25.1; source 10's proof is this proof. Source 08's Theorem 17.3 embeds the cone above $A'_{\oplus}$ by a different map, $H \mapsto F_H$, with the parameter in constant rows inside the code, and proves its uniform half through the normal form; by Remark 17.7 (a merge observation) that map agrees, up to uniform coarse equivalence, with this theorem for $X = Q(\vec{A})$.

The lower endpoint $\mathbf{x}'$ of the domain is not claimed to map to $[X]_r$. The theorem is an embedding into the region above $[X]_r$, not a bottom-preserving isomorphism with that entire region. Likewise it identifies a join-closed *subcollection* of the fiber; it does not show that the full fiber is closed under arbitrary binary coarse joins.

9.3 Consequences for all countable ideals

Corollary 9.3 (Continuum many spectra with one prescribed core). *For every countable Turing ideal $\mathcal{I}$ and either coarse reducibility, $\mathfrak{F}_r(\mathcal{I})$ contains a join-preserving copy of a Turing upper cone. Above each of its members there are continuum many members with pairwise distinct representative spectra.*

Proof. Theorem 5.1 makes the fiber nonempty. Apply Theorem 9.2 to any representative X of a chosen member. Every Turing upper cone has continuum many degrees: there are continuum many oracles of the form $X' \oplus Z$, and each Turing degree contains only countably many oracles. Distinctness of the image spectra is part of that theorem. $\square$

[merge] Source 08 (Corollary 17.6) and source 10 (Theorem 26.2) state the companion fact explicitly: every countable ideal is the core of continuum many coarse degrees, distinct for both reducibilities, having *no least Turing-degree representative*. For the construction of this part the absence of a least member follows from Corollary 8.5 whenever $H \not\leq_T X'$.

Corollary 9.4 (No maximal member and no countable cofinal family). *For every countable Turing ideal $\mathcal{I}$ and $r \in \{\text{uc}, \text{nc}\}$, the ordered collection $\mathfrak{F}_r(\mathcal{I})$ has no maximal member and no countable cofinal subset.*

Proof. For any X with core $\mathcal{I}$, the set $T_X(X'')$ has the same core and is strictly above X even for nonuniform coarse reducibility, by Corollary 8.5. This proves the first assertion.

For the second, let (X_n) be any nonempty countable sequence of representatives with core $\mathcal{I}$, repeating entries if the proposed family is finite. Put

$$L = \bigoplus_n X'_n, \quad H = L', \quad Z = T_{X_0}(H).$$

Then $\text{Core}(Z) = \mathcal{I}$. If $Z \leq_{\text{nc}} X_n$, the exact oracle X_n would compute a coarse description of Z . Formula (8.4) would imply $H \leq_T X'_n \leq_T L$, contradicting $L' \not\leq_T L$. Thus Z is not below any listed member even nonuniformly, so the family is not cofinal for either reducibility. The empty family is not cofinal because the fiber is nonempty. $\square$

Cofinality here is inside the *coarse-degree collection with a fixed core*. It must not be confused with the coinitality of Turing degrees inside one representative spectrum, which is a different question studied in [1].

10 Interpretation, limitations, and formalization

10.1 What has been determined

There is no additional realizability restriction on a countable Turing ideal arising from being the information common to all coarse descriptions. Every such ideal occurs, with a code whose exact degree is the degree of its chosen presentation. In particular, cores need not be principal, and the lack of a least representative can reflect either an unattained infimum or the complete absence of an infimum.

At the same time, this classification is far from a classification of coarse degrees. Theorem 9.2 locates a full Turing upper cone inside every one of the classes grouped together by the core invariant. Those embedded problems can be ordered, joined, and separated by their spectra while preserving every member of their common lower cone. The failure of completeness of the core invariant is therefore structural, not just an isolated counterexample.

The construction also separates two kinds of uniformity. A coarse description of $X_{\vec{A}}$ uniformly produces an array approximating all its rows. It need not uniformly identify the stabilized row for each i . That missing selection information is a genuine source of spectral complexity, already visible when every individual A_i is computable.

10.2 Claims not made

The paper does not show that every upward-closed collection of Turing degrees with a prescribed common lower cone is a coarse spectrum. Formulas (6.2) and (8.5) describe the particular constructions used here. They do not imply that every possible spectrum has one of these forms.

A spectrum without a least member can still have minimal members. None of the arguments proves the absence of all minimal elements in a representative spectrum. Similarly, the upper-cone embedding uses no claim that Turing jumps preserve arbitrary meets or that the Turing degrees form a lattice.

No computable rate of convergence is supplied for arbitrary sequences of generators, and no arithmetical complexity bound follows from countability alone. The explicit bound $X_{\vec{A}} \equiv_T \bigoplus_{i \in \mathbb{N}} A_i$ is relative to a selected presentation, not an intrinsic canonical degree assigned to the ideal.

Finally, density-zero modifications preserve coarse descriptions because errors are permitted. Effective-dense descriptions permit omissions but not erroneous answers. Sparse replacement and the coding constructions cannot be reused for that setting without a separate proof. Gerdes's effective-dense question [6, Question 7], also marked C2 in [1], is not answered here.

10.3 A formalization route

The results have been proved as conventional mathematics in this manuscript; no accompanying Lean theorem file has been compiled. A useful formalization order would isolate three modules.

First, formalize finite density estimates and the partition $p(i, m)$: the exact tail count, restriction

of null errors to a positive-density column, the finite-head/small-tail assembly lemma, and block-majority recovery. These are the parts directly supported by the finite checks in this package, although finite checks do not establish their universal statements.

Second, formalize the common-information invariant and its robust-radius theorem. This requires the prefix-density metric, its completeness on $CD(X)$, continuity of finite oracle computations, and the local decoder. The dependence on Baire category should be exposed rather than hidden in an admitted “compactness” statement. With that module available, the ideal-realization and neutrality theorems are short deductions.

Third, add use-bounded oracle computations, the limit lemma, and relative jump inversion. The proof of Theorem 9.2 then separates into a uniform positive reduction and a single inverse-jump witness for the converse. A clean API should distinguish a Turing reduction of an individual real from an oracle procedure returning a convergent array. That distinction is essential in Theorem 6.2.

[merge] The manuscript did not consult the Lean library. At the pinned commit (and now) `Computability/TuringDegrees/Lean/CoarseDegrees` already proves much of the first module: the exact tail count, restriction of density-zero errors to a column, the bounded-column assembly lemma and block-majority decoding (Section 1.4). The second and third modules are absent: the library has no prefix-density metric and no Baire-category argument, states the dyadic criterion only in limit form because it has no jump operator, and has no use-bounded model of Turing functionals. Sources 08 (Section 18.4) and 10 (Section 28.3) propose similar layered decompositions.

11 Further research questions

The questions below are proposed continuations of the proved results. They are not all asserted to be independently verified open problems in the published literature. Each identifies a boundary not settled by this manuscript.

Question 11.1 (Spectra over a fixed ideal). Fix a countable ideal $\mathcal{I}$. Which upward-closed sets S of Turing degrees occur as $\text{Spec}(X)$ with $\text{Core}(X) = \mathcal{I}$? Beyond the common-lower-cone condition, what effective or descriptive-set-theoretic restrictions are necessary and sufficient?

Theorem 8.4 gives exact closure operations on already realized spectra. A first concrete target is to find a necessary condition not implied merely by upward closure and the prescribed common lower cone.

[merge] Source 08 (Question 19.1) and source 10 (Question 29.1) ask the same question. It is the spectra half of the synthesis’s “Classification” item and of plan target C5(b), and it stays open. Theorem 8.4, Theorem 17.1 and Theorem 24.1 give three families of realized spectra.

Question 11.2 (Simplifying the whole-row condition). For which presentations $\vec{A}$ is the right inclusion in (6.4) an equality? Can whole-row stabilization be characterized by a useful index-selection or domination condition on a presentation?

The constant-row example rules out dropping the jump condition altogether. It does not settle the converse with both necessary conditions present. Canonical increasing sequences of finite jumps provide concrete test cases.

[merge] For source 08's code the corresponding spectrum has the row-finite normal form of Corollary 13.3; Remark 13.5 compares the two exact spectra.

Question 11.3 (Exact characterization of neutral joins). Is a set Y neutral for all cores, in the sense that $\text{Core}(X \oplus Y) = \text{Core}(X)$ for every X , exactly when $\gamma(Y) = 1$? If not, what is the correct characterization?

Only the sufficient direction is proved here. Taking X computable shows that universal neutrality implies a trivial core, but this necessary condition is weaker than the hypothesis used in Theorem 8.2.

Question 11.4 (When common information increases under joins). Given $\text{Core}(X) = \text{Core}(Y) = \mathcal{I}$, under what additional assumptions does $\text{Core}(X \oplus Y) = \mathcal{I}$ follow? Which larger ideals can arise from their join?

The embedded families in Theorem 9.2 stay in one core under binary joins. This does not settle the behavior of arbitrary pairs from the same fixed-core collection.

[merge] Re-scoped. For $\mathcal{I} = \text{Comp}$ the second half is answered in report 11 of this programme (batch-37 manuscript 07; its two-share realization theorem and the corollary after it, Theorem 6.1 and Corollary 6.2 in its delivered numbering): there are X, Y with $\text{Core}(X) = \text{Core}(Y) = \text{Comp}$ and $\text{Core}(X \oplus Y)$ equal to any prescribed countable ideal, so no operation on cores computes the core of a join. The first half, and the second for nontrivial $\mathcal{I}$, remain open here.

Question 11.5 (Uniform degree structure beyond spectra). Within a fixed core, how much can uniform coarse degrees vary while the nonuniform coarse degree, and therefore the representative spectrum, is fixed? Can one give an exact presentation-theoretic criterion for that variation?

Equation (2.1) completely captures the nonuniform order, but not the requirement for a single functional. The explicit uniform reductions in Theorem 9.2 do not collapse the two global degree structures.

[merge] Source 08 (Proposition 18.1) shows that the *uniform exact* core, the sets produced by one functional from every description, is Comp ; report 11 relativizes this (its Proposition 5.3 in the delivered numbering). That concerns uniform extraction of a fixed set, not uniform coarse reductions, so this question stays open, as does plan target C8.

Question 11.6 (Minimal solving degrees). For the spectra $\text{Row}(\vec{A}) \cap \mathcal{Q}_H$, when do minimal elements exist, and can one give a natural presentation whose spectrum has none?

The existence of an infimum is already decided by the core. Minimality is a separate local property of the upward-closed spectrum and calls for a different argument, likely involving controlled jump inversions or refinements of presentations.

[merge] Sources 08 and 10 ask the same for their constructions (Questions 19.2 and 29.2). The research plan (target C5) records that for the dyadic codes minimal elements do exist, by Cooper's jump inversion for minimal degrees; for the spectra of this report the question is open.

Question 11.7 (Effective realizations and formal assurance). For ideals presented at specified arithmetical or hyperarithmetical complexity, what are the sharp degrees of realizing codes and of cone-avoiding descriptions? Which parts of the radius and whole-row theorems admit economical Lean formalizations over the repository's existing oracle-computability framework?

An effective index for a sequence is substantially more information than a nonuniform collection of individually simple generators. Sharp results should state explicitly which presentation data are supplied to the construction.

[merge] Parts II and III give presentation-relative bounds: source 08's pairs have jumps below $\Theta_{\bar{A}} \leq_T A'_{\oplus}$, and its fiber pairs have any prescribed common jump above $A'_{\oplus}$ (Theorems 12.2 and 17.5); source 10's splitting systems are computable in $(A_{\oplus} \oplus H)'$ (Theorem 20.3). Sharpness is open (Questions 19.3 and 29.5).

Question 11.8 (The error-free boundary). What replaces the core and the whole-row spectrum theorem for effective-dense descriptions, and can an error-free coding operation preserve their common information while imposing an independent jump condition?

A successful answer would have to address the prohibition on erroneous values, not merely shrink their density. It could provide a genuinely new route to the remaining effective-dense least-representative problem.

[merge] All three sources leave the effective-dense questions open (source 08's Question 19.7, source 10's Question 29.8). Report 14 of this programme (batch-37 manuscript 11) answers the least-representative question C2 negatively for 1-generics with computable agreement; it does not treat cores, spectra of codes, or this question.

Part II

Source 08: row-finite normal form, effective perfect exact pairs, and prescribed-jump fibers

12 Source 08: the repository question and the precise contribution

[merge] Part II is source 08, *Every Countable Turing Ideal Is a Coarse Core: Effective perfect exact pairs and rich fibres of the core map* (research draft prepared for Vladimir Reshetnikov, 28 September 2026). Its Sections 1 and 2 are this section, and its Section n is Section $n + 10$ here for $n = 3, \dots, 9$; every numbered statement keeps its place in its section, so its Theorem $n.m$ is Theorem $(n + 10).m$ here, except that its Theorems 1.1–1.3 are Theorems 12.1–12.3. Its Section 10 (conclusion) is Section 19.1, and its Appendix C is Section 18.5. Its Section 2 (Lemmas 2.2 and 2.3, Corollary 2.4) and its Appendices A and B duplicate Lemma 2.2, Proposition 2.3, Corollary 2.5, Lemma 7.1 and Lemma 9.1 of Part I with the same proofs, and are printed there (see the merge notes in Part I). Symbols follow Section 1.5: source 08’s $A, \mathcal{I}_A, B_m, B, H_A, P_n, J, R, F_C, W^C$ and $\mathcal{J}(\mathbf{c})$ are $\vec{A}, \vec{\mathcal{I}}_A, A_{<m}, A_\oplus, \Theta_{\vec{A}}, C_n, \mathcal{J}^{\text{sh}}, \mathcal{R}, F_H, \vec{W}^H$ and Q_H here, and its fiber parameters C, E are H, K . Its spelling “fibre” is normalized to “fiber”.

Source 08’s abstract. For a binary sequence X , let $\text{Core}(X)$ consist of the sets computable from every coarse description of X . We prove that the possible cores are exactly the countable Turing ideals. The construction uses positive-density columns containing robust block codes, and has an explicit computable normal form: its coarse descriptions are Medvedev equivalent to arrays whose individual rows are finite variants of prescribed generators. A finite-extension argument gives a perfect family in a single density-zero equality class whose distinct members have precisely the prescribed ideal as their common lower cone. For generators A_n , putting $A_{<m} = \bigoplus_{n < m} A_n$ and $\Theta_{\vec{A}} = \bigoplus_m A'_{<m}$, two members can be chosen with jumps below $\Theta_{\vec{A}}$; the entire family has a $\Theta_{\vec{A}}$ -computable common null envelope and a $\Theta_{\vec{A}}$ -computable density modulus.

The core does not determine the coarse degree. For each countable ideal, one fiber of the core map contains an order-embedded upper cone of the Turing degrees, simultaneously for uniform and nonuniform coarse reducibility. In these fibers we construct exact pairs with any prescribed common jump H above the jump of a uniform join of the generators. In particular, every countable ideal is the core of continuum many coarse degrees having no least Turing-degree representative.

These statements answer the countable-core realization question explicitly left open in the inspected ProvelT synthesis. The classification is also a short consequence of published cone-avoiding compactness. Ordinary exact-pair existence and jump inversion are classical, not discoveries of this paper. Global priority for the combined refinements is unverified; the results are proved here in ordinary mathematics, not certified in Lean or independently refereed.

12.1 The repository question

The inspected source is the ProvelT repository at commit

1a1396d4d3a2ac6812692df520517d86ba3a4785.

Its coarse-degree synthesis explicitly asks which countable ideals can be cores, while realizing principal ideals and particular jump-cone spectra [1]. We answer the first question completely. We also give a normal form, effective witnesses, and a uniform way of changing the coarse degree

without changing the core. This is different from reproving the already-recorded negative answer to the least-representative question.

[merge] As in Part I, the question is also plan target C4, with a proof sketch [7], and the column mechanism has the antecedent [4, Lemma 3.10 and Theorem 3.11]; source 08 cites neither.

12.2 Three main conclusions

All ideals in this part are nonempty. We regard an ideal either as a collection of sets closed under Turing equivalence, or as its corresponding collection of degrees. Let $\vec{A} = (A_n)_{n \in \mathbb{N}}$ generate the countable ideal

$$\mathcal{I}_{\vec{A}} = \{Z : (\exists m) Z \leq_T A_{<m}\}, \quad A_{<m} = \bigoplus_{n < m} A_n, \quad A_{<0} = \emptyset. \quad (12.1)$$

Use fixed effective codings for all finite and countable joins, and set

$$A_{\oplus} = \bigoplus_n A_n, \quad \Theta_{\vec{A}} = \bigoplus_m A'_{<m}. \quad (12.2)$$

The principal results are as follows; all constructions and reductions are given later.

Theorem 12.1 (Core realization; proved in Section 16). *A collection of sets is $\text{Core}(X)$ for some $X \in 2^{\mathbb{N}}$ if and only if it is a countable Turing ideal. For every presentation $\vec{A}$ there is an explicit $X = Q(\vec{A}) \equiv_T A_{\oplus}$ realizing $\mathcal{I}_{\vec{A}}$.*

Theorem 12.2 (Effective perfect exact pairs; proved in Section 15). *For every $\vec{A}$ and every m_0 , there is a $\Theta_{\vec{A}}$ -computable continuous injection $P \mapsto D_P$ from $2^{\mathbb{N}}$ into $\text{CD}(Q(\vec{A}))$ such that, for distinct P, Q ,*

$$\text{Low}(D_P) \cap \text{Low}(D_Q) = \mathcal{I}_{\vec{A}}, \quad D'_P \leq_T \Theta_{\vec{A}} \oplus P. \quad (12.3)$$

There is one $\Theta_{\vec{A}}$ -computable set V of density zero, with a $\Theta_{\vec{A}}$ -computable density modulus, covering all $D_P \triangle Q(\vec{A})$. Moreover

$$|V \cap [0, N]| \leq 2^{-m_0} N \quad (N \geq 1).$$

In particular, two computable branch parameters give an exact pair with both jumps below $\Theta_{\vec{A}}$.

Theorem 12.3 (Rich fibers and prescribed common jumps; proved in Section 17). *For each $\vec{A}$, there are binary sets F_H , all with core $\mathcal{I}_{\vec{A}}$, such that for $H, K \geq_T A'_{\oplus}$,*

$$F_H \leq_{\text{nc}} F_K \iff F_H \leq_{\text{uc}} F_K \iff H \leq_T K. \quad (12.4)$$

For every such H , the class $\text{CD}(F_H)$ contains an exact pair for $\mathcal{I}_{\vec{A}}$ with both jumps Turing equivalent to H . Restricting the parameters to $H \geq_T A''_{\oplus}$ gives continuum many distinct uniform coarse degrees, and also continuum many distinct nonuniform coarse degrees, with this core and with no least Turing-degree representative.

12.3 What is classical, what is established here

The information set called $\text{Core}(X)$ here is the X^c of Hirschfeldt, Jockusch, Kuyper and Schupp [2, Definition 3.1]. Their Theorem 3.7 supplies the short alternative proof of realization in Section 16.1. The finite-extension mechanism used below is an adaptation of classical exact-pair reasoning; the row-finite form is visible, for example, in Franklin and Solomon's presentation of an exact-pair construction [10, Section 2]. Neither ordinary exact-pair existence nor the underlying no-splitting argument is claimed as new.

The contribution of this draft is the explicit package: every countable ideal inside a single coarse-description class; a computable retraction normal form; a presentation-sensitive jump bound and common null envelope; and an upper-cone embedding with prescribed-jump exact pairs inside each fixed core. The literature check establishes the relevant precedents, but does not establish priority for this package. Even the bare classification should not be advertised as a previously unknown general theorem: its deduction from published compactness is short once the coding is chosen.

Scope of the claimed resolution

The countable-core realization question in the inspected repository is resolved. The classification of *all representative spectra* is not. We do not settle whether the spectra in question have minimal elements, the general effective-dense least-representative problem, or a general equivalence of uniform and nonuniform coarse reducibility. The equivalence in (12.4) holds only for the displayed family.

12.4 Cores, spectra, and the uniformity boundary

Source 08 uses the definitions of Section 2: $\rho_N, \bar{\rho} = \bar{\rho}, \text{CD}(X), \leq_{nc}, \leq_{uc}, \text{Low}, \text{Core}$ and Spec , the standard coarse reducibilities [2, 6], with a Turing ideal a nonempty collection closed downward under $\leq_T$ and under finite joins. Its Lemma 2.2 is Lemma 2.2 with the monotonicity recorded after it, its Lemma 2.3 is Proposition 2.3 together with the lower-cone identity of Lemma 2.2, and its Corollary 2.4 is Corollary 2.5; its proofs are the same.

The distinction between individual and simultaneous computation is essential throughout. From $A_n \leq_T D$ for every n , one may *not* conclude $\bigoplus_n A_n \leq_T D$. The reduction indices can depend noncomputably on n . Our construction exploits exactly this distinction, rather than hiding it in an implicit uniformization step.

13 A two-level code and a computable normal form

13.1 Columns and blocks

Use the computable bijection

$$p(n, t) = 2^n(2t + 1) - 1 \quad (n, t \in \mathbb{N}). \quad (13.1)$$

Its n th column is $C_n = \{p(n, t) : t \in \mathbb{N}\}$. The tail of the column partition is

$$T_m = \bigcup_{n \geq m} C_n = \{x : 2^m \mid x + 1\}.$$

Direct counting gives

$$|C_n \cap [0, N]| = \left\lfloor \frac{N + 2^n}{2^{n+1}} \right\rfloor, \quad |T_m \cap [0, N]| = \left\lfloor \frac{N}{2^m} \right\rfloor. \quad (13.2)$$

In particular, C_n has density 2^{-n-1} , whereas the tail has density 2^{-m} . These are exact finite counts, not merely asymptotic estimates.

Inside each column, partition the row coordinate t into blocks

$$I_k = [2^k - 1, 2^{k+1} - 1), \quad |I_k| = 2^k, \quad \ell(t) = \lfloor \log_2(t + 1) \rfloor. \quad (13.3)$$

For $U \in 2^{\mathbb{N}}$ define the repetition code $\mathcal{J}^{\text{sh}}(U)(t) = U(\ell(t))$. For an array $Y = (Y_n)_{n \in \mathbb{N}}$ define

$$Q(Y)(p(n, t)) = Y_n(\ell(t)). \quad (13.4)$$

Thus every bit of every row is repeated on a finite block, while every row occupies a positive-density column. The array itself is regarded as a set via a fixed computable pairing of its two coordinates.

[merge] $\mathcal{J}^{\text{sh}}$ is Part I's $\mathcal{J}$ shifted by one position, $\mathcal{J}^{\text{sh}}(U)(t) = \mathcal{J}(U)(t + 1)$, so its blocks are $[2^k - 1, 2^{k+1} - 1)$ rather than $[2^k, 2^{k+1})$; the Lean library's `blockCode` is $\mathcal{J}$, not $\mathcal{J}^{\text{sh}}$. The code $Q(\vec{A})$ is the plan's C4 code with $\mathcal{J}^{\text{sh}}$ in place of $\mathcal{J}$, and it is not Part I's $X_{\vec{A}}$, which uses $\mathcal{E} = \mathcal{J} \circ \mathcal{R}$.

Lemma 13.1 (Restriction and majority). *If $E \subseteq \mathbb{N}$ has density zero, then for each fixed n the row set $\{t : p(n, t) \in E\}$ has density zero. If D is a coarse description of $\mathcal{J}^{\text{sh}}(U)$, majority decoding on the blocks I_k , with ties decoded as 0, agrees with U except at finitely many k .*

Proof. For $s \geq 1$, all $p(n, t)$ with $t < s$ are below $2^{n+1}s$. Therefore the row error fraction is at most

$$\frac{|E \cap [0, 2^{n+1}s]|}{s} = 2^{n+1} \rho_{2^{n+1}s}(E),$$

which tends to zero for fixed n .

If a block majority is wrong, at least 2^{k-1} of the 2^k positions in that block are wrong; this lower bound is harmless also at $k = 0$. At the endpoint $N_k = 2^{k+1} - 1$, the total error fraction is at least $2^{k-1}/N_k > 1/4$. Infinitely many wrong block majorities would therefore contradict density zero. $\square$

13.2 The row-finite mass problem

For a prescribed sequence $\vec{A} = (A_n)$, put

$$\mathcal{F}_{\vec{A}} = \{Y : (\forall n) Y_n =^* A_n\}, \quad (13.5)$$

where $=^*$ means finite symmetric difference. The finite exceptional set can depend on n , with no uniform bound or effective selector.

Theorem 13.2 (Computable retraction normal form). *There are total computable functionals Γ and Δ , independent of $\vec{A}$, with the following properties:*

- (i) $\Gamma(Y) = Q(Y)$ and $\Delta\Gamma(Y) = Y$ for every array Y .
- (ii) If $D \in \text{CD}(Q(\vec{A}))$, then $\Delta(D) \in \mathcal{F}_{\vec{A}}$.

(iii) If $Y \in \mathcal{F}_{\vec{A}}$, then $\Gamma(Y) \in \text{CD}(\mathbf{Q}(\vec{A}))$ and $\Gamma(Y) \equiv_{\mathbf{T}} Y$, uniformly.

Consequently $\text{CD}(\mathbf{Q}(\vec{A}))$ and $\mathcal{F}_{\vec{A}}$ are Medvedev equivalent: a single computable functional translates solutions in either direction. The map $\Gamma\Delta$ is idempotent and normalizes coarse descriptions to row-block codes.

Proof. Let Γ be (13.4), and define

$$\Delta(D)(n, k) = \begin{cases} 1, & \sum_{t \in I_k} D(p(n, t)) > 2^{k-1}, \\ 0, & \text{otherwise.} \end{cases}$$

Every output bit uses finitely many oracle bits. For a code $\Gamma(Y)$ the relevant block is constant, so $\Delta\Gamma(Y) = Y$. Alternatively $Y_n(k)$ can be recovered by sampling $\Gamma(Y)(p(n, 2^k - 1))$; thus coding and decoding preserve the degree uniformly.

For (ii), restrict the error set $D \Delta \mathbf{Q}(\vec{A})$ to each column. Lemma 13.1 gives density-zero row errors and then finitely many majority errors in each row. This is precisely membership in $\mathcal{F}_{\vec{A}}$.

For (iii), let $S_n = Y_n \Delta A_n$, a finite set. The error set of the coded sequences has exactly

$$K_m = \sum_{n < m} \sum_{k \in S_n} 2^k \quad (13.6)$$

points in the first m columns. Every other error belongs to T_m . Hence for all $N \geq 1$,

$$\rho_N(\mathbf{Q}(Y) \Delta \mathbf{Q}(\vec{A})) \leq \frac{K_m}{N} + 2^{-m}. \quad (13.7)$$

For fixed m , take the limit superior in N , and then let m increase. The result is density zero. Finally, $\Delta\Gamma$ is the identity on all arrays, so $(\Gamma\Delta)^2 = \Gamma\Delta$. $\square$

Corollary 13.3 (Spectrum normal form).

$$\text{Spec}(\mathbf{Q}(\vec{A})) = \{\deg_{\mathbf{T}} Y : Y \in \mathcal{F}_{\vec{A}}\} = \{\deg_{\mathbf{T}} D : (\exists Y \leq_{\mathbf{T}} D) Y \in \mathcal{F}_{\vec{A}}\}. \quad (13.8)$$

Every member of this spectrum bounds every member of $\mathcal{I}_{\vec{A}}$, and $\mathbf{Q}(\vec{A}) \equiv_{\mathbf{T}} A_{\oplus}$.

Proof. Theorem 13.2 identifies which oracles compute a solution. To check equality of the *exact* degree spectra, note that $\mathcal{F}_{\vec{A}}$ has an upward-closed spectrum: given $Y \in \mathcal{F}_{\vec{A}}$ and $Z \geq_{\mathbf{T}} Y$, replace $Y_n(0)$ by $Z(n)$ for each n . The resulting array is still a finite variant in each row and has degree exactly that of Z .

For $Y \in \mathcal{F}_{\vec{A}}$, each A_n is Turing reducible to Y_n , by hard-coding its finite exceptional set. For any fixed finite collection of rows these reductions can be combined. This proves the lower-bound claim but does not uniformly recover the infinite join. Exact coding of the original array does recover its infinite join, proving $\mathbf{Q}(\vec{A}) \equiv_{\mathbf{T}} A_{\oplus}$. $\square$

Remark 13.4 (Why both coding levels are needed). The outer positive-density partition lets each individual row survive coarse errors. The growing inner blocks let the row itself survive with only finitely many decoded errors. Merely putting $A_n(t)$ at $p(n, t)$ would yield a coarse description of A_n , not a finite variant of A_n . Conversely, assigning rows to density-zero sets would allow an entire generator to be erased. The two levels solve different problems.

Remark 13.5 (Merge: two exact spectrum theorems compared). **[merge]** This remark was written for the merge. For the same presentation $\vec{A}$, Part I's code $X_{\vec{A}}$ and source 08's code $Q(\vec{A})$ both have exactly computed spectra: $\text{Row}(\vec{A})$ (Theorem 6.2) and the row-finite degrees of (13.8). The first is contained in the second. If B computes f as in (6.1), then $Y_i(n) = f(i, n, n)$ is B -computable and $Y_i(n) = A_i(n)$ for all $n \geq s_i$, so $Y \in \mathcal{F}_{\vec{A}}$. The converse is not claimed: a finite variant of each row does not obviously yield a threshold after which whole rows are correct, and the two codes use different inner codes ($\mathcal{E}$ and $\mathcal{J}^{\text{sh}}$).

14 The finite-extension mechanism

We now work inside $\mathcal{F}_{\vec{A}}$. A useful topology on this mass problem is not its inherited Cantor-space topology. Instead, give the countable collection of finite variants of each row A_n the discrete topology, and take the product. Equivalently, identify a row with its finite error set. The resulting space is homeomorphic to Baire space.

A condition is a finite sequence $p = (S_0, \dots, S_{m-1})$ of finite subsets of $\mathbb{N}$. Its length is $|p| = m$, and its cylinder is

$$[p] = \{Y \in \mathcal{F}_{\vec{A}} : (\forall n < m) Y_n = A_n \triangle S_n\}. \quad (14.1)$$

Thus a condition fixes finitely many *complete rows*, not a finite initial segment of the coded set. An extension fixes more rows and never alters a previously fixed row. Every cylinder is nonempty.

[merge] Here and in (13.6) the letters S_n are finite error masks and p is a condition, as in source 08; $p(n, t)$ remains the pairing.

14.1 Finite computations are searchable in a finite join

Lemma 14.1 (Possible convergence). *For a condition p of length m , the relation*

$$(\exists Y \in [p]) \Phi_e^Y(k) \downarrow = b \quad (14.2)$$

is computably enumerable in $A_{<m}$, uniformly in the finite parameters. For two conditions p, q , the existence of a cross-disagreement

$$(\exists k)(\exists Y \in [p])(\exists Z \in [q]) \quad \Phi_e^Y(k) \downarrow \neq \Phi_f^Z(k) \downarrow \quad (14.3)$$

is computably enumerable in $A_{<\max(|p|, |q|)}$.

The oracle $\Theta_{\vec{A}}$ decides these existential questions and, in a positive case, computes extending conditions that preserve the displayed finite computations.

Proof. On the fixed rows, oracle answers are computable from $A_{<m}$ and the finite masks in p . On all other rows, any finite consistent assignment of queried bits is possible. Indeed, finitely many prescribed bits can always be realized by a finite variant of A_n , regardless of how complicated the unqueried part of that row is.

Enumerate finite computation traces together with their finitely many guesses for unfixed-row answers. Reject inconsistent traces and traces conflicting with a fixed row. This is an $A_{<m}$ -effective enumeration and enumerates exactly the possible convergences. Two independent copies of the same procedure enumerate (14.3), using the larger finite join for their fixed rows.

The relevant finite-join jump is a component of $\Theta_{\vec{A}}$, so this oracle decides whether such a trace exists. When it exists, search for one. The oracle $\Theta_{\vec{A}}$ also uniformly computes every A_n : it

computes $A_{<n+1}$ from $A'_{<n+1}$. For each as-yet-unfixed row touched by the trace, choose the finite error mask making its queried answers correct. Fill any skipped rows with the empty mask and fix all rows through the largest touched index. These extensions preserve the convergences. For a cross-disagreement do this separately on the two sides. $\square$

The extension oracle is used in two distinct ways. A finite-join jump decides whether a trace exists; the original rows allow the chosen finite trace to be completed to whole rows. No decision about totality of an arbitrary functional is made.

14.2 No cross-disagreement means common information is already finite

Lemma 14.2 (No-cross-disagreement lemma). *Let p, q be conditions and e, f indices. If (14.3) fails, then whenever $Y \in [p]$ and $Z \in [q]$ make Φ_e^Y and Φ_f^Z total, those outputs are equal and their common value is computable in $A_{<|p|}$. In particular any common total output belongs to $\mathcal{I}_{\tilde{A}}$.*

Proof. Fix a total pair. On input k , search, using $A_{<|p|}$, for any finite trace witnessing a possible convergence of Φ_e on input k inside $[p]$. The search terminates because the actual computation from Y is one such trace. Let the found output be b .

The actual computation of $\Phi_f^Z(k)$ is a possible convergence on the other side. Since there is no cross-disagreement anywhere in the rectangle, its output must be b . The same argument applies to the actual left computation, and hence both total outputs equal the function computed by this search. The finite condition and program index are fixed constants in the reduction, not an additional oracle. $\square$

Proposition 14.3 (Dense exact-pair requirements). *For every p, q, e, f , there are extensions $p^* \supseteq p$ and $q^* \supseteq q$ such that either the two functionals are forced to disagree at some input, or every common total output on $[p^*] \times [q^*]$ belongs to $\mathcal{I}_{\tilde{A}}$. Such extensions and the relevant case can be chosen with oracle $\Theta_{\tilde{A}}$.*

Proof. Decide (14.3). In the positive case, force a witnessing disagreement using Lemma 14.1. In the negative case, leave both conditions unchanged and apply Lemma 14.2. Both conclusions persist under further extensions. $\square$

This is the classical exact-pair dichotomy in the form needed here. Crucially, a negative answer does not require deciding that a functional is partial. If total common output eventually occurs, the no-cross-disagreement search supplies its finite-join computation. This avoids an unnecessary extra jump in the construction oracle.

15 Perfect exact pairs with jump control and one null envelope

We prove Theorem 12.2. A family indexed by all of Cantor space cannot consist entirely of sets below one oracle. The branch parameter in the bound $D'_P \leq_T \Theta_{\tilde{A}} \oplus P$ is therefore essential; only selected computable branches have the unparameterized bound.

15.1 Construction of the finite tree

Fix m_0 . At level zero use the single condition of length m_0 with every mask empty. It requires the first m_0 rows to agree exactly with $\vec{A}$. We build conditions p_σ for binary strings σ , extending their predecessor conditions. At the end of each level all conditions at that level have a common finite length, and that length tends to infinity with the level. Store finite records for the jump decisions made below.

Suppose the conditions at level s have common length L . To construct level $s + 1$, perform the following finite operations, all with oracle $\Theta_{\vec{A}}$.

1. **Split.** For each level- s node, create two extending conditions fixing the fresh row L . Make its bit 0 equal to 0 in one child and 1 in the other, and keep the rest of that row equal to A_L . The required masks are either empty or $\{0\}$, selected using $A_L(0)$. Descendants of distinct children will remain different.
2. **Meet cross requirements.** For every ordered pair of distinct nodes now at level $s + 1$, and every pair of indices $e, f \leq s$, apply Proposition 14.3, replacing the two conditions by the extensions obtained. The list is finite. A later replacement cannot destroy an earlier forced disagreement or an earlier absence of cross-disagreements.
3. **Decide the next jump bit.** For each node, decide whether some array in its current cylinder makes $\Phi_s^Y(s)$ converge. In a positive case extend the condition to preserve a finite convergent computation and record 1; in a negative case leave it and record 0. Lemma 14.1 makes this decision possible using $\Theta_{\vec{A}}$. Again these decisions persist under all further extensions.
4. **Pad and synchronize lengths.** Extend all conditions with empty masks to a common length at least $s + 1$ and at least as large as every length produced so far. Retain every finite record.

There is no injury: rows, disagreement witnesses, and negative existence decisions are preserved. The complete finite tree and its records are computable in $\Theta_{\vec{A}}$.

15.2 Verification of exactness and effectivity

For $P \in 2^{\mathbb{N}}$, every row is eventually fixed along the branch. Let Y_P be the unique array in every cylinder along that branch, and put $D_P = \Gamma(Y_P)$. Each row of Y_P is a finite variant of the corresponding A_n , so Theorem 13.2 gives $D_P \in \text{CD}(\text{Q}(\vec{A}))$.

Every member of $\mathcal{I}_{\vec{A}}$ is computable from every D_P . For the reverse inclusion, take distinct P, Q and a set Z with

$$Z = \Phi_e^{Y_P} = \Phi_f^{Y_Q}.$$

Choose a level after the two branches have separated and with $s \geq e, f$. The construction processes the corresponding pair of nodes. A forced disagreement would contradict the displayed equality, so the no-cross-disagreement alternative must apply. Lemma 14.2 then puts Z in $\mathcal{I}_{\vec{A}}$. Since D_P and Y_P have uniformly equivalent degrees, this proves

$$\text{Low}(D_P) \cap \text{Low}(D_Q) = \mathcal{I}_{\vec{A}}.$$

To compute Y'_P from $\Theta_{\vec{A}} \oplus P$, locate the node on P at the level where the construction decided its e th jump bit, and read the stored answer. A positive answer preserved a halting trace. A negative answer excluded every possible extension in that cylinder. Thus the answer is correct for the final

branch, and the procedure is uniform in e . The equivalence $D_P \equiv_T Y_P$ is uniform, hence so is $D'_P \equiv_T Y'_P$. We obtain

$$D'_P \leq_T \Theta_{\vec{A}} \oplus P. \quad (15.1)$$

To compute a bit of D_P , compute enough of the finite tree for its row to have been fixed, and use the corresponding finite prefix of P . This proves that $P \mapsto D_P$ is a $\Theta_{\vec{A}}$ -computable continuous map. The preserved marker bits make it injective. A continuous injection from compact Cantor space into Cantor space is a homeomorphism onto its image: images of closed sets are compact and therefore closed. The image is consequently a nonempty perfect set.

In particular, $P = 000 \dots$ and $Q = 100 \dots$ give two descriptions whose jumps are below $\Theta_{\vec{A}}$. The same is true for every computable branch parameter, and any two distinct such parameters give an exact pair.

15.3 A common error set with an effective modulus

For each n , compute a level at which every node has fixed row n . Let R_n be the union of the finitely many error masks appearing in that row among those nodes. Later extension cannot alter them. Thus R_n is a finite set computable uniformly in n from $\Theta_{\vec{A}}$, and

$$Y_{P,n} \Delta A_n \subseteq R_n \quad \text{for every } P.$$

Define

$$V = \{p(n, t) : \ell(t) \in R_n\}. \quad (15.2)$$

Membership in V is $\Theta_{\vec{A}}$ -computable. All errors $D_P \Delta Q(\vec{A})$ are covered by V . With

$$K_m^V = \sum_{n < m} \sum_{k \in R_n} 2^k,$$

the exact counting argument from (13.7) gives

$$\rho_N(V) \leq \frac{K_m^V}{N} + 2^{-m}. \quad (15.3)$$

This proves density zero. More explicitly,

$$M(r) = 2^{r+1}(K_{r+1}^V + 1) \quad (15.4)$$

is $\Theta_{\vec{A}}$ -computable, and $N \geq M(r)$ implies $\rho_N(V) < 2^{-r}$. This is a density modulus, not merely a proof that a limit exists.

The initial condition imposed $R_n = \emptyset$ for $n < m_0$. Therefore $V \subseteq T_{m_0}$, and the exact tail count gives the asserted global bound $|V \cap [0, N]| \leq 2^{-m_0}N$. This completes the proof of Theorem 12.2.

Remark 15.1 (What the bound does not say). The oracle $\Theta_{\vec{A}}$ is the countable join of the jumps of *finite joins*. One must not replace it by $\bigoplus_n A'_n$ without additional hypotheses. The jump need not distribute over finite joins. On the other hand, $\Theta_{\vec{A}} \leq_T A'_\oplus$ because each finite projection $A_{<m} \leq_T A_\oplus$ is uniform in m . The smaller bound $\Theta_{\vec{A}}$ can matter substantially; Section 16.3 gives an example.

Corollary 15.2 (Computable masks would force a principal core). *Suppose $D, E \in \text{CD}(X)$ satisfy $\text{Low}(D) \cap \text{Low}(E) = \text{Core}(X) = \mathcal{I}$, and a computable density-zero set V covers both $D \Delta X$ and $E \Delta X$. Then $\mathcal{I}$ is principal and its generator lies in $\text{Spec}(X)$. In particular, for a nonprincipal ideal the common envelope in the preceding construction cannot be computable.*

Proof. Let W agree with X outside V and be 0 on V . Since D and E agree with X outside V , the computable mask lets either oracle compute W . Thus $W \in \mathcal{I}$. Also $W \in \text{CD}(X)$, so every member of $\text{Core}(X) = \mathcal{I}$ is below W . Downward closure gives $\mathcal{I} = \text{Low}(W)$, and its generator is attained. For the construction above, equality of its core with $\mathcal{I}_{\vec{A}}$ is proved in the next section. $\square$

16 The complete core-realization theorem

Proof of Theorem 12.1. The necessity is Lemma 2.2. Conversely, let $\mathcal{I}$ be a countable Turing ideal. Choose a countable sequence of sets generating it under downward closure and finite joins. If the ideal is presented as a countable collection of degrees, first choose representatives. In classical ZFC these are ordinary countable choices; no effective enumeration of the ideal is assumed.

Let $\vec{A}$ be the chosen sequence and $X = \text{Q}(\vec{A})$. Corollary 13.3 gives $\mathcal{I} \subseteq \text{Core}(X)$. The construction of Section 15 supplies $D, E \in \text{CD}(X)$ with $\text{Low}(D) \cap \text{Low}(E) = \mathcal{I}$. Since the core is contained in the common lower cone of any two descriptions, $\text{Core}(X) \subseteq \mathcal{I}$. Thus equality holds. The degree identity $X \equiv_{\text{T}} \bigoplus_n A_n$ was proved in Corollary 13.3. $\square$

[merge] This is route (c) of Section 1.4: it uses neither the Baire category theorem on $\text{CD}(X)$ nor [2, Theorem 3.7].

16.1 A shorter independent route through published compactness

For perspective, the classification itself does not require our perfect-family construction. We state the particular published result needed for the shorter route.

Theorem 16.1 (Cone-avoiding compactness, HJKS). *If, for every $\varepsilon > 0$, there is an E_ε with $\bar{\rho}(X \triangle E_\varepsilon) \leq \varepsilon$ and $Z \not\leq_{\text{T}} E_\varepsilon$, then some $D \in \text{CD}(X)$ satisfies $Z \not\leq_{\text{T}} D$.*

This is [2, Theorem 3.7]; it is not proved anew here. To apply it, fix $Z \notin \mathcal{I}_{\vec{A}}$ and truncate $X = \text{Q}(\vec{A})$ to its first m columns, putting 0 on all remaining columns. Call the result E_m . Then

$$E_m \leq_{\text{T}} A_{<m}, \quad Z \not\leq_{\text{T}} E_m, \quad \bar{\rho}(X \triangle E_m) \leq 2^{-m}.$$

The published compactness theorem yields a description avoiding Z , so $Z \notin \text{Core}(X)$. Together with the already-proved reverse inclusion this yields the same classification.

This argument explains the literature status precisely: a question explicitly unclassified in the repository has a positive answer obtainable from an existing general theorem. The longer construction is retained because it supplies joint witnesses, perfect families, jump bounds, and an effective null envelope, none of which follows just by separately avoiding one set at a time in this short argument.

[merge] Second route (source 08). Theorem 16.1 is the contrapositive form of Theorem 3.1, which Part I proves; with it this argument is the upper inclusion of Theorem 5.1, applied to $\text{Q}(\vec{A})$ via Corollary 3.4.

16.2 Nonprincipal ideals and missing infima

Corollary 16.2. *Every countable nonprincipal Turing ideal is the common lower cone of each pair of distinct members of a perfect set contained in one density-zero equality class. Such pairs have no greatest lower bound.*

Their common coarse class has neither a greatest lower bound for its representative spectrum nor a least representative.

Proof. Use Theorem 12.2 and core realization. The lower cone of a greatest lower bound would be principal, contradicting the chosen ideal. The representative-spectrum assertions follow from Corollary 2.5. $\square$

This does not imply that the spectrum has no *minimal* members. A partially ordered set can lack a least member and still have many minimal members. No construction in this part silently replaces the least-member question by the minimal-member question.

16.3 The ideal of all arithmetical sets

Take $A_n = \emptyset^{(n)}$, using the usual finite iterates of the Turing jump, with $A_0 = \emptyset$. Then $\mathcal{I}_{\vec{A}}$ is exactly the ideal of arithmetical sets, and the uniform join is $\emptyset^{(\omega)}$. With standard uniform choices of the finite-jump reductions,

$$A_{<m} \equiv_T \emptyset^{(m-1)} \quad (m \geq 1), \quad \Theta_{\vec{A}} \equiv_T \emptyset^{(\omega)}.$$

Thus there is a binary set $X \equiv_T \emptyset^{(\omega)}$ with

$$\text{Core}(X) = \{Z : (\exists n) Z \leq_T \emptyset^{(n)}\},$$

and two coarse descriptions D, E such that

$$\text{Low}(D) \cap \text{Low}(E) = \text{Core}(X), \quad D', E' \leq_T \emptyset^{(\omega)}. \quad (16.1)$$

Neither D nor E is arithmetical. If, for example, $D \leq_T \emptyset^{(n)}$, then the fact that D computes every generator would give $\emptyset^{(n+1)} \leq_T \emptyset^{(n)}$, contradicting strictness of the jump. Nevertheless both descriptions, and even their jumps, lie below the displayed limit join.

Equation (16.1) illustrates the nonuniform nature of recovering the generators: each description computes every finite jump, but the construction does not give a uniform computation of the entire infinite join from either description. Nor does it assert that their jumps are *equal* to the limit join. The conclusion is the stated upper bound.

[merge] Compare Example 5.4 (Part I's code, spectrum without infimum) and Example 27.2 (source 10's carrier). Here $\Theta_{\vec{A}} \equiv_T A_{\oplus} = \emptyset^{(\omega)}$, strictly below $A'_{\oplus}$, which is the example promised in the remark after (15.4).

16.4 Principal ideals and dependence on presentation

For a principal ideal $\text{Low}(A)$ one may take every row equal to A . Then $\mathcal{Q}(\vec{A}) \equiv_T A$ and its spectrum has least degree $\deg_T A$. The ideal is not enough, however, to determine the mass problem in (13.5). A sequence of individually computable rows can still carry noncomputable information in the way those rows are assembled.

For example, for a set H , let the n th row be the constant sequence $\overline{H(n)}$. Every row is computable, so the generated ideal is Comp . But solving the row-finite mass problem produces a simultaneous family of sequences converging to the bits of H . The uniform difficulty of that task can be arbitrarily large. The next section makes this observation quantitative for every core, not just Comp .

[merge] Source 08 writes C for this set; it is renamed H to match Example 6.4 and Proposition 27.3, which record the same phenomenon for the other two codes.

17 Changing the coarse degree while keeping the core fixed

Fix the sequence $\vec{A}$ for the rest of this section. For each set H , define an interleaved sequence of rows by

$$W_{2n}^H = A_n, \quad W_{2n+1}^H = \underline{H(n)}, \quad F_H = Q(\vec{W}^H). \quad (17.1)$$

The odd rows are constant binary sequences. As individual sets they have degree zero, regardless of whether H is computable. Consequently

$$\mathcal{I}_{\vec{W}^H} = \mathcal{I}_{\vec{A}}, \quad \text{Core}(F_H) = \mathcal{I}_{\vec{A}} \quad \text{for every } H. \quad (17.2)$$

There is no appeal here to a density-preserving arbitrary permutation of $\mathbb{N}$: F_H is literally the code of the displayed interleaved array.

[merge] Do not read F_H as Part I's $T_{Q(\vec{A})}(H) = Q(\vec{A}) \oplus \mathcal{R}(H)$: here the parameter occupies the odd rows *inside* one code, not a second half. The two sets differ, but their coarse classes coincide up to uniform equivalence; see Remark 17.7.

17.1 An exact spectrum formula

For a degree $\mathbf{h}$, write

$$Q_{\mathbf{h}} = \{\mathbf{d} : \mathbf{h} \leq \mathbf{d}'\},$$

so $Q_H = Q_{\deg_T(H)}$ is the jump cone of Part I. The elementary relativized limit lemma used below is Lemma 7.1. It says that $H \leq_T D'$ exactly when there is a D -computable binary approximation $h(n, s)$ converging to $H(n)$ for every n .

Theorem 17.1 (Spectrum factorization). *For every set H ,*

$$\text{Spec}(F_H) = \text{Spec}(Q(\vec{A})) \cap Q_H. \quad (17.3)$$

Equivalently, D computes a coarse description of F_H if and only if it computes a coarse description of $Q(\vec{A})$ and $H \leq_T D'$.

Proof. Suppose D computes a description of F_H . Apply Δ from Theorem 13.2 to obtain a D -computable array Y with $Y_j =^* W_j^H$ for each j . The even rows form a member of $\mathcal{F}_{\vec{A}}$. Hence D computes a coarse description of $Q(\vec{A})$. The odd rows give the D -computable approximation

$$h(n, s) = Y_{2n+1}(s) \longrightarrow H(n),$$

so $H \leq_T D'$ by the limit lemma.

Conversely, suppose D computes $Y \in \mathcal{F}_{\vec{A}}$ and a convergent approximation $h(n, s) \rightarrow H(n)$. Form the D -computable array Z with $Z_{2n} = Y_n$ and $Z_{2n+1}(s) = h(n, s)$. Each odd row is a finite variant of the relevant constant sequence, and each even row is a finite variant of A_n . Thus $Z \in \mathcal{F}_{\vec{W}^H}$, and $\Gamma(Z)$ is a coarse description of F_H . Proposition 2.3 translates this statement about oracles computing descriptions into the exact degree-spectrum equality. $\square$

The constant-row special case is the familiar dyadic jump-cone phenomenon. The code $\mathcal{R}(H)(p(n, t)) = H(n)$ has a D -computable coarse description exactly when $H \leq_T D'$; compare Jockusch–Schupp [3] and the repository's dyadic development [1]. Theorem 17.1 attaches that phenomenon to an arbitrary prescribed core without changing the core.

17.2 Uniform reductions from ordinary Turing reductions

Lemma 17.2 (Uniform monotonicity). *If $H \leq_T K$, then $F_H \leq_{uc} F_K$.*

Proof. Fix an oracle program Ψ with $\Psi^K = H$. From an arbitrary $D \in CD(F_K)$ compute $Y = \Delta(D)$. Its even rows are finite variants of A_n ; its odd rows supply the convergent approximation

$$k_s(n) = Y_{2n+1}(s) \longrightarrow K(n).$$

Define a total approximation $h_s(n)$ by running $\Psi^{k_s}(n)$ for at most s steps, using its binary output if it halts with output 0 or 1, and using 0 otherwise. For each fixed n , the true computation $\Psi^K(n)$ has finite time and finite oracle use. Eventually s exceeds that time and k_s is correct on every queried input. Consequently $h_s(n) \rightarrow H(n)$.

Keep the even rows of Y , and replace its odd rows by the sequences $s \mapsto h_s(n)$. The resulting array lies in $\mathcal{F}_{\vec{W}H}$. Apply Γ . All steps use one total computable functional determined by the index of Ψ , and no convergence modulus or extra jump is used. This proves the uniform reduction. $\square$

17.3 Reflection of order on an upper cone

Theorem 17.3 (Order-embedded upper cone in every fiber). *Let $A_\oplus = \bigoplus_n A_n$. For $H, K \geq_T A'_\oplus$,*

$$F_H \leq_{nc} F_K \iff F_H \leq_{uc} F_K \iff H \leq_T K.$$

Hence both coarse-degree structures contain an order-embedded copy of the Turing upper cone above $\deg_T A'_\oplus$ all of whose members have core $\mathcal{I}_{\vec{A}}$.

Proof. The right-to-left implication, including uniformity, is Lemma 17.2; every uniform coarse reduction is also nonuniform.

For the remaining implication assume $F_H \leq_{nc} F_K$. Relativized Friedberg jump inversion gives a set $D \geq_T A_\oplus$ with $D' \equiv_T K$, since $K \geq_T A'_\oplus$. A proof of exactly this form is Lemma 9.1 (with $X = A_\oplus$). Because D computes the complete generator array, D computes $Q(\vec{A})$. The spectrum factorization therefore says that D computes a coarse description of F_K . Nonuniform reducibility transfers this to a D -computable coarse description of F_H . Applying the same factorization again yields

$$H \leq_T D' \equiv_T K.$$

This proves order reflection. If $H \equiv_T K$, the uniform monotonicity lemma in both directions gives $F_H \equiv_{uc} F_K$, so the map factors through Turing degrees. Reflection shows that it is injective and has the asserted order. $\square$

The upper-cone hypothesis is a sufficient condition allowing the reflecting oracle D to compute the original array. No claim is made that this threshold is optimal for every presentation. The spectrum identity alone would not prove the positive *uniform* reduction; the separate construction in Lemma 17.2 is needed.

[merge] For the relation of this theorem to Part I's Theorem 9.2 see Remark 17.7 at the end of this section.

17.4 Exact pairs with a prescribed common jump

Let $\Theta_{\tilde{W}^H}$ denote the finite-join-jump oracle for the interleaved sequence in (17.1). The following computation is important: adding individually computable rows need not preserve this *uniform* oracle, because their constant values can encode H .

Lemma 17.4 (Oracle accounting). *Uniformly in the presentations,*

$$\Theta_{\tilde{W}^H} \equiv_T \Theta_{\tilde{A}} \oplus H, \quad \Theta_{\tilde{A}} \leq_T A'_{\oplus}. \quad (17.4)$$

In particular $H \geq_T A'_{\oplus}$ implies $\Theta_{\tilde{W}^H} \equiv_T H$.

Proof. Every finite initial join of the A_n is a computable projection of a finite initial join of the W_j^H , uniformly in its length. Jump monotonicity gives $\Theta_{\tilde{A}} \leq_T \Theta_{\tilde{W}^H}$. Also $H(n)$ is the first bit of the $(2n+1)$ st row, so a sufficiently large finite-join jump computes it uniformly; hence $H \leq_T \Theta_{\tilde{W}^H}$.

Conversely, a finite initial join of the interleaved rows consists of finitely many A_n together with finitely many constant rows. Given those finitely many bits of H , an oracle program using this join can be translated into one using the appropriate $A_{<m}$: replace queries to an odd row by its known constant answer and queries to an even row by the corresponding finite-join query. Therefore its halting problem is decidable in $A'_{<m}$ using these finite parameters. The translation is uniform from the prefix length and the requisite bits of H , proving $\Theta_{\tilde{W}^H} \leq_T \Theta_{\tilde{A}} \oplus H$.

Finally the projection $A_{<m} \leq_T A_{\oplus}$ is uniform in m , so the induced reduction of $A'_{<m}$ to $A'_{\oplus}$ is uniform in m . Taking their countable join proves $\Theta_{\tilde{A}} \leq_T A'_{\oplus}$. This uses monotonicity of the jump, not distribution over finite joins. $\square$

Theorem 17.5 (Prescribed-jump exact pairs in a single coarse class). *Fix $H \geq_T A'_{\oplus}$ and m_0 . There is an H -computable continuous injection $P \mapsto D_P$ into $\text{CD}(\mathbb{F}_H)$ such that, for $P \neq Q$,*

$$\text{Low}(D_P) \cap \text{Low}(D_Q) = \mathcal{I}_{\tilde{A}}, \quad H \leq_T D'_P \leq_T H \oplus P. \quad (17.5)$$

There is one H -computable null envelope, with an H -computable density modulus and prefix-density bound 2^{-m_0} , covering all errors against $\mathbb{F}_H$. Every computable branch parameter yields $D'_P \equiv_T H$. Thus any two distinct computable parameters give an exact pair with prescribed common jump H .

Proof. Apply Theorem 12.2 to the sequence $\vec{W}^H$. Its generated ideal is $\mathcal{I}_{\tilde{A}}$, and Lemma 17.4 identifies its construction oracle with H . This proves all assertions except the lower bound on each jump. But $D_P \in \text{CD}(\mathbb{F}_H)$, so the spectrum factorization gives $H \leq_T D'_P$. For computable P the upper bound reduces to H , proving equality. $\square$

Notice that $\mathbb{F}_H \equiv_T A_{\oplus} \oplus H \equiv_T H$ under this hypothesis, whereas a computable-parameter description with $D'_P \equiv_T H$ is strictly below H . This is compatible with belonging to the same coarse class: coarse descriptions need not preserve the original Turing degree. Pairwise common information has been held fixed at $\mathcal{I}_{\tilde{A}}$ while the exact jump of selected descriptions is prescribed independently above the stated threshold.

17.5 Continuum many classes without least representatives

Corollary 17.6. *Every countable Turing ideal is the core of continuum many different nonuniform coarse degrees, and of continuum many different uniform coarse degrees, each with no least Turing-degree representative.*

Proof. Fix $\vec{A}$ and restrict the embedding of Theorem 17.3 to $H \geq_T A''_{\oplus}$. If the ideal is nonprincipal, Corollary 2.5 already excludes a least representative. If it is principal, say $I_{\vec{A}} = \text{Low}(Z)$, any least representative would have degree $\deg_T Z$. Since Z is below some finite join, $Z \leq_T A_{\oplus}$ and $Z' \leq_T A'_{\oplus}$. But if $\deg_T Z$ belonged to $\text{Spec}(F_H)$, Theorem 17.1 would give

$$A''_{\oplus} \leq_T H \leq_T Z' \leq_T A'_{\oplus},$$

contradicting strictness of the jump. Thus the principal case also has no attained least degree. Its infimum is $\deg_T Z$; in the nonprincipal case there is no infimum.

There are continuum many Turing degrees above $A''_{\oplus}$: the sets $A''_{\oplus} \oplus R$ for $R \in 2^{\mathbb{N}}$ form a continuum-sized collection, and each degree contains only countably many sets. In ZFC a countable-to-one partition of a continuum-sized set has continuum many classes. The order embedding preserves distinctness in both coarse-degree structures. There cannot be more than continuum many binary coarse degrees, so the cardinality is exactly continuum. $\square$

This completes Theorem 12.3. The result is a failure of classification by common information, not a failure of core realization: every allowable core occurs, but fixing it leaves enough room for an entire upper cone of degree-theoretic complexity.

Remark 17.7 (Merge: the interleaved code and the filter of Part I). **[merge]** This remark is a merge observation, in no source, and unrefereed. For every H , $F_H \equiv_{\text{uc}} T_{Q(\vec{A})}(H)$. Given a description D of F_H , let $Y = \Delta(D)$; applying Γ to its even rows gives a description of $Q(\vec{A})$, and writing $Y_{2n+1}(m)$ at $p(n, m)$ gives a description of $\mathcal{R}(H)$, since each column then has finitely many errors (Lemma 4.1). Conversely, from a description $D_0 \oplus D_1$ of $T_{Q(\vec{A})}(H)$ take the rows of $\Delta(D_0)$ as even rows and the column majorities $h(n, s) = \text{Maj}\{D_1(p(n, 0)), \dots, D_1(p(n, s))\}$ as odd rows; the array lies in $\mathcal{F}_{\vec{W}_H}$, and Γ of it describes F_H . Both directions are fixed functionals. Hence (17.3) is also a case of (8.4), and Theorem 17.3 is, up to uniform coarse equivalence, Theorem 9.2 for $X = Q(\vec{A})$, on the same cone because $Q(\vec{A}) \equiv_T A_{\oplus}$ (Corollary 13.3); in particular source 08's map also preserves binary joins, which source 08 does not state. Source 08's proofs through the normal form remain an independent route, and its prescribed-jump pairs (Theorem 17.5) use the interleaved form essentially.

18 Uniformity, comparison, and proof boundaries

18.1 A uniform core would collapse

There are three different notions of uniformity in the preceding arguments: one functional converting descriptions to arrays; a uniform join of the original generators; and one functional extracting a fixed set from all descriptions. Only the first is supplied by the normal-form theorem. The second is deliberately not recovered from an arbitrary solution, and the third is impossible for a noncomputable set.

Proposition 18.1 (No uniform extraction of noncomputable common information). *For any $X, C \in 2^{\mathbb{N}}$, the following are equivalent:*

$$C \text{ is computable; } \quad (\exists e)(\forall D \in \text{CD}(X)) \Phi_e^D = C.$$

Proof. A computable C can be produced without using its oracle. Conversely, suppose one functional produces C from every description. Every finite binary string extends to a member of $\text{CD}(X)$, simply by modifying a finite prefix of X . Consequently any finite convergent trace of that functional on input n must output $C(n)$: extend its finite oracle answers to a description and use the assumed correctness. Search computably for any such trace. One exists because the functional is total on X itself. The search computes $C(n)$ without an oracle. $\square$

Thus $\text{Core}(X)$ is genuinely a *nonuniform* intersection of lower cones. This proposition also explains why the finite exceptional sets in majority decoding cannot be uniformly corrected to recover an arbitrary generator from every coarse description. The uniform coarse reductions of Section 17 avoid that impossible step: they only transform one family of converging approximations into another.

[merge] This answers the last task of plan target C4 (“decide whether the uniform core, defined with a single functional, behaves differently”): it is always Comp. Report 11 of this programme proves the relativized form (its Proposition 5.3 in the delivered numbering).

18.2 Comparison with the inspected repository

The following comparison records the extension, rather than treating the entire argument as a new discovery. The source is the synthesis and the Turing-degree project README at the pinned revision [1, 8].

Repository material	Present continuation
Principal ideals realized as cores	All countable Turing ideals realized by one explicit two-level code.
Perfect exact partners for arbitrary targets, without general effective bounds	A concrete normal form for the new targets, followed by a $\Theta_{\bar{A}}$ -effective perfect-family construction and a common effective null envelope. Perfect exact-pair existence by itself is not claimed as a first result.
Dyadic spectra described by jump cones	An exact intersection formula attaching the jump-cone constraint to an arbitrary row-finite mass problem, while leaving its core unchanged.
Examples sharing a principal core but having different least-representative behaviour	An embedded full upper cone in every core fiber, and continuum many no-least classes for each countable ideal.
Jump control in specialized reservoir constructions	Presentation-sensitive finite-join-jump control, and selected exact pairs with a prescribed common jump throughout every displayed fiber.

[merge] The table omits the research plan, which at the same revision already proposed the first row as target C4 with a proof sketch [7].

The project README describes both formalized results and admitted inputs. This manuscript does not use any admitted Lean declaration as an axiom. The main route is the elementary coding and finite-extension proof given here. The alternative short proof explicitly invokes a published theorem instead. The archive contains no purported Lean formalization and no claim that the new statements have passed a proof-assistant kernel.

18.3 Dependency and correctness checks

The proof has a small number of sensitive points, each made explicit in the text.

- (1) **Finite row errors are not a uniform finite error set.** The passage to density zero uses a fixed finite set of columns followed by a small-density tail; it never takes the union of all row errors to be finite.
- (2) **Positive column density is used before decoding.** A global density-zero error set induces density-zero errors in every fixed row because the column enumeration has bounded linear growth. No arbitrary sparse subsequence is used.
- (3) **The search oracle is a finite join.** Conditions fix complete rows; possible convergence is enumerable in the finite join of those rows. A negative existence answer is decided by that finite join's jump. No unproved effective totality test occurs.
- (4) **Jump accounting retains the presentation.** $\Theta_{\bar{A}}$ uses jumps of finite joins. Adding individually computable but nonuniformly assembled rows can increase it. Lemma 17.4 tracks that increase exactly for the fiber construction.
- (5) **A perfect family requires a branch parameter.** Only its finite tree is computed by the construction oracle. A general branch is allowed its own P , yielding the bound in (15.1); computable branch parameters give the two low-complexity witnesses.
- (6) **Density transport is explicit.** The fiber construction is an actual interleaving of rows inside the same code, not an appeal to preservation of asymptotic density under arbitrary computable permutations.
- (7) **Uniform and nonuniform reductions are proved separately.** Spectrum inclusion supplies the necessary condition for order reflection. A total functional on descriptions supplies the sufficient uniform reduction.
- (8) **The mask obstruction needs exactness with the core.** Corollary 15.2 assumes $\text{Low}(D) \cap \text{Low}(E) = \text{Core}(X)$. Without this equality, two descriptions could share additional information written only on the null mask.

18.4 A practical formalization decomposition

The repository's existing set-based Turing-reducibility and coarse-density infrastructure is a natural starting point [8]. A formal development should separate the purely combinatorial code from the oracle semantics and from the infinitary construction.

Coding layer. Formalize the pairing p , its inverse via the power of two dividing $x + 1$, the exact column and tail counts, the block partition, and the finite-majority error implication. These are arithmetic and finite-set statements. Next prove the density transfer, estimate (13.7), and the two directions of the normal-form theorem. The executable finite checks in the archive exercise these definitions, but are not substitutes for these proofs.

Oracle layer. Use a model of partial oracle computations with finite use. Prove a trace-extension theorem and an effective enumeration of traces compatible with finitely many fixed rows. Package Lemma 14.1 with its exact finite-join oracle, rather than assuming a generic forcing decision procedure. Jump monotonicity and uniform index translations must be supplied explicitly.

Construction layer. Represent a condition by a finite list of finite error masks. State the invariants of the level construction: predecessor extension, common end-of-level row length, permanent distinct markers, persistent cross requirements, and recorded correct jump decisions. Define branches from the row-stabilization property. Derive the continuous injection and the common null envelope from the same finite tree, so that their effectivity is not hidden behind an application of an abstract perfect-set theorem.

Degree-theoretic layer. After quotienting by Turing equivalence, prove core realization, spectrum factorization, and the two implications in the fiber embedding. The only extra classical constructions needed for order reflection are the limit lemma and relative jump inversion, both given in Part I (Lemmas 7.1 and 9.1; source 08 gives them in its appendices). The common-jump theorem should then be a short corollary of the explicit oracle-accounting lemma.

This is a proposed implementation plan, not a report of completed formalization. It deliberately keeps machine-level uniformity statements separate from existence statements in the degree order.

[merge] Of the coding layer, the Lean library already proves the tail count, the restriction of density-zero errors to a column and the bounded-column assembly lemma, but for Part I's block code $\mathcal{J}$, not for $\mathcal{J}^{\text{sh}}$ (Section 1.4). The oracle and construction layers are absent: the library has no use-bounded model of Turing functionals.

18.5 Finite verification and reproducibility

The accompanying script, shipped as `code/08-effective-exact-pairs-verify.py`, uses only the Python standard library. It checks the following finite statements by direct enumeration: the pairing inverse; exact column and tail counts; the block-majority error implication; coding and sampling identities; finite versions of the first-columns-plus-tail estimate; and the finite Cartesian-product fact behind no cross-disagreement. It also checks the dyadic density modulus against finite masks, and a finite model of row interleaving and block normalization.

The machine-readable output is shipped as `data/08-effective-exact-pairs-verification.json`. The script is deterministic and records the scope of each check. Running it verifies the finite calculations, not the existence of the infinitary sets, the oracle-jump identities, or any global novelty assertion. Those statements are justified by the mathematical proofs above. No amount of finite enumeration of these tests would independently establish them.

[merge] Source 08 names these files `code/verify.py` and `data/verification.json` and describes an archive with its own source, PDF, `build.sh`, proof audit and source ledger. In this report its proof audit and source ledger are `08-effective-exact-pairs-PROOF_AUDIT.md` and `08-effective-exact-pairs-SOURCES.md`, its build script is `code/08-effective-exact-pairs-build.sh` (which still uses the delivery paths), its document-QA record is `data/08-effective-exact-pairs-document_qa.json`, and its manuscript and PDF are not shipped: this part is the manuscript. The recorded run passed 552,927 finite cases.

19 Source 08: further research questions

The following are proposed next problems for this programme. They are not blanket assertions that every formulation below is absent from the published literature; each should receive a dedicated literature check before a priority claim.

Question 19.1 (Which upward-closed spectra occur?). Characterize the sets of Turing degrees that occur as $\text{Spec}(X)$. Core realization determines their common lower bounds, but this is far from a spectrum classification. Even with a fixed core $\mathcal{I}$, Theorem 17.3 leaves an upper cone’s worth of different possibilities. Can the row-finite normal form be extended to capture all coarse spectra, or can one prove that some coarse spectra are not of this form?

Question 19.2 (Minimal representatives, not least representatives). For a nonprincipal $\mathcal{I}$ and the explicit classes F_H , when does $\text{Spec}(F_H)$ contain a minimal degree? Can one identify a presentation and parameter for which it has no minimal element? The present proof excludes a least element but neither establishes nor excludes minimality. The mass problem $\mathcal{F}_{\vec{A}}$ offers a concrete place to test minimal-upper-bound methods.

Question 19.3 (Optimal jump thresholds). The common-jump theorem works for $H \geq_T A'_{\oplus}$, whereas the unparameterized construction already gives pair jumps bounded by $\Theta_{\vec{A}} \leq_T A'_{\oplus}$. Determine the least hypotheses on H permitting an exact pair for $\mathcal{I}_{\vec{A}}$ in a prescribed coarse-description class with both jumps exactly H . Is there a presentation-invariant characterization, rather than one stated using a particular uniform join?

Question 19.4 (Sharper effective error control). The family has a common $\Theta_{\vec{A}}$ -computable null envelope and can be confined to a tail of arbitrarily small global prefix density. Can one simultaneously impose an arbitrary prescribed unbounded sublinear counting budget while retaining an arbitrary nonprincipal core and the same jump bound? The present tail argument does not do this: the sizes of individual finite block corrections are unrestricted.

Question 19.5 (When do two presentations give the same coarse degree?). Find useful necessary and sufficient conditions on generator sequences $\vec{A}$ and $\vec{E}$ for $Q(\vec{A}) \leq_{uc} Q(\vec{E})$ or $Q(\vec{A}) \leq_{nc} Q(\vec{E})$. Inclusion of generated ideals is necessary but, by the fiber theorem, not sufficient. A natural target is an effective transformation of finite-variant arrays that does not require finding their unknown finite corrections.

Question 19.6 (Structure internal to one core fiber). The fiber contains an order-embedded upper cone, but this alone does not identify which joins, meets, antichains, or intervals in the coarse-degree order remain inside the fiber. Determine whether the embeddings can preserve additional operations, or whether common information is necessarily created by some coarse joins.

Question 19.7 (Other reducibilities and ideals). Which parts of the construction survive for effective-dense or generic descriptions, for bounded-error variants, or with hyperarithmetic reducibility replacing Turing reducibility? The majority normalization is specific to permitted errors. A description notion that allows omissions but forbids errors requires a different normal form, and no answer to its least-representative question follows automatically from this paper.

Question 19.8 (Formal and foundational economy). Implement the finite-trace and row-mask argument in the repository and determine the weakest classical choice and comprehension principles needed for the realization and effective-family statements. The parameterized theorem for a given sequence avoids selecting an enumeration of an arbitrary ideal; separating these formulations may clarify the foundational strength.

[merge] Status of these questions in the merged report. Question 19.1 is Part I’s Question 11.1 and Part III’s Question 29.1. Question 19.5 is Part III’s Question 29.4 for another code. For Question 19.6, Part I’s Theorem 9.2

gives a *join-preserving* upper-cone embedding into every fiber, which by Remark 17.7 covers source 08's map as well, and Corollary 9.4 shows that no fiber has a maximal member or a countable cofinal family; meets, antichains, intervals and joins of arbitrary fiber members remain open. For hyperarithmetic reducibility in Question 19.7, report 10 of this programme shows that cone-avoiding compactness fails, so no transfer is automatic. The others remain open.

19.1 Source 08's conclusion

The core-realization problem has a clean answer: there is no additional restriction beyond being a countable Turing ideal. Positive-density columns protect individual generators, and robust finite blocks reduce coarse errors to independent finite errors in the rows. Classical exact-pair reasoning can then be performed inside this row-finite mass problem, with every search charged to a finite join and every decision to its jump.

The second conclusion is equally important for classification. The core records unavoidable individual information, but not the cost of assembling solutions uniformly. By interleaving constant rows carrying a separate parameter, one preserves the core while changing that assembly problem. This produces an upper-cone embedding in every fiber, exact spectrum constraints, and exact pairs with controlled common jumps. These are proved extensions of the repository programme; claims of historical priority or proof-assistant certification are separate questions and are not made here.

Part III

Source 10: layered realization, ideal-protected fusion, and quantitative recovery

20 Source 10: research target and principal results

[merge] Part III is source 10, *Every Countable Turing Ideal Is a Coarse Core: Layered realization, exact pairs, and order embeddings with fixed common information* (research manuscript prepared for Vladimir Reshetnikov, 28 September 2026). Its Sections 1 and 2 are this section, and its Section n is Section $n + 18$ here for $n = 3, \dots, 11$; numbered statements keep their places, so its Lemma $n.m$ is Lemma $(n + 18).m$ here, except that its Theorems 1.1–1.3 are Theorems 20.1–20.3. Its Section 12 (conclusion) is Section 29.1. Its Section 2 (Lemmas 2.1–2.2, Proposition 2.3), its Lemma 3.4, Theorem 3.5 and Lemma 3.6 (limit lemma, dyadic criterion, uniform dyadic transport) and its Appendix A (relativized Friedberg inversion) duplicate Part I with the same proofs and are printed there; its Appendix B (notation) is folded into Section 1.5. Symbols: source 10’s carrier $X_{\tilde{A}}$ is $\Lambda_{\tilde{A}}$ here, **not** Part I’s $X_{\tilde{A}}$; its $\mathcal{J}, I_m, h_m, \ell(m)$ are $\mathcal{J}^!, I_m^!, (m + 1)!, \lambda(m)$; its $c_n(k)$ is $p(n, k)$; its Z is $A_{\oplus}$; its dyadic parameters B, C are H, K and $F_{\tilde{A}, B}$ is $T_{\Lambda_{\tilde{A}}}(H)$; its $\Sigma(X)$ is $\text{Spec}(X)$; its truncations $T_s, U_s(B), V_s$ are $\Lambda_{\tilde{A}, < s}, U_s(H), \Lambda_{\tilde{A}, < s}^H$; and in its robust-recovery lemma its eventually correct sets A_s are $\tilde{A}_s$; its common function H in Lemma 23.2 is G .

Source 10’s abstract. For a set X , its coarse core consists of the sets computable from every density-zero perturbation of X . We prove that the possible coarse cores are exactly the countable Turing ideals. The construction places robust codes for a cofinal sequence of the ideal on positive-density columns whose tail densities tend to zero. A short proof uses the cone-avoiding compactness theorem of Hirschfeldt, Jockusch, Kuiper, and Schupp; a separate finite-extension proof constructs a perfect family of descriptions with pairwise common lower cone exactly the prescribed ideal, and does not use that compactness theorem.

We then separate common information from coarse computational strength. For each ideal $\mathcal{I}$, a carrier X and an oracle $Z \geq_T X$ are constructed such that

$$\text{Core}(X \oplus \mathcal{R}(H)) = \mathcal{I}, \quad \text{Spec}(X \oplus \mathcal{R}(H)) = \text{Spec}(X) \cap \{\mathbf{d} : H \leq_T D' \text{ for } D \in \mathbf{d}\}.$$

On the parameter cone $\deg_T(H) \geq \deg_T(Z')$, this construction is an upper-semilattice embedding for both uniform and nonuniform coarse reducibility. Consequently, each countable ideal is the common core of continuum many distinct coarse degrees having no least Turing-degree representative. We also give quantitative bounds on the information recoverable at a fixed positive error tolerance and an explicit demonstration that the carrier can depend on the presentation of the ideal. The paper resolves the core-realization question recorded in the inspected ProveIt synthesis, but does not classify all representative spectra or settle the existence of minimal elements in them.

Status. These are conventional mathematical proofs, not a Lean-certified development. The extensions below are distinguished from the repository’s existing results and from published background. Priority in the wider literature has not been established, and the manuscript has not been independently refereed.

[merge] In this abstract X is the carrier $\Lambda_{\tilde{A}}$ and Z is $A_{\oplus}$; the parameter is renamed H as throughout Part III.

20.1 The gap in the repository

The inspected snapshot of ProveIt/Computability/TuringDegrees is the commit

1a1396d4d3a2ac6812692df520517d86ba3a4785.

Its research synthesis [1], in the subsection “What is not settled”, explicitly leaves open which countable ideals occur as coarse cores and which upward-closed degree sets occur as representative spectra. The same synthesis already contains the negative answer to the basic least-representative question, principal-core examples with unattained infimum, dyadic jump-cone spectra, and several exact-pair constructions. Repeating those results would not answer the present research request.

This part answers the *first* classification question completely. It then proves that fixing the resulting invariant leaves substantial freedom: an entire Turing upper cone can be embedded in a single core fiber. Here a *core fiber* means the collection of coarse degrees whose core is one specified ideal; it is not a new reducibility. All objects constructed below are binary sets.

[merge] As in Part I, the classification question is also plan target C4, with a proof sketch [7], and the column mechanism has the antecedent [4, Lemma 3.10 and Theorem 3.11]; source 10 cites neither.

20.2 Statements at a glance

We identify a Turing ideal with its collection of set representatives. Thus a nonempty family $\mathcal{I} \subseteq 2^{\mathbb{N}}$ is a Turing ideal when it is downward closed under Turing reducibility and closed under finite joins. Write $\text{Low}(A) = \{C : C \leq_T A\}$.

Theorem 20.1 (Core realization). *A family of sets is the coarse core of some binary set if and only if it is a countable Turing ideal. More explicitly, if*

$$A_0 \leq_T A_1 \leq_T \cdots, \quad \mathcal{I} = \bigcup_{n \in \mathbb{N}} \text{Low}(A_n),$$

then the layered carrier $\Lambda_{\vec{A}}$ defined in Section 22 satisfies

$$\text{Core}(\Lambda_{\vec{A}}) = \mathcal{I}, \quad \Lambda_{\vec{A}} \leq_T A_{\oplus} := \bigoplus_n A_n.$$

No computable enumeration of the ideal is assumed.

Theorem 20.2 (Fixed-core embeddings). *For the carrier and $A_{\oplus}$ above, let $T_{\Lambda_{\vec{A}}}(H) = \Lambda_{\vec{A}} \oplus \mathcal{R}(H)$, where $\mathcal{R}$ is the dyadic code. For all H ,*

$$\text{Core}(T_{\Lambda_{\vec{A}}}(H)) = \mathcal{I}, \quad \text{Spec}(T_{\Lambda_{\vec{A}}}(H)) = \text{Spec}(\Lambda_{\vec{A}}) \cap \{\mathbf{d} : H \leq_T D', D \in \mathbf{d}\}.$$

For $H, K \geq_T A'_{\oplus}$,

$$H \leq_T K \iff T_{\Lambda_{\vec{A}}}(H) \leq_{\text{nc}} T_{\Lambda_{\vec{A}}}(K) \iff T_{\Lambda_{\vec{A}}}(H) \leq_{\text{uc}} T_{\Lambda_{\vec{A}}}(K). \quad (20.1)$$

The induced maps on degrees preserve finite joins. The canonical representative $T_{\Lambda_{\vec{A}}}(H)$ has Turing degree $\deg_T(H)$ on this parameter cone.

Theorem 20.3 (Perfect exact-pair realization). *For every $\vec{A}, H$, every $\varepsilon > 0$, and every $L \in \mathbb{N}$, there is a perfect set $\mathcal{P} \subseteq \text{CD}(T_{\Lambda_{\vec{A}}}(H))$ such that every $D \in \mathcal{P}$ extends $T_{\Lambda_{\vec{A}}}(H) \upharpoonright L$, satisfies $d(D, T_{\Lambda_{\vec{A}}}(H)) < \varepsilon$, and, for distinct $D, E \in \mathcal{P}$,*

$$\text{Low}(D) \cap \text{Low}(E) = \mathcal{I}.$$

A binary splitting system parametrizing $\mathcal{P}$ is computable in $(A_{\oplus} \oplus H)'$. In particular, there are two such distinct descriptions below $(A_{\oplus} \oplus H)'$. The whole perfect set is not asserted to consist of sets below that oracle.

The proofs of Theorems 20.1–20.3 appear in Sections 22–25. Section 26 proves the continuum-many/no-least consequence, and Section 27 controls robustness quantitatively.

[merge] Theorem 20.2 is the case $X = \Lambda_{\bar{A}}$ of Theorems 8.4 and 9.2 of Part I, on the subcone above $A'_{\oplus}$ of the cone above Λ'_A ; source 10’s proof of its core clause is a second route (Theorem 24.1).

20.3 What is inherited and what is extended

The published foundations are asymptotic-density computability [3], robust reducibilities [5], and coarse coarsenings and cone-avoiding compactness [2]. The dyadic description criterion is the relativized form of the Jockusch–Schupp criterion, also developed explicitly in the repository. Friedberg jump inversion [11] supplies the order-reflection argument; Lemma 9.1 gives the precise relativized proof used here (source 10 gives the same proof in its Appendix A).

The new work relative to the inspected synthesis is the unrestricted ideal-realization theorem, the varying-ideal version of reservoir fusion, and the fixed-core upper-cone embeddings. The fusion method itself is adapted from the synthesis’s fixed-oracle reservoir argument: a common computation is now forced below *some finite level of a cofinal ideal presentation*, rather than below one fixed oracle. We give the argument in full instead of treating the synthesis’s unrefereed witness theorems as black boxes.

Gerdes’s recent preprint [6] places coarse and effective-dense reducibilities in a broader setting. Our theorem is not a solution of the corresponding effective-dense least-representative problem. Likewise, the hyperdegree investigations in report 10 [9] are separate: none of the proofs below silently replaces Turing reducibility by hyperarithmetic reducibility.

[merge] “Report 10” is research report 10 of this programme, *Coarse Classes, Hyperdegrees, and the Cone Filter*, not source 10 (batch-37 manuscript 10), which is this part.

20.4 Notation and coarse-information bookkeeping

Source 10 uses the density notation and reducibilities of Section 2 and the metric d of Section 3: $\rho_m, \bar{\rho} = \bar{\rho}, \delta(X, Y) = \bar{\rho}(X \triangle Y), d(X, Y) = \sup_{m \geq 1} \rho_m(X \triangle Y), X \approx Y, \text{CD}(X), \text{Core}(X), \leq_{\text{nc}}, \leq_{\text{uc}}, \equiv_{\text{nc}}, \equiv_{\text{uc}}$. Its oracle spectrum $\Sigma(X)$, the set of degrees of oracles computing a member of $\text{CD}(X)$, is $\text{Spec}(X)$ by Proposition 2.3. Its Lemma 2.1 is Proposition 2.3 with (2.1) and the lower-cone identity of Lemma 2.2; its Lemma 2.2 is Proposition 2.4 with core monotonicity; its Proposition 2.3 is the first assertion of Lemma 2.2. Two conventions are its own:

A density-zero change is allowed to be noncomputable, and no convergence modulus for its density is supplied. This omission of an effective error bound is essential.

A Turing functional Φ_e^D may be partial. A computation $\Phi_e^\sigma(n) \downarrow$ with a finite oracle string means that it halts without querying any oracle bit outside that string. Such convergence persists under extension. All procedures claimed to output a coarse description below are total on their stipulated inputs.

21 Two codes and two kinds of recovery

21.1 Positive-density columns

For $n, k \in \mathbb{N}$, let

$$p(n, k) = 2^n(2k + 1) - 1, \quad C_n = \{p(n, k) : k \in \mathbb{N}\}. \quad (21.1)$$

These are a computable partition of $\mathbb{N}$, with

$$\rho(C_n) = 2^{-n-1}, \quad \left| \bigcup_{n \geq s} C_n \cap [0, m] \right| = \left\lfloor \frac{m}{2^s} \right\rfloor. \quad (21.2)$$

The tail identity follows because k lies in the union exactly when 2^s divides $k + 1$.

For a set D , denote its pullback along the n th column by $D_n(k) = D(p(n, k))$.

Lemma 21.1 (Restriction and finite heads). *For all X, D, n ,*

$$\delta(X_n, D_n) \leq 2^{n+1} \delta(X, D). \quad (21.3)$$

In particular, a coarse description restricts to a coarse description on each column. Conversely, if each $(X \triangle D) \cap C_n$ is finite, then $D \approx X$.

Proof. The first t elements of C_n lie below $p(n, t - 1) + 1$, and the ratio of that endpoint to t tends to 2^{n+1} . Bounding their errors by all errors below that endpoint and taking a limsup proves (21.3). For the converse, fix s . Errors on the first s columns form a finite set, and all remaining errors lie in a tail of upper density 2^{-s} . Thus $\delta(X, D) \leq 2^{-s}$ for every s . $\square$

21.2 A robust repeated-block code

Let $I_m^! = [(m + 1)!, (m + 2)!)$ and let $\lambda(m) = v_2(m + 1)$. The intervals partition the positive integers. Define

$$\mathcal{J}^!(A)(0) = 0, \quad \mathcal{J}^!(A)(k) = A(\lambda(m)) \quad (k \in I_m^!). \quad (21.4)$$

This is a factorial-block coarsening with a computable repetition schedule, a variant of the constructions in [2, Section 2]. The repetition will distinguish recovery of an entire set from pointwise convergence of an approximation.

[merge] $\mathcal{J}^!$ is neither Part I's $\mathcal{J}$ (blocks $[2^k, 2^{k+1})$, each label once) nor source 08's $\mathcal{J}^{\text{sh}}$: its blocks are factorial and every label k recurs on the infinitely many blocks $I_m^!$ with $v_2(m + 1) = k$, which is what makes the recovery radius $1/2$ below.

Lemma 21.2 (Robust recovery). *The code $\mathcal{J}^!$ is computable from its argument. If*

$$\delta(D, \mathcal{J}^!(A)) < \frac{1}{2},$$

then $A \leq_{\text{T}} D$. In particular, every coarse description of $\mathcal{J}^!(A)$ computes A . There is also a single procedure which, from such a D , produces total sets $\tilde{A}_s$ with $\tilde{A}_s = A$ for every sufficiently large s .

Proof. Let $v_D(m)$ be the majority bit of D on $I_m^!$, breaking ties by choosing 0. If this bit is wrong, at least half the block consists of errors, and therefore

$$\rho_{(m+2)!}(D \triangle \mathcal{J}^!(A)) \geq \frac{(m+2)! - (m+1)!}{2(m+2)!} = \frac{m+1}{2(m+2)}.$$

The right side tends to $1/2$. Under the strict error hypothesis, only finitely many blocks can have a wrong majority. Fix M beyond them. Given k , find an $m \geq M$ with $\lambda(m) = k$ and output $v_D(m)$. Every label occurs infinitely often, so this computes A . The integer M is a finite parameter in the reduction; it is not claimed to be recoverable uniformly from D .

For the last assertion, let $m(k, s)$ be the least $m \geq s$ with label k and put $\tilde{A}_s(k) = v_D(m(k, s))$. This is total and uniform in D, k, s . For all $s \geq M$, it equals A on *every* input simultaneously. $\square$

Remark 21.3 (What uniformity does not say). The lemma does not give one functional that outputs A exactly from every description of $\mathcal{J}^1(A)$. The uniformly available object is an eventually correct sequence of whole sets. The exact computation of A may use a different finite stabilization parameter for each description. This is sufficient for membership in a Turing core.

21.3 The dyadic code and the jump

The dyadic code $\mathcal{R}(H)(p(n, k)) = H(n)$ has a different recovery property. It repeats one bit on each positive-density column. Unlike $\mathcal{J}^1$, it gives pointwise eventual information, with no single stage at which all decoded bits must be correct.

Source 10 then proves the relativized limit lemma (its Lemma 3.4), the dyadic description criterion “ D computes a coarse description of $\mathcal{R}(H)$ iff $H \leq_T D$ ” with $\text{Spec}(\mathcal{R}(H)) = \mathcal{Q}_H$ (its Theorem 3.5), and uniform dyadic monotonicity and joins, $H \leq_T K \Rightarrow \mathcal{R}(H) \leq_{\text{uc}} \mathcal{R}(K)$ and $\mathcal{R}(H \oplus K) \equiv_{\text{uc}} \mathcal{R}(H) \oplus \mathcal{R}(K)$ (its Lemma 3.6). These are Lemma 7.1, Theorem 7.2 and Proposition 7.3 of Part I, with the same proofs. Source 10 notes that the unrelativized criterion is due to Jockusch and Schupp [3, Theorem 2.19], that the relativized spectrum formulation is already in the ProveIt synthesis, and that the proofs are included to expose the exact uniformity needed next: in the monotonicity proof each oracle query requires only a finite majority computation, and every procedure is total and fixed independently of the supplied description.

22 Realizing every countable ideal

22.1 The approximation principle

We first record the published compactness theorem in the exact form used.

Theorem 22.1 (Cone-avoiding compactness, HJKS). *Let A, X be sets. If for every $\varepsilon > 0$ there is a set Y such that $\delta(X, Y) \leq \varepsilon$ and $A \not\leq_T Y$, then some $D \in \text{CD}(X)$ satisfies $A \not\leq_T D$.*

This is [2, Theorem 3.7]. We use it as published background, not as a new result. Its contrapositive gives

$$A \in \text{Core}(X) \iff (\exists \eta > 0)(\forall Y) [\delta(X, Y) < \eta \implies A \leq_T Y]. \quad (22.1)$$

The switch between strict and non-strict error tolerances is harmless after shrinking a positive radius. The reverse implication follows simply because every description has error zero.

[merge] Second route (source 10) to Theorem 3.1: (22.1) is (3.1), which Part I proves from Baire category.

Lemma 22.2 (Ideal approximation). *Suppose $\mathcal{I}$ is a Turing ideal and, for every $\varepsilon > 0$, there is a set $T \in \mathcal{I}$ with $\delta(X, T) < \varepsilon$. Then $\text{Core}(X) \subseteq \mathcal{I}$. If in addition $\mathcal{I} \subseteq \text{Core}(X)$, equality holds.*

Proof. If $A \notin \mathcal{I}$, downward closure implies $A \not\leq_T T$ for every $T \in \mathcal{I}$. The approximants therefore satisfy the hypothesis of Theorem 22.1. Thus $A \notin \text{Core}(X)$. Equivalently, one can apply (22.1) to a core member and choose a sufficiently accurate approximant in the ideal. $\square$

This generalizes the familiar argument that arbitrary accuracy by computable approximants forces a trivial core. Countability is not required in the lemma, though it is necessary for an ideal to equal a core.

22.2 The layered carrier

Let $\mathcal{I}$ be a countable Turing ideal. Enumerating its members set-theoretically and taking finite joins produces a cofinal sequence $\vec{A} = (A_n)$ with

$$A_n \leq_T A_{n+1}, \quad \mathcal{I} = \bigcup_n \text{Low}(A_n). \quad (22.2)$$

No effectiveness is built into this enumeration. Define

$$\Lambda_{\vec{A}}(p(n, k)) = \mathcal{J}^1(A_n)(k). \quad (22.3)$$

A standard computable pairing is used for the presentation oracle $A_\oplus = \bigoplus_n A_n$, so the carrier is uniformly computable from $A_\oplus$.

[merge] The increasing-chain hypothesis (22.2) is used below only to place each truncation in $\text{Low}(A_{s-1})$ and each protected pattern in $\text{Low}(A_s)$. For an arbitrary sequence one may pass to $\hat{A}_n = A_{<n+1}$, which is increasing and generates the same ideal; Part I works with arbitrary sequences directly.

Proof of Theorem 20.1. Necessity is Lemma 2.2. For sufficiency, let $D \in \text{CD}(\Lambda_{\vec{A}})$. Its n th column describes $\mathcal{J}^1(A_n)$, so $A_n \leq_T D$ by Lemmas 21.1 and 21.2. It follows that $\mathcal{I} \subseteq \text{Core}(\Lambda_{\vec{A}})$.

For $s \geq 1$, define the truncation

$$\Lambda_{\vec{A}, <s}(p(n, k)) = \begin{cases} \mathcal{J}^1(A_n)(k), & n < s, \\ 0, & n \geq s. \end{cases} \quad (22.4)$$

It is computable from the finite join of $A_0, \dots, A_{s-1}$, hence from A_{s-1} , and belongs to $\mathcal{I}$. By (21.2),

$$d(\Lambda_{\vec{A}}, \Lambda_{\vec{A}, <s}) \leq 2^{-s}. \quad (22.5)$$

Lemma 22.2 now gives $\text{Core}(\Lambda_{\vec{A}}) \subseteq \mathcal{I}$. The computation from $A_\oplus$ was explicit in (22.3). $\square$

Remark 22.3 (Finite versus infinite uniformity). For each fixed s , finitely many reduction indices suffice to compute $\Lambda_{\vec{A}, <s}$ from A_{s-1} . No algorithm is asserted to obtain those indices from s . Likewise, a description computes each A_n , but this does not imply that it computes the infinite join $A_\oplus$: the reductions may depend on n noncomputably. Confusing these two assertions would destroy the distinction between a nonprincipal ideal and a principal lower cone.

Section 23 gives a second proof of the same core equality, by constructing two descriptions whose common lower cone is exactly $\mathcal{I}$. That proof does not appeal to cone-avoiding compactness.

23 An independent perfect exact-pair construction

23.1 Protected regions and reservoirs

The following lemma permits the finite-information oracle to rise through an ideal. Its assumptions are chosen to make all finite compatibility checks explicit.

Definition 23.1 (An ideal-protected presentation). Let $\vec{A}$ satisfy (22.2). A set X has an ideal-protected presentation if there are uniformly computable increasing sets P_s and positive numbers $\epsilon_s \rightarrow 0$ such that:

- (i) every $\mathbb{N} \setminus P_s$ is infinite, and $\rho_m(\mathbb{N} \setminus P_s) \leq \epsilon_s$ for all $m \geq 1$;
- (ii) the total function which is X on P_s and zero elsewhere is computable in A_s , for each fixed s ;
- (iii) $I \subseteq \text{Core}(X)$.

An oracle W with $X \leq_T W$ will bound the construction, but is not used as the bound on common information.

For a finite string σ , set

$$\text{Res}(\sigma, s) = \{Y \geq \sigma : (\forall k \geq |\sigma|) [k \in P_s \implies Y(k) = X(k)]\}.$$

A finite extension $\tau \geq \sigma$ is admissible if it obeys the same restriction below $|\tau|$. Admissibility is decidable in A_s : only finitely many protected values are queried. It is also decidable in W .

Lemma 23.2 (Cross-disagreement dichotomy). Fix s, σ, τ, e, j . Either there are admissible extensions $\alpha \geq \sigma, \beta \geq \tau$, and an input n for which

$$\Phi_e^\alpha(n) \downarrow \neq \Phi_j^\beta(n) \downarrow, \tag{23.1}$$

or every common total function of the form

$$G = \Phi_e^D = \Phi_j^E, \quad D \in \text{Res}(\sigma, s), \quad E \in \text{Res}(\tau, s),$$

is computable in A_s . The existence of witnesses to (23.1) is a $\Sigma_1^0(W)$ question.

Proof. Suppose there are no witnesses and a common total G exists. With oracle A_s , enumerate all admissible α and dovetail $\Phi_e^\alpha(n)$. Return the first value obtained. Some prefix of D supplies a convergent computation, so this search terminates. A sufficiently long prefix of E supplies the value $G(n)$ on the other side. An incorrect returned value would therefore give (23.1), a contradiction. The finite strings, s , and the finitely many indices needed to compute protected values are constants in this reduction. This proves $G \leq_T A_s$, not necessarily uniformly in those constants.

Using W instead, admissibility is decidable and convergence on finite strings is effectively enumerable. Thus existence of the displayed finite witnesses is c.e. in W and decidable in W' . $\square$

[merge] Source 10 writes H for the common function; it is G here because H is the dyadic parameter.

Remark 23.3 (Why the product is legitimate). The two extensions in (23.1) are chosen independently. There is no extra requirement that their disagreement set already have density zero, or satisfy a joint budget. Each eventual path will approximate X because the protected regions increase. This separation of the compatibility condition from the limiting density condition is what makes the no-disagreement search valid.

23.2 Fusion with a rising ideal bound

Theorem 23.4 (Ideal-protected fusion). *Suppose X has an ideal-protected presentation and $X \leq_T W$. Fix a starting level t and a length L . There is a binary splitting system $(\sigma_u)_{u \in 2^{<\mathbb{N}}}$, computable in W' , whose path map*

$$D_Q = \bigcup_k \sigma_Q \upharpoonright k \quad (Q \in 2^{\mathbb{N}})$$

is an injection into $\text{CD}(X)$. Its image is perfect, every path extends $X \upharpoonright L$, and distinct paths satisfy

$$\text{Low}(D_Q) \cap \text{Low}(D_R) = \mathcal{I}. \quad (23.2)$$

The system can be arranged so that $D_Q \in \text{Res}(\sigma_Q \upharpoonright k, t + k)$ and

$$d(D_Q, X) \leq \epsilon_t, \quad Q \upharpoonright k = R \upharpoonright k \implies d(D_Q, D_R) \leq \epsilon_{t+k}. \quad (23.3)$$

In particular, $\text{Core}(X) = \mathcal{I}$.

Proof. Start with the root string $X \upharpoonright L$ at protection level t . At level k , the current leaves are indexed by $u \in 2^k$, each with reservoir parameter $s = t + k$. Process every ordered pair of distinct leaves and every pair of functional indices $e, j \leq k$. There are only finitely many tasks at that level.

For a task, ask W' whether (23.1) has witnesses in the two current reservoirs. If it does, find such witnesses using W and replace the two leaves by those extensions. If it does not, leave them unchanged and retain the no-disagreement conclusion of Lemma 23.2. All subsequent reservoirs for these leaves are subsets of the current ones, so a forced disagreement persists, and so does the relevance of a no-disagreement conclusion. Tasks already processed cannot be injured.

After the finite task list, pad every leaf to a common larger length using the values of X at newly appended positions. For each leaf choose a fresh position outside P_s , beyond this length, and form two extensions which agree with X at intervening positions and assign different bits at that position. Such a position exists because the complement of P_s is infinite. These siblings are incompatible. Pad further if needed so that the next level has a common length strictly larger than the preceding one. Regard the children as conditions with protection level $s + 1$. Their reservoirs are subsets of their parents' reservoirs: on previously protected positions the finite extensions were admissible, and beyond the new prefixes $P_s \subseteq P_{s+1}$. New errors on positions newly protected at level $s + 1$ are confined to the finite prefix and are permitted.

This describes a W' -computable construction. More formally, intermediate leaves can be extended while tasks are processed, and the strings for a level are finalized before its children are built. Descendants always extend the finalized strings. The strictly increasing common lengths ensure total paths; sibling incompatibility ensures that the map $Q \mapsto D_Q$ is injective.

Density. For any fixed level k , let M_k bound the lengths of its finalized strings. Every path agrees with X on P_{t+k} beyond M_k . Thus, for every m ,

$$\rho_m(D_Q \triangle X) \leq \frac{M_k}{m} + \epsilon_{t+k}.$$

Taking a limsup and then letting $k \rightarrow \infty$ gives $D_Q \approx X$. Since all paths remain in the root reservoir and its prefix agrees with X , their errors are contained in $\mathbb{N} \setminus P_t$, proving the first part of (23.3). If two paths share a level- k node, they agree below its prefix length and both agree with X on P_{t+k} thereafter. Their disagreement is contained in $\mathbb{N} \setminus P_{t+k}$, proving the second part.

Exactness. Let $Q \neq R$, and let a set G be computed from both paths by functionals e, j . Choose k sufficiently large that $e, j \leq k$ and the paths have distinct level- k nodes. Their task was processed at that level. A forced cross-disagreement would contradict equality of the two total outputs. Hence the no-disagreement alternative held, and Lemma 23.2 gives $G \leq_T A_{t+k}$, so $G \in \mathcal{I}$. Conversely, every path is a description of X , and assumption (iii) implies that every member of $\mathcal{I}$ is below every path. This proves (23.2).

Perfectness and core. Equation (23.3) gives continuity into the prefix-density metric. A continuous injection from compact Cantor space into a metric space is a homeomorphism onto its image, so the image is compact and has no isolated points. It is also perfect in the usual Cantor topology, by the same injection and the growing prefix lengths. Finally, a core member is computed from any two distinct paths; (23.2) puts it in $\mathcal{I}$. The other inclusion was an assumption. $\square$

23.3 Applying fusion to the carrier

For $X = \Lambda_{\vec{A}}$, take

$$P_s = \bigcup_{n < s} C_n, \quad \epsilon_s = 2^{-s}, \quad W = A_{\oplus}.$$

All hypotheses hold. The protected values are computable from A_s , the complement is an infinite arithmetic progression, and the lower core inclusion was proved using robust recovery only. Therefore Theorem 23.4 independently proves $\text{Core}(\Lambda_{\vec{A}}) = \mathcal{I}$ and gives an $A'_{\oplus}$ -computable splitting system of pairwise exact descriptions.

This establishes that the ideal-realization theorem does not depend on any admitted theorem in the repository or on our using Theorem 22.1 as a black box. The compactness proof is shorter and yields the useful general approximation principle; the fusion proof supplies additional witnesses and a presentation-relative complexity bound.

[merge] This is route (d) of Section 1.4. Source 08's Theorem 12.2 is the analogous construction for its code, with a lower bound: its splitting tree is computable in $\Theta_{\vec{A}} \leq_T A'_{\oplus}$ and the jumps of its computable-parameter paths are below $\Theta_{\vec{A}}$. The two constructions differ in their conditions: source 08 fixes complete rows of an array, source 10 fixes finite strings inside protected reservoirs. Report 13 of this programme (batch-37 manuscript 09) proves a layered-reservoir theorem under closely related hypotheses.

24 Varying coarse strength without changing the core

24.1 Core-preserving insertion

For an arbitrary parameter set H , define

$$T_{\Lambda_{\vec{A}}}(H) = \Lambda_{\vec{A}} \oplus \mathcal{R}(H). \quad (24.1)$$

The intended roles of its two halves are different. The carrier protects the information in $\mathcal{I}$. The dyadic half constrains the jump of an oracle computing a description, while being approximable arbitrarily well by individually computable sets.

Theorem 24.1 (Core-preserving spectral cuts). *For every H ,*

$$\text{Core}(T_{\Lambda_{\vec{A}}}(H)) = \mathcal{I}, \quad (24.2)$$

$$\text{Spec}(T_{\Lambda_{\vec{A}}}(H)) = \text{Spec}(\Lambda_{\vec{A}}) \cap \{\mathbf{d} : H \leq_T D', D \in \mathbf{d}\}. \quad (24.3)$$

Proof. Every description of $T_{\Lambda_{\vec{A}}}(H)$ projects to a description of $\Lambda_{\vec{A}}$, so $\mathcal{I} \subseteq \text{Core}(T_{\Lambda_{\vec{A}}}(H))$.

Let $U_s(H)$ agree with $\mathcal{R}(H)$ on columns C_n , $n < s$, and be zero on the others. For each fixed s , this is a computable set: only the finitely many bits $H(0), \dots, H(s-1)$ are needed, and they can be built into its program. A uniform computable map from s to such programs is not being asserted. Its error satisfies $d(U_s(H), \mathcal{R}(H)) \leq 2^{-s}$. The set

$$\Lambda_{\vec{A}, < s}^H = \Lambda_{\vec{A}, < s} \oplus U_s(H)$$

belongs to $\mathcal{I}$ and satisfies $d(\Lambda_{\vec{A}, < s}^H, T_{\Lambda_{\vec{A}}}(H)) \leq 2^{-s}$. To see the last inequality also for odd initial-segment lengths, apply the bound separately to the first $\lceil m/2 \rceil$ and $\lfloor m/2 \rfloor$ positions in the two halves; their lengths sum to m . Lemma 22.2 proves (24.2).

Equation (24.3) is Proposition 2.4 together with Theorem 7.2. $\square$

[merge] Second route (source 10) to the case $X = \Lambda_{\vec{A}}$ of Theorem 8.4. Part I derives (24.2) from core neutrality (Theorem 8.2) for an arbitrary base set; this proof uses approximants lying in the ideal itself.

Remark 24.2 (No general additivity of cores is assumed). The proof did not infer the core of a join from the cores of its halves. Such an inference would discard possible joint information. Here the exclusion of extra common information comes from explicit approximants lying in the same ideal. The exact-pair proof below gives a second exclusion argument.

24.2 Exact pairs for every parameter

Define the protected regions in the joined space by

$$P_s^* = \{2k : k \in P_s\} \cup \{2k+1 : k \in P_s\}, \quad P_s = \bigcup_{n < s} C_n.$$

They are uniformly computable and increasing, with infinite complements. For every $m \geq 1$,

$$\begin{aligned} |(\mathbb{N} \setminus P_s^*) \cap [0, m]| &= \left\lfloor \frac{\lceil m/2 \rceil}{2^s} \right\rfloor + \left\lfloor \frac{\lfloor m/2 \rfloor}{2^s} \right\rfloor \\ &\leq 2^{-s} m. \end{aligned}$$

The values of $T_{\Lambda_{\vec{A}}}(H)$ on P_s^* are computable from A_s and the finitely many bits $H(0), \dots, H(s-1)$. Hence they are computable in A_s for each fixed s . Those finitely many bits do not enlarge its Turing degree. The whole target is computable in $W = A_{\oplus} \oplus H$.

Proof of Theorem 20.3. Apply Theorem 23.4 to these protected regions and choose t with $2^{-t} < \varepsilon$. The lower inclusion $\mathcal{I} \subseteq \text{Core}(T_{\Lambda_{\vec{A}}}(H))$ uses only projection and robust recovery, so this is also an independent proof of (24.2). The construction gives all the asserted properties, including the $(A_{\oplus} \oplus H)'$ bound on its splitting system. Taking two distinct computable paths through the binary index tree yields two descriptions below that oracle. $\square$

A useful distinction follows. The location of a perfect family can be controlled by one jump of a presentation oracle, but its uncountably many members cannot all be computable in that oracle: each oracle computes only countably many sets. The correct uniform bound is

$$D_Q \leq_T (A_{\oplus} \oplus H)' \oplus Q. \quad (24.4)$$

This is exactly what the splitting-system construction supplies.

25 Upper-semilattice embeddings in a fixed core fiber

25.1 The order-reflection argument

The relevant form of Friedberg jump inversion is

$$K \geq_T A'_\oplus \implies (\exists D \geq_T A_\oplus) D' \equiv_T K. \quad (25.1)$$

This is a classical relativized theorem, not a new construction claimed here. It is Lemma 9.1 with $X = A_\oplus$; source 10 proves it in its Appendix A by the same argument.

Theorem 25.1 (Order and join preservation). *Fix $\vec{A}$ and $A_\oplus = \bigoplus_n A_n$. For each $r \in \{\text{nc}, \text{uc}\}$, the assignment*

$$\deg_T(H) \mapsto [T_{\Lambda_{\vec{A}}}(H)]_r, \quad \deg_T(H) \geq \deg_T(A'_\oplus),$$

is a well-defined upper-semilattice embedding. Its image lies above $[\Lambda_{\vec{A}}]_r$ and entirely inside the fiber with core $\mathcal{I}$.

Proof (as merged). This is Theorem 9.2 for $X = \Lambda_{\vec{A}}$, restricted to parameters above $A'_\oplus$; since $\Lambda_{\vec{A}} \leq_T A_\oplus$, these parameters lie above $\Lambda'_{\vec{A}}$. Source 10's proof is the same argument: order preservation and well-definedness by the uniform dyadic transformations of Proposition 7.3 (keep the carrier half, convert the other half, interleave); order reflection by (25.1) (a $D \geq_T A_\oplus$ with $D' \equiv_T K$ computes $\Lambda_{\vec{A}}$ exactly and a description of $\mathcal{R}(K)$, hence a description of $T_{\Lambda_{\vec{A}}}(H)$, whence $H \leq_T D' \equiv_T K$); and joins by

$$T_{\Lambda_{\vec{A}}}(H) \oplus T_{\Lambda_{\vec{A}}}(K) \equiv_{\text{uc}} \Lambda_{\vec{A}} \oplus \mathcal{R}(H) \oplus \mathcal{R}(K) \equiv_{\text{uc}} T_{\Lambda_{\vec{A}}}(H \oplus K). \quad (25.2)$$

The reassociations here are implemented by projections and interleavings of finitely many components, each of positive density. Thus they preserve density-zero errors; no invariance under arbitrary computable permutations is invoked. Projection places each image above the carrier, and Theorem 24.1 fixes its core. $\square$

Completion of Theorem 20.2. The core and spectrum formulas are Theorem 24.1, and the order equivalences and join preservation are Theorem 25.1. If $H \geq_T A'_\oplus$, then $T_{\Lambda_{\vec{A}}}(H) \leq_T A_\oplus \oplus H \equiv_T H$. Conversely, reading the exact dyadic half computes H , so $H \leq_T T_{\Lambda_{\vec{A}}}(H)$. Thus $T_{\Lambda_{\vec{A}}}(H) \equiv_T H$ on the stated cone. $\square$

25.2 Why an upper cone, rather than all degrees?

The threshold $A'_\oplus$ is used in a specific place: order reflection needs a jump inverse of K which also computes the carrier. Unrelativized inversion only supplies a jump inverse somewhere, and it need not compute $\Lambda_{\vec{A}}$. The base-oracle clause in (25.1) closes this gap.

Accordingly, the theorem does not assert an embedding of the entire Turing-degree structure into every fiber. Nor does it assert that the parameter-to-class map is injective below $A'_\oplus$. For example, when the carrier is computable, all parameters $H \leq_T \emptyset'$ have coarsely computable dyadic codes by Theorem 7.2 and collapse to the least coarse degree.

It is also unnecessary, and generally invalid, to distribute the Turing jump over joins. The join argument uses only the elementary equivalence

$$H \oplus K \leq_T D' \iff H \leq_T D' \text{ and } K \leq_T D',$$

and the explicit uniform code transformations of Proposition 7.3.

[merge] Part I's Theorem 9.2 lowers the threshold for this carrier from $A'_\oplus$ to Λ'_A (still an upper cone): the jump inverse only needs to compute Λ_A , not the presentation. Whether it can be lowered further is Question 29.3.

26 Consequences for representative spectra

26.1 Least representatives and principal cores

Lemma 26.1. *If a coarse-equivalence class has a representative of least Turing degree $\mathbf{a}$, its core is the principal ideal $\downarrow \mathbf{a}$. Conversely, a principal core does not alone imply that its generator occurs in the representative spectrum.*

Proof. By Proposition 2.3, the representative spectrum is $\text{Spec}(X)$. If $\mathbf{a}$ is its least element, its common lower cone is exactly $\downarrow \mathbf{a}$, which equals $\text{Core}(X)$ by Lemma 2.2. The final assertion will follow from the examples in the next theorem, and also appears in the repository synthesis. $\square$

Theorem 26.2 (Continuum many classes with the same core and no least degree). *Every countable Turing ideal $\mathcal{I}$ is the core of continuum many distinct nonuniform coarse degrees, also distinct as uniform coarse degrees, whose representative spectra have no least element.*

Proof. Use the family $T_{\Lambda_A}(H)$ with $\deg_T(H) > \deg_T(A'_\oplus)$. If $\mathcal{I}$ is nonprincipal, Lemma 26.1 rules out a least representative.

If $\mathcal{I} = \text{Low}(A)$ is principal, its generator satisfies $A \leq_T A_\oplus$: membership in the union (22.2) places it below some A_n . A least representative would have degree $\deg_T(A)$. But (24.3) would then imply $H \leq_T A' \leq_T A'_\oplus$, contradicting the parameter choice. Thus no member of the family has a least representative in either case.

There are continuum many degrees strictly above any fixed degree. For completeness, the reals $A'_\oplus \oplus Q$ range over continuum many distinct sets as Q varies; each Turing degree contains only countably many sets, so their degrees have cardinality continuum in ZFC. Removing the single degree of $A'_\oplus$ leaves continuum many. Theorem 25.1 makes their images distinct already in the nonuniform coarse degrees. $\square$

Corollary 26.3 (Perfect antichains of descriptions). *For the parameters in Theorem 26.2, the perfect families supplied by Theorem 20.3 have pairwise incomparable Turing degrees. In particular, every countable ideal is the exact common lower cone of a perfect family of pairwise incomparable, density-equal sets.*

Proof. No degree of a description lies in $\mathcal{I}$. Indeed, such a degree would, because every core member is below that description, be a greatest member of $\mathcal{I}$ and a least element of the spectrum. If D, E from the perfect family were comparable, say $D \leq_T E$, exactness would put D in $\text{Low}(D) \cap \text{Low}(E) = \mathcal{I}$, a contradiction. $\square$

26.2 What these statements do not imply

A *least* element is below every element of the spectrum. A *minimal* element merely has no strictly smaller element within the spectrum. Theorem 26.2 proves absence of the former, not the latter.

Even a perfect antichain of descriptions does not by itself rule out other minimal elements of the same spectrum.

Likewise, (24.3) is an exact formula for the constructed family relative to $\text{Spec}(\Lambda_{\vec{A}})$. It is not a characterization of every upward-closed subset of degrees that can be a coarse spectrum. The carrier spectrum is not silently replaced by the set of all upper bounds of $\mathcal{I}$. Every description degree is such an upper bound, but an arbitrary oracle computing each A_n separately need not uniformly assemble a description of (22.3).

27 Quantitative recovery and presentation sensitivity

27.1 A filtration by error tolerance

For $r > 0$, define

$$\mathcal{K}_r(X) = \bigcap_{\delta(X,Y) < r} \text{Low}(Y). \quad (27.1)$$

This is the information which survives every error pattern of upper density less than r , with no uniformity demanded of the recovery programs. It is a countable Turing ideal, since X itself occurs among the oracles being intersected. The ideals grow as r decreases. Equation (22.1) says

$$\text{Core}(X) = \bigcup_{r>0} \mathcal{K}_r(X).$$

For our carriers, their relation to the presentation levels is quantitative.

Theorem 27.1 (Fourfold localization of the ideal filtration). *For every $n \in \mathbb{N}$,*

$$\mathcal{K}_{2^{-n}}(\Lambda_{\vec{A}}) \subseteq \text{Low}(A_n) \subseteq \mathcal{K}_{2^{-n-2}}(\Lambda_{\vec{A}}). \quad (27.2)$$

Proof. If $\delta(Y, \Lambda_{\vec{A}}) < 2^{-n-2}$, the n th column has error less than

$$2^{n+1} 2^{-n-2} = \frac{1}{2}$$

relative to $\mathcal{J}^!(A_n)$, by (21.3). Robust recovery yields $A_n \leq_T Y$. This proves the right inclusion.

For the left inclusion, the truncation $\Lambda_{\vec{A}, < n+1}$ is computable in A_n and satisfies

$$\delta(\Lambda_{\vec{A}, < n+1}, \Lambda_{\vec{A}}) \leq 2^{-n-1} < 2^{-n}.$$

A set in $\mathcal{K}_{2^{-n}}(\Lambda_{\vec{A}})$ is therefore computable in $\Lambda_{\vec{A}, < n+1}$ and hence in A_n . $\square$

Thus the decreasing spatial weights of the layers have an exact conceptual purpose: each finite level of the ideal becomes robust at a definite positive tolerance, while the tolerance needed for all levels simultaneously tends to zero. The factor four is the bound proved by this particular choice of column weights and majority decoder; no optimality claim is made.

[merge] The filtration uses the quantitative radius $1/2$ of $\mathcal{J}^!$ (Lemma 21.2); neither Part I's code nor source 08's has a stated analogue.

27.2 Arithmetical information as a nonprincipal core

Example 27.2. Take $A_n = \emptyset^{(n)}$, with $\emptyset^{(0)} = \emptyset$. Then

$$\text{Core}(\Lambda_{\vec{A}}) = \{C : (\exists n) C \leq_T \emptyset^{(n)}\}.$$

This is the ideal of arithmetical sets. Every description computes every finite jump, separately. The presentation oracle $A_{\oplus} = \bigoplus_n \emptyset^{(n)}$ is not in this ideal: if $A_{\oplus} \leq_T \emptyset^{(m)}$, then its $(m+1)$ st column would contradict strictness of the jump. By the exact-pair construction, there are descriptions $D, E \leq_T A'_{\oplus}$ such that their common lower cone is precisely the arithmetical ideal. At least one fails to compute $A_{\oplus}$, since otherwise $A_{\oplus}$ would lie in their common lower cone. Hence the distinction between separate computations and one computation of the infinite join is witnessed inside a single density-zero class.

The construction uses only the usual finite-jump characterization of arithmetical sets; it does not supply uniform reduction indices from a description to all finite jumps.

[merge] For source 08's code the corresponding pair has jumps below $\emptyset^{(\omega)}$ (equation (16.1)), a stronger bound than $D, E \leq_T A'_{\oplus} = (\emptyset^{(\omega)})'$ here.

27.3 Even the zero ideal has presentation-dependent carriers

Proposition 27.3 (Noncanonical carriers). *A fixed cofinal sequence of degrees need not determine the coarse degree of its layered carrier. This fails already for the constant sequence of computable degrees.*

Proof. Given an arbitrary set H , let A_n be the constant binary function with value $H(n)$. Each A_n is computable, so the generated ideal is Comp and the sequence is increasing in Turing degree. For positive arguments, $\mathcal{J}^!(A_n)$ has constant value $H(n)$; only its argument 0 is set to zero by definition. Therefore $\Lambda_{\vec{A}}$ agrees with $\mathcal{R}(H)$ except possibly at the positions

$$\{p(n, 0) : n \in \mathbb{N}\} = \{2^n - 1 : n \in \mathbb{N}\},$$

a density-zero set. Thus $\Lambda_{\vec{A}} \approx \mathcal{R}(H)$, even though every level of its cofinal degree sequence is $\mathbf{0}$.

For $H = \emptyset$ the carrier is computable. For $H = \emptyset''$, Theorem 7.2 says that it has no computable coarse description. These carriers have the same core and the same level-by-level Turing degrees, but different coarse degrees. $\square$

This phenomenon is not an accident of the proof. The ideal records which sets are necessarily recoverable one at a time. It does not record the nonuniform information used to choose or present infinitely many such sets. The spectral insertion theorem exploits precisely that separation.

27.4 Principal ideals recover the known special cases

For $\mathcal{I} = \text{Low}(A)$, choose $A_n = A$ for all n . Then $\Lambda_{\vec{A}} \leq_T A$, while robust recovery gives $A \leq_T D$ for every description D . Consequently $\text{Spec}(\Lambda_{\vec{A}})$ is exactly the cone above $\deg_T(A)$. The insertion formula becomes

$$\text{Spec}(T_{\Lambda_{\vec{A}}}(H)) = \{\mathbf{d} : \deg_T(A) \leq \mathbf{d} \text{ and } H \leq_T D', D \in \mathbf{d}\}.$$

These principal-core constructions specialize the unattained-infimum and jump-cone mechanisms already present in the repository. The unrestricted nonprincipal realization and the resulting general fixed-core embeddings are the additional content of this manuscript.

[merge] Part I's Example 8.6 is the same formula for $\mathcal{E}(A)$.

28 Proof assurance and a formalization route

28.1 Dependency ledger

The strongest claims rest on a small number of separately checkable components.

Claim	Dependencies and status in this manuscript
Every core is a countable ideal	Countability of oracle programs; proof in Lemma 2.2 (source 10's Proposition 2.3).
Every countable ideal is a core	Explicit block/column coding plus published HJKS compactness; independently, explicit ideal-protected fusion.
Perfect exact descriptions	Finite-use persistence, decidable protected values, cross-disagreement search, and binary fusion; full proof given.
Core-preserving insertion	Finite-head approximants in the ideal; independently, the same fusion theorem on joined protected regions.
Exact spectral cut	Projection/interleaving of descriptions and the relativized limit lemma; both proved.
Upper-cone embedding	Uniform dyadic transformations and relativized Friedberg inversion; explicit inversion proof in source 10's Appendix A (Lemma 9.1 here).
No least representatives	Spectrum identity and the fixed-core theorem, with the distinction between least and minimal kept explicit.

No result of report 10 about hyperdegrees or forcing over admissible sets is used. No large-cardinal assumption, determinacy assumption, randomness theorem, or minimal-degree existence theorem is needed. Ordinary classical computability theory in ZFC suffices. The countable ideal is a given set; selecting a cofinal presentation is not an effective enumeration theorem.

28.2 Critical quantifier checks

There are four places where exchanging quantifiers would produce a false or unsupported statement.

First, the conclusion that a description computes A_n is for each fixed n , with its own finite stabilization parameter. It is not a uniform reduction to $\bigoplus_n A_n$. Second, the finite truncations of $\mathcal{R}(H)$ are individually computable even when no computable sequence of their program indices exists. Third, a no-disagreement argument produces a program for a common function only under the hypothesis that both designated functionals are total and equal; it does not decide totality. Fourth, the perfect family's splitting system is below one oracle, but its paths have the bound (24.4), not a uniform bound by that oracle alone.

The join proofs combine only finitely many outputs. The density arguments first fix a finite number of protected columns, take a limsup, and only then let the number of columns grow. They never interchange an infinite sum and a limsup without a tail bound.

28.3 Mapping the development to ProveIt

The current repository separates its foundational Turing-degree library from its coarse-degree library; its README and audit document admitted published inputs on the coarse side [8]. This manuscript does not claim to have rebuilt either library or certified new Lean files.

A practical formalization can be divided into three layers. The first consists of finite combinatorics: the exact column counts, density pullbacks, factorial-block majority bounds, and interleaving bounds. The existing files for columns, majority, blocks, and dyadic coding provide natural integration points, although the repeated-label block convention here should be reconciled explicitly with the repository’s convention rather than identified by name.

The second layer is the semantic one: an ideal of set degrees, a cofinal family, the carrier, finite truncations, and the implication from ideal-valued approximation to a core bound. A development accepting HJKS compactness as an explicit hypothesis can obtain the first proof quickly. Its theorem statement should expose that hypothesis and not hide it behind a global axiom.

The third layer formalizes the independent construction. It needs finite oracle strings, a use-bounded convergence predicate, persistence under extension, and a proof that admissibility is computable relative to a finite ideal level. The key interface is Lemma 23.2, not a generic appeal to forcing. A recursive family of finite task lists then gives the splitting system, followed by separate proofs of density, exactness, and continuity. This route removes dependence on HJKS compactness for the explicit realizations. The upper-cone embedding further requires a genuine jump and the relativized inversion argument.

[merge] The files meant are `Columns.lean`, `Majority.lean`, `Block.lean` and `Dyadic.lean` in `Computability/TuringDegrees/Lean/CoarseDegrees`. The repository’s convention is $\mathcal{J}$ (blockCode), which is not $\mathcal{J}^!$; nothing about $\mathcal{J}^!$ is formalized. See Section 1.4 for what those files prove.

28.4 Finite checks supplied with the paper

The accompanying Python script, shipped as `code/10-layered-realization-check_finite.py`, checks exact arithmetic and finite combinatorial components: the column partition and tail formula, the joined-tail formula, finite-head disagreement bounds, the block-majority lower bound, finite-prefix recovery cases, and a finite model of the cross-disagreement dichotomy. It uses rational arithmetic and exhaustive small cases where practical. The recorded output, shipped as `data/10-layered-realization-results.json`, states the number and range of checks actually run.

Those checks are not a verification of the infinitary theorems. In particular, they cannot establish a Turing nonreducibility, an infinite density limit, or the existence of the perfect family. Their purpose is to catch off-by-one mistakes and mismatches between the displayed coding conventions and the finite bookkeeping. The PDF was separately compiled and visually inspected; successful typesetting is also not a proof certificate.

[merge] “The PDF” is source 10’s delivered 22-page PDF, which is not shipped. The recorded run passed 616,836 finite cases in eight groups.

29 Source 10: further research questions

The following questions concern extensions not proved in this manuscript. They should not be read as assertions that every formulation has been established as open throughout the literature.

Question 29.1 (Classify spectra over a prescribed core). For a fixed countable ideal $\mathcal{I}$, characterize the upward-closed collections $\mathcal{S}$ of Turing degrees for which

$$\mathcal{S} = \text{Spec}(X), \quad \text{Core}(X) = \mathcal{I}$$

for some X . Necessarily $\mathcal{I}$ is the common lower cone of $\mathcal{S}$. Is there a useful additional descriptive or effective condition sufficient for realization? The family (24.3) supplies many examples but is not a full answer.

Question 29.2 (Minimal elements of a spectrum). Which of the spectra $\text{Spec}(T_{\Lambda_{\mathcal{I}}}(H))$ have minimal elements? Can one choose an ideal presentation and a parameter for which the spectrum has no minimal element at all? A proof must lower a degree while retaining a description, not merely exhibit an incomparable exact partner.

Question 29.3 (Remove or lower the embedding threshold). Can the upper-cone embedding for a fixed ideal be extended to an embedding of the entire Turing-degree structure? More modestly, can the threshold $A'_{\oplus}$ be lowered for useful classes of presentations, or replaced by an invariant of the ideal rather than of the chosen presentation? The present proof needs a jump inverse that computes the carrier.

Question 29.4 (Effective comparison of presentations). Give sufficient uniformity hypotheses on two cofinal presentations of the same ideal that force their carriers to be uniformly or nonuniformly coarsely equivalent. Proposition 27.3 shows that merely having the same sequence of Turing degrees is insufficient. Uniformly supplied cofinal maps and reduction indices are a natural starting point, but the totality requirements in a coarse transformation must still be proved.

Question 29.5 (Sharper witness complexity). For a carrier computable in $A_{\oplus}$, when can the exact descriptions in Theorem 23.4 be placed strictly below $A'_{\oplus}$? Can one simultaneously control their jumps, degrees, or definability while retaining the prescribed nonprincipal common lower cone? The present algorithm makes $\Sigma_1^0(A_{\oplus})$ extension decisions and therefore gives an $A'_{\oplus}$ bound, not an optimality result.

Question 29.6 (Quantitative error profiles). Can one prescribe the entire family $r \mapsto \mathcal{K}_r(X)$, rather than only its union? Theorem 27.1 realizes a cofinal filtration within a factor four in radius. Determine sharp constants, realizable discontinuities, and the effect of changing column weights or block codes. A stronger version would characterize which decreasing families of ideals arise in this way.

Question 29.7 (Small disagreement budgets with nonprincipal cores). Can the perfect exact-pair family be made to agree off one set obeying a prescribed unbounded sublinear counting budget, while retaining an arbitrary nonprincipal ideal as its common lower cone? The present protected-column construction gives density zero and arbitrarily small prefix-density distance, but not an arbitrarily slow prescribed bound on the number of disagreements. The repository's one-bit-reserve budget constructions suggest a different method, whose compatibility with rising ideal bounds remains to be established.

Question 29.8 (Restrictions on representatives and reducibilities). Which ideals can be realized when X is required to lie in a fixed definability class, or to satisfy an effective randomness condition? Which parts of the realization and fixed-fiber results survive for effective-dense descriptions, where wrong answers are forbidden? A separate question concerns hyperarithmetic common information: report 10 identifies a failure of the Turing compactness mechanism there, so a transfer would require a genuinely different proof.

Question 29.9 (A checked construction rather than an admitted compactness theorem). Implement the finite-use oracle interface and ideal-protected fusion in Lean, then derive the core-realization theorem and perfect exact-pair theorem without importing an admitted cone-avoidance result. This is a specific mechanization project with a small mathematical interface; it would strengthen assurance without changing the statements proved here.

[merge] Status of these questions in the merged report. Question 29.1 is Question 11.1 and Question 19.1; it stays open. For Question 29.3, Theorem 9.2 lowers the threshold for this carrier from $A'_{\oplus}$ to Λ'_A (and for any base set X to X'); an embedding of all degrees, or a threshold invariant of the ideal, remains open. For Question 29.5, source 08 answers the analogue for its own code: its computable-parameter descriptions have jumps below $\Theta_A \leq_T A'_\oplus$ (Theorem 12.2), so they are strictly below $A'_\oplus$; for the layered carrier and general protected presentations the question stays open. Question 29.7 is source 08's Question 19.4. For Question 29.9, the Lean library still has no use-bounded model of Turing functionals (Section 1.4). The others remain open.

29.1 Source 10's conclusion

The countable-ideal part of the repository's coarse-core classification problem has a complete answer: every countable Turing ideal occurs, and no other kind of family can occur. The construction has two complementary explanations. Each ideal level is encoded robustly on a positive-density region, so it survives every coarse description. At the same time, finite initial collections of levels approximate the carrier arbitrarily well, preventing extra information from becoming common to all descriptions. Independent reservoir fusion realizes this exclusion as a perfect exact-pair family and gives a presentation-relative effective bound.

The core is nevertheless a coarse invariant of limited discriminatory power. Dyadic insertion leaves it unchanged while imposing an arbitrary jump-cone cut on the description spectrum. Relativized jump inversion makes those cuts order-reflecting above a presentation-dependent threshold. Thus every countable core fiber contains a join-preserving copy of a Turing upper cone, with continuum many examples having no least representative. The distinction between common information and the full computational strength of an approximation problem is therefore structural, not merely a feature of the computable-core examples.

Appendices

A Proof audit and quantitative checks

A.1 Critical logical checkpoints

The following checkpoints summarize the issues most likely to create a false stronger statement.

Finite constants are not a uniform algorithm. The finite patch in Theorem 3.1, the stable row in Proposition 4.3, and the finite truncations in Theorem 7.2 are used as witnesses to individual Turing reductions. The paper never claims that their indices or stabilization stages are computable uniformly from the displayed oracles.

Whole-row stabilization is stronger than pointwise convergence. The decoding formula (6.3) uses the value at a coordinate that itself changes with the stage m . It is justified by $\exists s_i \forall s \geq s_i \forall n$, not by $\forall n \exists s_{i,n} \forall s \geq s_{i,n}$. The block-of-repetitions code was chosen specifically to supply the stronger order of quantifiers.

Countably many error sets require the tail estimate. Nullity of every column's error set does not by itself follow from a countable union theorem. The proof first handles finitely many columns, then bounds *all* remaining errors by 2^{-k} , and only then lets k grow. The exact floor formula in (4.2) is the quantitative certificate.

Core realization needs both inclusions. Robust recovery of each generator proves only $\mathcal{I}_{\vec{A}} \subseteq \text{Core}(X_{\vec{A}})$. The upper inclusion is a separate application of robust radius to the finite-column truncations. Omitting it would leave open the possibility of unintended common information across the columns.

The reverse embedding uses a relative jump inverse. The witness B must satisfy both $B \geq_T X$ and $B' \equiv_T K$. These properties ensure that B actually solves the target base-plus-code problem before its jump separates H from K . Lemma 9.1 proves exactly this pair of requirements.

Different partial orders are kept separate. The notions of least member, minimal member, infimum, cofinal family, and coinital family are not interchanged. In particular, Corollary 9.4 concerns coarse degrees in a fixed-core collection; Corollary 2.5 concerns Turing degrees within one spectrum.

A.2 Finite verification artifacts

The companion program, shipped as `code/12-ideal-realization-finite_checks.py`, uses only Python's standard library. It checks the column partition and exact tail counts, majority-error thresholds, the index bound used for whole-real stabilization, and finite simulations of the assembly and presentation examples. It also checks the finite-prefix triangle and patch bounds used in the local decoder. A machine-readable result is provided, shipped as `data/12-ideal-realization-results.json`.

[merge] Source 12 names these files `checks/finite_checks.py` and `checks/results.json`; they are shipped under the names above, byte-identical to the delivery. Parts II and III describe their own checks (Sections 18.5 and 28).

These are finite combinatorial checks. They do *not* verify the Baire category theorem, Turing reducibility claims, convergence at unbounded stages, jump inversion, global novelty, or any full theorem of this paper. The mathematical assurance for those assertions is the written proof, pending independent review or formalization.

B Source ledger and reproducibility

The source-level comparison used the following repository snapshot:

Repository: <https://github.com/VladimirReshetnikov/ProveIt>
Commit: 1a1396d4d3a2ac6812692df520517d86ba3a4785
Primary research source: Computability/TuringDegrees/Research/CoarseDegrees/research-synthesis/Turing_Degrees_Synthesis.tex

The relevant source locations are its section on spectra, its category section on robust radius, and its final subsection on unsettled questions. In particular, the classification question is explicitly present in that final subsection. The repository is a research source, not a substitute for independent verification. All ingredients used here have been proved in this manuscript, apart from standard basic facts about oracle computation, the Turing jump, and Baire category.

The targeted literature check covered the original coarse-reducibility paper [2], the dyadic criterion [3], the countable-column antecedent [4], the robust-coding framework [5], and the motivating question [6]. The comparison does not certify absence of these conclusions from all other papers, preprints, theses, or unpublished notes.

The archive includes the self-contained source `article.tex`, the compiled `article.pdf`, a three-pass `build.sh`, a concise proof-status file, a source ledger, and the finite checks. The bibliography is embedded in the TeX source; no external `.bib` file or network access is needed to rebuild. The build requires a standard L^AT_EX installation with the packages listed in the source. No Lean verification is included or implied.

[merge] The paragraph above describes source 12’s delivered archive. In this report its proof-status file and source ledger are `12-ideal-realization-PROOF_AUDIT.md` and `12-ideal-realization-SOURCE_LEDGER.md`, its build script is `code/12-ideal-realization-build.sh`, and its build record is `data/12-ideal-realization-validation.json`; its `article.tex` is the base of this report’s `article.tex`, and its PDF is not shipped. The source-level comparison above did not consult the programme’s research plan, which states the classification as target C4 [7], or the Lean library (Section 1.4).

C Provenance of this report

[merge] This appendix was written for the merge.

Sources. The report is built from three manuscripts of batch 37 of ProveIt’s intake, all delivered in the arrival commit 37e61c1fd and placed in 0e53d1034 (“Place twelve manuscripts: ...”), which staged source 12’s `article.tex` and `README.md` unprefixes and every other delivered code, data and audit file under the prefixes below. Each manuscript is pinned to ProveIt commit 1a1396d4d3a2ac6812692df520517d86ba3a4785; the programme’s plan, synthesis and Lean library are unchanged between that commit and the placement commit. All three are dated 28 September 2026, prepared for Vladimir Reshetnikov, and were delivered with a 22-page PDF each.

Source	Archive (inner directory)	Prefix	Printed as
12	ProveIt_Coarse_Ideal_Realization (same)	12-ideal-realization-	Part I: Sections 1–11, Appendices A–B
08	ProveIt_Coarse_Core_Realization (same)	08-effective-exact-pairs-	Part II: Sections 12–19
10	coarse_core_realization (same)	10-layered-realization-	Part III: Sections 20–29

The directory number 12 follows the programme’s rule (numbering by archive modification time, using the earliest member of the merge, source 08); it is not a manuscript number.

What each source contributed. Source 12: the robust radius proved from Baire category; ideal realization for an arbitrary generator sequence with $X_{\vec{A}} \equiv_T A_{\oplus}$; the whole-row spectrum and its bounds; core neutrality for joins with sets of coarse computability bound 1 over an arbitrary base; the exact jump filter; strict extensions and unattained infima; the join-preserving upper-cone embedding over an arbitrary base; continuum many spectra per core; no maximal member and no countable cofinal family in a fiber; eight questions. Source 08: the two-level code $Q(\vec{A})$ and its computable retraction normal form (Medvedev equivalence with the row-finite mass problem); the finite-extension mechanism with oracle $\Theta_{\vec{A}}$; a perfect family of pairwise exact descriptions with jump bound $\Theta_{\vec{A}} \oplus P$ and one common null envelope with an explicit modulus; the computable-mask obstruction; a self-contained proof of realization and a short proof from [2, Theorem 3.7]; the arithmetical pair with jumps below $\emptyset^{(\omega)}$; the interleaved fiber codes F_H with spectrum factorization, uniform monotonicity and order reflection; oracle accounting and prescribed-jump exact pairs; continuum many no-least classes per core; no uniform extraction of noncomputable common information; eight questions. Source 10: the layered carrier $\Lambda_{\vec{A}}$ with the quantitative recovery radius $1/2$ and the restriction bound; the ideal approximation lemma; ideal-protected fusion, a second proof of realization without compactness, with perfect families close in the prefix metric and splitting systems computable in $(A_{\oplus} \oplus H)'$; core-preserving insertion by approximants in the ideal; perfect antichains of descriptions; the factor-four filtration $\mathcal{K}_{2^{-n}}(\Lambda_{\vec{A}}) \subseteq \text{Low}(A_n) \subseteq \mathcal{K}_{2^{-n-2}}(\Lambda_{\vec{A}})$; noncanonical carriers; nine questions.

Where the merge had to choose.

- *Base.* Source 12, for the reasons in Section 1.4: its neutrality and embedding theorems hold over an arbitrary base set, it takes arbitrary generator sequences, and it proves the robust radius. Its title and numbering are kept; Parts II and III are appended and renumbered by a fixed shift.
- *Duplicates printed once.* Sources 08 and 10 reprove, with the same arguments, the elementary properties of cores and spectra, the binary-join lemma, the infimum criterion, the relativized limit lemma and relative Friedberg inversion, and source 10 also the dyadic criterion and uniform dyadic transport. These are printed only in Part I, with merge notes. Source 10’s embedding theorem is printed as a statement with its proof summarized, because its proof is Part I’s. Source 10’s Appendix B (notation) is absorbed into Section 1.5.
- *Second routes kept.* The black-box routes from [2, Theorem 3.7] (sources 08 and 10), row-finite forcing (source 08), ideal-protected fusion (source 10), and source 10’s ideal-approximant proof of core preservation are printed in full and marked.

- *Codes kept apart.* The three codes $X_{\vec{A}}$, $Q(\vec{A})$ and $\Lambda_{\vec{A}}$ are different sets, so their realization theorems, arithmetical examples and constant-row examples are all kept, each in its part.
- *Notation.* The renamings of Section 1.5; in particular source 10's carrier $X_{\vec{A}}$ became $\Lambda_{\vec{A}}$, and the three block codes are $\mathcal{J}$, $\mathcal{J}^{\text{sh}}$ and $\mathcal{J}^!$.
- *Priority.* All three sources present realization as answering an open question of the synthesis without mentioning plan target C4, and sources 08 and 10 omit [4]. The merge credits both (Sections 1 and 1.4); every source's own priority caveat is kept.
- *Merge observations.* Remark 13.5 (the whole-row spectrum is contained in the row-finite one) and Remark 17.7 ($F_H \equiv_{\text{uc}} T_{Q(\vec{A})}(H)$) are new, unrefereed, and appear in no source.

Shipped files. Besides this `article.tex`, its build `article.pdf` and the report `README.md`, the report directory ships each source's proof audit and source ledger (six files), its build script and finite checks in `code/` (six files), and its recorded outputs and build records in `data/` (five files), all byte-identical to the delivery. The report `README.md` maps delivery names to shipped names. Not shipped: the three delivered PDFs; the delivered READMEs (source 12's was staged at placement and is replaced by the report `README`); the manuscripts of sources 08 and 10, printed here as Parts II and III; and source 08's checksum manifest. They survive in the arrival commit.

References

- [1] Vladimir Reshetnikov's Provel repository. *Coarse Classes Without Least Turing Degrees: Deduplicated Results of the Nine Research Reports on Target C1*. Research synthesis, September 2026. Inspected at commit 1a1396d4d3a2, 28 September 2026 (Pacific time). Source path and full revision identifier in Appendix B. <https://github.com/VladimirReshetnikov/Provel/tree/1a1396d4d3a2ac6812692df520517d86ba3a4785/Computability/TuringDegrees/Research/CoarseDegrees>.
- [2] Denis R. Hirschfeldt, Carl G. Jockusch, Jr., Rutger Kuiper, and Paul E. Schupp. Coarse reducibility and algorithmic randomness. *Journal of Symbolic Logic* **81** (2016), 1028–1046. <https://doi.org/10.1017/jsl.2015.70>. Preprint: <https://arxiv.org/abs/1505.01707>.
- [3] Carl G. Jockusch, Jr., and Paul E. Schupp. Generic computability, Turing degrees, and asymptotic density. *Journal of the London Mathematical Society* **85**(2) (2012), 472–490. <https://doi.org/10.1112/jlms/jdr051>. Preprint: <https://arxiv.org/abs/1010.5212>.
- [4] Denis R. Hirschfeldt, Carl G. Jockusch, Jr., and Paul E. Schupp. *Coarse computability, the density metric, Hausdorff distances between Turing degrees, perfect trees, and reverse mathematics*. Preprint, 2021. References to Lemma 3.10 and Theorem 3.11 use the inspected arXiv text. <https://arxiv.org/abs/2106.13118>.
- [5] Damir D. Dzhamalov and Gregory Igusa. Notions of robust information coding. *Computability* **6**(2) (2017), 105–124. <https://doi.org/10.3233/COM-160059>. Preprint: <https://arxiv.org/abs/1406.2982>.
- [6] Peter M. Gerdes. *Comparing Notions of Dense Computability on ω^ω and 2^ω* . arXiv:2508.06925v1, 2025. Question 7. <https://arxiv.org/abs/2508.06925v1>.
- [7] Vladimir Reshetnikov's Provel repository. *Turing Degrees: A Unified Research Report*, second edition, 18 September 2026 (the programme's research plan), target C4 “Is every countable Turing ideal a core?” and its Proposition (proposed), label `prop:idealcore`. `Computability/TuringDegrees/Research/CoarseDegrees/research-plan/turing_degrees_unified.tex`, at commit 1a1396d4d3a2 and unchanged since.
- [8] Vladimir Reshetnikov's Provel repository. *Turing degrees in Lean and Rocq*, project `README`, `Computability/TuringDegrees/README.md`, inspected by sources 08 and 10 at commit 1a1396d4d3a2.

- <https://github.com/VladimirReshetnikov/ProveIt/blob/1a1396d4d3a2ac6812692df520517d86ba3a4785/Computability/TuringDegrees/README.md>.
- [9] Vladimir Reshetnikov’s ProveIt repository. *Coarse Classes, Hyperdegrees, and the Cone Filter*, CoarseDegrees research report 10, README and status notes, at commit 1a1396d4d3a2. Cited by source 10 to delimit, not support, its Turing-theoretic claims.
<https://github.com/VladimirReshetnikov/ProveIt/blob/1a1396d4d3a2ac6812692df520517d86ba3a4785/Computability/TuringDegrees/Research/CoarseDegrees/research-reports/10/README.md>.
- [10] Johanna N. Y. Franklin and Reed Solomon. *Lowness for isomorphism, countable ideals, and computable traceability*. Preprint arXiv:1909.06003v1 (2019), Section 2, especially the row-finite exact-pair requirements and the no-disagreement verification. <https://arxiv.org/abs/1909.06003v1>.
- [11] Richard Friedberg. A criterion for completeness of degrees of unsolvability. *Journal of Symbolic Logic* **22** (1957), 159–160. <https://doi.org/10.2307/2964177>. The original research article, not the later review with a different DOI.
- [12] David R. Belanger. *Conservation theorems for the Cohesiveness Principle*. Preprint arXiv:2212.13011 (2022). Cited by source 08 for contemporary context on jump-inversion methods, not as a dependency of the inversion proof. <https://arxiv.org/abs/2212.13011>.